BSCS FINAL PROJECT

Intelligent Secure Meditation Counselor Patient Care System with AI Virtual Assistant

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Abstract

BENZI.AI addresses the growing need for secure, accessible, and intelligent digital mental healthcare by reducing the manual workload of therapists and improving continuous support for patients. Traditional counselling systems often depend on phone based scheduling, paper records, limited progress tracking, and separate communication tools, which can lead to appointment conflicts, weak patient engagement, privacy concerns, and inefficient care delivery. This project develops a secure web based therapist patient care platform with three role based portals for patients, therapists, and administrators, integrating appointment scheduling, encrypted communication, medical record management, goal tracking, subscription handling, payment processing, and AI assisted support. The system is built using a MERN stack architecture with React.js, Node.js, Express.js, MongoDB Atlas, JWT authentication, RBAC, AES 256 encryption, Stripe payments, notification services, and a fine tuned Ollama Llama AI model connected through a context aware RAG pipeline. Testing was performed across nine major use cases, including login, registration, appointment booking, encrypted record access, goal completion, notifications, payment failure handling, therapist verification, and AI assistant guardrails. The project reports 100% implementation of planned modules, with all core objectives completed, including secure access control, AI powered virtual assistance, analytics dashboards, and responsive multi role interfaces. The result is a scalable and privacy focused mental wellness platform that supports therapists in managing care more efficiently while helping patients access personalized guidance, track progress, attend secure consultations, and remain engaged throughout their therapy journey.

Dedication

I dedicated this project to my mentor and senior at Single Solution Software House, who inspired me to work on an AI based mental health project. Their guidance helped me understand how artificial intelligence can support future mental healthcare systems and create meaningful real world impact.

I also dedicated this work to my project advisor, who supported my idea and encouraged me to develop it as an industry level project. Their feedback and motivation helped me improve the concept and focus on security, patient care, and AI powered assistance.

Finally, I dedicate this project to everyone facing mental health challenges. I hope this work becomes a small step toward making mental healthcare more accessible, private, and supportive through technology.

Acknowledgements

First, I am thankful to Allah Almighty for giving me the strength, knowledge, and patience to complete this final year project successfully. This project would not have been possible without His blessings and guidance throughout the development process.I would like to express my sincere gratitude to my project advisor, Mam Misha Asif, for her continuous support, valuable feedback, and encouragement. Her guidance helped us improve our idea and develop it into a practical, secure, and industry level mental health care system.

I am also thankful to my mentor and senior, whose suggestion inspired me to choose an AI based mental health project. Their professional guidance helped me understand the future scope of artificial intelligence in healthcare and motivated me to work on a meaningful real world solution.I would also like to thank my project team members for their dedication, teamwork, and cooperation throughout the project. Their efforts in development, testing, documentation, and problem solving played an important role in completing this system.

Finally, I am grateful to my family, friends, and teachers for their constant motivation, prayers, and support. Their encouragement helped me stay focused and complete this project with confidence.

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Revision History

Name	Date	Reason For Changes	Version

Chapter 1. Introduction

The Intelligent Therapist Patient Management System (ITPMS) is a web based, highly secure management system that includes an AI Virtual Assistant and was designed specifically to streamline therapist/patient communication, help manage therapy sessions, and belongs to a secure platform. The importance of mental health care is growing, yet managing mental health care as an effective therapy is a big challenge. There are still many therapists who are using cumbersome and inefficient methods (phone calls, paper records etc.) which are time consuming and labor intensive and are easily prone to scheduling mistakes. It can be hard for patients to schedule appointments, keep on top of their progress, and ensure privacy during therapy. To address these concerns, this project aims to develop a comprehensive and intelligent online system that allows therapists to register their patients, maintain comprehensive medical records, schedule sessions flexibly, and conduct them in an encrypted manner. The results of this will be allowing patients to find a therapist in no time, patients (and therapists) will be able to see the availability of any appointment almost instantly, patients will easily be able to schedule and pay for appointments, and patients will be able to receive care anonymously, without stigma. One way the app is expected to support patients in maintaining their treatment is its AI driven virtual assistant, which will offer patients continuous support via personalized recommendations and reminders, all based on data from their medical history, allergies, and therapist notes collected under each profile. The combination of automation, Artificial Intelligence driven personalization, and privacy protections that are at the core of the system results in a reduction of the administrative burden on therapists as well as greater patient engagement in therapy, improved adherence to treatment plans, and higher overall treatment outcomes. Not only will it reduce the demand for digital, accessible and private mental health services, a growing trend, it also will produce a more effective, efficient and reliable tool for patients to get therapy. The complete software specifications that follow will be defined in this document.

1.1 Product (Problem Statement)

Today's leadership in mental health care delivery faces a multitude of overarching challenges that hinder access to effective therapy and hinder involvement in mental health care by patients. Despite manually used methods, most therapists are stuck with phone scheduling and paper records to follow patients' progress, and this can lead to inefficiencies, constant time consuming manual entries, incorrect appointment schedules, and even some patients avoiding treatment. In addition to that, if patients have difficulty scheduling their own appointments, tracking their progress towards recovery, finding privacy during sensitive appointments, etc; many patients will avoid any type of mental health care completely. Plus, most of the existing digital platforms typically lack robust privacy features (e.g., anonymous communication and personalized support for each individual user's medical history). This results in missed appointments, reduced compliance with therapy, treating doctor burnout and reduced mental health outcomes for patients. The Intelligent Therapist Patient Management System will offer a safe and automated intelligent web based platform, it will be used to support real time scheduling of appointments and communication between patients and therapists will be encrypted, while communication with patients will be supported by use of AI to provide each patient with a customised support experience. By doing so, the Intelligent Therapist Patient Management System will improve the accessibility, effectiveness and reliability of mental health therapy by not only providing enhanced privacy and improving patient engagement but also eliminating all the inefficiencies associated with manual scheduling procedures.

1.2 Background

This project aims to highlight the emerging area of modern digital mental healthcare, which merges telehealth and medication management to tackle a major inefficiency in traditional approaches: the treatment of patients who are often simultaneously suffering from multiple chronic illnesses, including mental health issues. While platforms have been more accessible during remote sessions, these platforms are often isolated, and there aren't always seamless, connected solutions that support an ongoing trend of care for the patient and clinical solutions for a therapist. This disconnect results in dehumanized patient care, burdensome administrative tasks for providers, and ongoing privacy concerns impacting the patient care experience and ultimately deterring patients from seeking treatment. Our project aims to maximize the potential of teletherapy by giving AI intelligent capabilities that automatically handle the "Positional Task Force", offer a mechanism for tracking team progress and make all services truly customized with context aware AI, thereby increasing patient engagement and therapeutic outcomes within a single safe ecosystem.

1.3 Objective(s)/Aim(s)/Target(s)

The aim of the proposed system is to develop an AI based Secure Meditation Consultant Patient Support System which will help in enhancing the technological level of providing consultation services by Mental Health professionals and provide automation in Individualized Counselling with utmost security and protection of the confidential information of patients and counsellors. The Secure Meditation Consultant Patient Support System provides Counsellors with tools for creating, tracking and communicating with their clients via a secure online environment. The most up to date information about sessions and other progress in therapy items will be forwarded to the clients anonymously/anonymous with a unique virtual identity. In addition, clients receive AI powered suggestions for their service delivery process right from a single highly secure web based platform. Summarizing for the purposes of the Secure Meditation Consultant Patient Support System, the following are the key objectives:

1.3.1 Develop a Secure Therapist Patient Management Platform

To create a centralized portal with encrypted storage and restricted access that enables therapists to register, manage, and keep an eye on patient profiles and therapy sessions while ensuring total data confidentiality.

1.3.2 Implement Automated Appointment Scheduling

To reduce scheduling conflicts and no shows, a dynamic booking system that automatically verifies therapist availability, avoids double booking, and provides real time confirmations should be designed.

1.3.3 Integrate Secure Communication and Consultation Channels

To reduce scheduling conflicts and no shows, a dynamic booking system that automatically verifies therapist availability, avoids double booking, and provides real time confirmations should be designed.

1.3.4 Incorporate an AI Powered Virtual Assistant

To incorporate an intelligent assistant that can evaluate session data, therapist notes, and patient medical histories to offer tailored, context aware advice and reminders about mental health.

1.3.5 Enable Goal Assignment and Progress Tracking

To enable patients to track their progress using a gamified points based system that promotes engagement, and to let therapists assign customizable wellness or therapeutic goals.

1.3.6 Establish Secure Medical Record Storage

To keep a secure database of therapy notes, prescriptions, and clinical reports that is only accessible by authorized personnel and complies with accepted data security procedures.

1.3.7 Implement Robust Authentication and Access Control

To create a secure authentication system that protects user data from unwanted access by utilizing session management, role based access control (RBAC), and encrypted credentials.

1.3.8 Ensure Comprehensive System Security

Enforcing secure API communication, end to end encryption for data in transit and resting, and ongoing monitoring to preserve information integrity, confidentiality, and dependability.

1.3.9 Automate Notifications and Reminders

To enhance communication by using integrated email and SMS services to automatically notify people about therapy exercises, upcoming appointments, and medication schedules.

1.3.10 Provide Data Driven Progress Analytics

To produce AI based reports and visual progress dashboards that let patients and therapists assess progress, spot trends in behaviors, and modify treatment plans as necessary.

1.3.11 Enhance Accessibility and User Experience

To create a user friendly, responsive, and intuitive web interface that is optimized for a variety of devices, guaranteeing a smooth experience for patients and therapists of all technical backgrounds.

1.4 Scope

This document will be used to describe a broad based, web based platform to monitor, protect and support the entire therapist to the client continuum of care (including the interaction between the therapist and client), and to protect and augment the overall therapist to client connection. The Therapist Portal will enable therapists to make appointments, store secure medical records, enlist patients and more, while the Patient Portal is expected to offer patients the ability to view their medical records, book real time appointments, pay with integrated payment systems and monitor their progress through a gamified goal setting system. At the heart of the platform will be an AI Virtual Assistant (VAS) that will be able to analyze the patient’s profile and offer recommendations based on the context of the patient's treatment. The platform will also feature a joint feedback mechanism, where PCPs can give and receive feedback from patients; enable patients to connect anonymously with their therapist and/or PCP; and enable secure video calls for delivering therapy, with the video calls being end to end encrypted. The platform will not involve AI automated therapy, support for native mobile applications or native mobile interfaces (apart from a responsive web application), handling of third party medical devices, direct medical diagnosis (by phone etc.) and/or filing of claims to the insurance.

1.5 Business Goals

The AI Integrated Secure Meditation Counsellor Patient Care System aims to be not just a technical solution but a viable and scalable product for the digital health market that addresses specific business and corporate needs. The key business objective of the system is carrying out core business processes, including patient registration and appointment scheduling, progress tracking, session management, billing, etc., depending on the core business, in an automated way, allowing a healthcare organization and its mental health workers to manage and grow their business more efficiently. Therapists can save valuable time and effort on administrative tasks, allowing them to focus more on the quality of their services and their own productivity.

A second important objective for business is providing a subscription model for therapists and Counselling Professionals. The platform offers tiered subscription models with optional free trial periods to provide a steady stream of recurring revenues and enable professionals to test the value of the system before committing to implementation solutions. This model can help achieve long term financial sustainability and ongoing system improvements. In addition, built in payment processing of therapy sessions makes it easier to conduct payments, decreases payment delays, and enhances cash flow for the service supplier.

The system can also be used as a digital discovery platform and engagement platform, where patients can simply find licensed therapists in accordance with availability as well as expertise. This makes therapists more visible and accessible to clients, helping them to fill their pipelines beyond their initially defined geographic scope. The platform's corporate goals are to establish itself as a reliable and privacy centric mental health technological brand with focus on security, responsible AI application and regulatory adherence (HIPAA/GDPR). These features are decisive in building customer trust and are fundamental for credibility in the market.

In addition, the business plan is based on scalability and expansion. Future additions to the system include the possibility of developing an app and add on wellness services powered by superior AI analytics without a brute force redesign of the system. The platform is modular and leverages a cloud infrastructure, so it can grow to meet the demands of its growing user base while maintaining its performance and reliability. The overall vision of the business objectives is to develop an intelligent, commercially viable, and secure digital mental health ecosystem that will benefit both therapists and patients, along with the various stakeholders, while facilitating the progression of the digital mental health market in the long run.

1.6 Challenges

As BENZI is being developed.One of the primary challenges faced was designing a secure system to manage sensitive mental health information. Due to its handling of patient records, therapy notes, AI interaction logs, appointments, and patient information, robust security measures like encrypted communication, secure authentication, role based access control, and restricted data access had to be implemented on the platform. Accessibility controls, featuring user roles and permissions, were meticulously planned so that each user could access only their respective features, ensuring the confidentiality of patient information, therapists, and administrators.

Then the hurdle was to make various modules into one platform. The system features patient management, therapist verification, appointment scheduling, confidential chatting, video sessions, safe payment, notifications, progress tracking and AI virtual assistant. Proper architecture, testing and debugging were needed to link these modules together with the Frontend, Backend, Database, AI engine and third party services. Testing the AI module proved particularly difficult; it had to be able to make recommendations and provide helpful feedback, yet it could not give a clinical diagnosis or a response that was unsafe.

Creating a user friendly interface for the mental health platform was another challenge. The interface had to be simple, calm, responsive and easy to use for patient and therapist backgrounds both with and without tech. Real time capabilities were also a set of important features for the system: appointments updates, chat, video sessions, reminders all of them needed to be implemented and work reliably and efficiently. The integration of these various technical, design, and security considerations presented a challenge in the project, yet often added significant benefit.

1.7 Learning Outcomes

In this project, we got to know the process of designing and developing complete healthcare software system from initial design to implementation at the industry level. All the web app pages were designed manually in Figma prior to development, such as Patient, Therapist, Admin, authentication, dashboard, appointment, reports, goals, profile and public pages. The main takeaway from this was knowledge about user flow, screen structure, UI consistency, user navigation plans, and how we need to create calm screens for a mental health platform that makes it easy for the users.

Once the design process was finished, we started building the actual app using the MERN stack, understanding the nuances of taking the Figma design to a functioning app. We gained an understanding of how to leverage React JS, Node JS, and Express JS for Front End development, associating Back End services, and MongoDB Atlas as a Database. We also learned how to create various layers for the system like presentation layer, application layer, AI engine layer, data layer such that system will scalable, easy to maintain and can be extended very easily in the future.

Also, I gained an understanding about the necessary security and privacy in healthcare based applications. This project gave us insights into JWT authentication, role based access control, AES 256 encryption, secure API communication and access restrictions for patient sensitive data. We also explored the ethical applications of AI in evaluating and providing recommendations for mental health, specifically restricting the use of the AI to support mental health rather than providing a clinical diagnosis or medical prescription.

We also gained insights into how AI can be integrated into real world software systems. We were taught how a customised Ollama Llama model for summarization and a RAG pipeline that is built by using provided context can be used to deliver personalised support using patient records, therapist notes and the history of sessions. In addition to this, we learned how to document the software, draw diagrams using the UML and how to test, plan deployment, integrate the payment gateway and notification, teamwork and complete project management.

1.8 Nature of End Product

The result, BENZI.AI is a safe, online healthcare and mental health management system. Built with the three user roles of Patient, Therapist and Administrator in mind. Patients can sign up, schedule appointments, connect to therapists, take part in online therapy sessions, converse with the AI virtual assistant, monitor objectives assigned by therapists, access progress analytics, and adjust their profile. Through the therapists can book prospects, keep patient data, assign wellness goals, view reports, track development, and communicate securely with individuals.

The result is more than just a basic prototype, but a fully developed digital platform incorporating various modules. Not only does it have appointment planning, protected medical record management, AI assistance, role dependency dashboards, subscription management, Stripe payment processing, notifications, analytics, goal tracking, but also video consultation support. It is a responsive Web app, so it can be responded to through any present day Web browser on a mobile, tablet, or computer.

BENZI.Moreover, AI can also be scaled and future proofed. It also has a modular design, meaning that new additions, like apps, more advanced AI analyses, further wellness services or even integrations with other health care providers can be added in the future. It is ideal for psychotherapy centers, counselling centres, mental health clinics, and digital healthcare communication service providers that require a comprehensive and secure platform for therapist patient communication and care management.

1.9 Related Work/ Literature Survey/ Literature Review

In traditional systems of patient care, a therapist will rely on manual methods about patient care such as calling patients, using paper based records, paper patient lists and separate communication tools. These processes may cause scheduling problems, missed appointments, record management problems, and privacy problems. The same issues become more urgent in relation to mental health services, where clients usually need the need for confidentiality, ongoing support and quick access to professional assistance.

The access to telemedicine and online counselling has been enhanced with existing solutions, but many operate as standalone. Some platforms are geared toward making video calls, others deal only with booking appointments, chats, or records. Many don't offer a fully integrated system to match therapist management, patient progress tracking, secure medical records, AI powered recommendations and reminders, payments, and analytics all on a single platform. This gap is a problem in patient care, therapist workflow and ongoing monitoring of patient progress.

Another significant field of study and development is the implementation of AI in mental health, using chatbots and tools designed to meet mental health needs. Many AI tools, however, offer generic answers and may not be linked to an individual's therapy experience, goals, or therapist notes. BENZI.AI helps overcome this limitation by applying context aware AI virtual assistant powered by an RAG pipeline. It is not designed to replace therapists but can help support patients and therapists, offering individual guidance, reminders, and insight into progress while still active preserving security, privacy, and ethical boundaries in AI usage.

1.10 Document Conventions

This final document follows a structured and standardised format, which was modeled after that from IEEE Software Requirements Specification (IEEE 830 1998). It maintains its layout simple and uniform for readability, accuracy and convenience of reference. The following conventions have been used throughout the document:

1.10.1 Font Style and Size:

The graphics are used in bold 12 font with times heading, bold as professional and clarity for the main text using arial font for size 11 and regular (size 12) for subheading.

1.10.2 Numbering Scheme:

Each section and subsection is numbered hierarchically, e.g., 1.1, 1.2, 1.3 etc., to facilitate the use of the organization and flow of information.

1.10.3 Terminology:

Definitions of core elements or entities in the project, like Therapist, Patient, System, AI Assistant and Anonymous Mode, are defined using capital letters.

1.10.4 Symbols and Notations:

For comparison tables the ☑ (tic) and ☐ (cross) are used to show whether or not a specific feature is present. Acronyms like AI (Artificial Intelligence), API (Application Programming Interface) and RBAC (Role Based Access Control) are acronymically defined the first time they appear.

1.10.5 Language Style:

The language is formal, technical, and concise appropriate for academic and professional environment. Where appropriate passive voice has been used to highlight process and functionality.

1.10.6 Figures and Tables:

One or more figures and tables are presented sequentially with proper names (e.g., Figure 1, Table 1) and then cited in the text in logical order to provide continuity and help readers find them easily.

1.10.7 Emphasis:

Key words or phrases are in bold or italic, to highlight particular points or technical terms.These conventions help keep the document readable, interpretive, and help readers, developers, and evaluators to follow the information correctly.

1.11 Miscellaneous

This section is meant to support the on going success of the project by giving extra details and advice for organization, development and implementation of the project. During the project the following should be taken into account:

1.11.1 Project Management Practices:

Role playing within the team is important. To keep track of the project the regular progress review, milestone monitoring and documentation updates should be maintained to ensure timely delivery of project tasks.

1.11.2 Version Control and Collaboration:

A version control system (VCS) for efficient source code, documentation and collaborative development is recommended (e.g., Git). Patterns of good branching, code review and commit documentation should be adhered to.

1.11.3 Testing and Quality Assurance:

There should be comprehensive testing strategies such as unit testing, integration testing, system testing and user acceptance testing. To ensure reliability and performance of the system, it is necessary to document test plans, test cases and bug tracking procedures.

1.11.4 Documentation Standards:

Conventions used in this SRS must be followed for all the project artifacts such as design diagrams, code documentation and user manuals. This ensures, consistency, developers, evaluators, and end users find it readable and easy to use.

1.11.5 Communication Channels:

Team communication, communication with others and supervisors or managers are very important. Meetings, progress reports and feedback loops should be put in place to ensure alignment with project goals.

1.11.6 Risk Management:

Project risks, whether technical or due to resource constraints or security, should be considered at an early stage. Contingency plans and mitigation plans should be recorded and regularly evaluated.

1.11.7 Learning and Knowledge Sharing:

Project development lessons learned, innovative approaches, and insights should be captured by the members of the team. In order for maintenance, upgrading etc. to be carried out in the future, such knowledge will help during this process.

1.11.8 Ethical Considerations:

Complying with privacy laws, ethical standards, and security best practices (HIPAA/GDPR compliant) is essential during the system's development and deployment, since patient information is sensitive.
These miscellaneous guidelines will provide a well planned, professional, and safe framework for software development, enabling a seamless workflow, collaborative effort, and the creation of a quality, reliable, and user friendly Intelligent Therapist Patient Management System.

Chapter 2. Overall Description

1.12 Product Features

BENZI.AI Web Based Therapist Patient Care Management System is a secure Web based System with AI Robotic manual assistant. It has a separate portal for patients, therapists and administrators. While organized, secure and user friendly, each portal includes features relevant to specified user responsibilities.

Key functions of the system consist of customer sign up and login, role based access, therapist verification module, consultation appointment scheduling, secure chat functionality, on line video consultation, administration of medical records, AI powered help and support and reference module, goal assigning, monitoring progress, analytics, payments, and automated alerts. It allows patients to find therapists, schedule appointments, securely communicate, monitor mental health status, access reports, and enjoy AI assistance. Therapists can oversee sessions, client's wellness objectives, progress, treatments provided, repayments, and registrations.

The super admin panel for handling therapists, patients, subscription, revenue and administration of the platform has been provided also. One of the main products is the use of the AI virtual assistant, which offers context related guidance in accordance with the involvement in patient history, therapist notes, reports, and information related to treatment. Accessibility, privacy, automation and patient engagement in mental care are the product's focus points.

1.13 Functional Description

It is a fully online mental wellness and counselling platform that integrates counselling, administration and patient in a secure central application. Patients can sign up, log in, schedule appointments, join live sessions, chat with therapists, talk to the AI agent, access reports, and monitor set wellness objectives. Therapists can sign up, input their credentials, schedule appointments, upload patient information, set goals and see progress, and communicate with patients.

On the back end, it manages sign in, authorization, arranging the appointments, dealing with payments, sending notifications, artificial intelligence communication, and ensuring the data is safely stored. User data, appointments, medical records, goals, reports, payments, and logs of interactions with AI are stored in MongoDB Atlas. A fine tuned Ollama Llama model is used to generate personalized, context aware responses through a RAG pipeline, comprised of a knowledge graph and an embedding model. Integrated payment gateway, video consultations, email notifications, and SMS reminders via third party service providers like Stripe, Google Meet and Jitsi Meet, SendGrid, and Twilio.
All users interact with BENZI on a high level.AI Web app via browser. They process their requests with the backend services, asks the database and AI engine as appropriate and bring the appropriate output to the user dashboard.

1.14 User Classes and Characteristics

There are 4 types of users: Patient, Therapist, Admin and Public/Guest User. Each user class will be given different access rights and responsibilities.

Patient:
The main users of the system are the patients. Through the use of the platform, they visit the pot to acquire therapists, book appointments, participate in online treatments, communicate securely, engage with the AI assistant, track assigned wellness objectives, see development analytics, and download records. Patients may not be technically literate themselves and hence their interface should be simple, clear, calm and user friendly. This is a crucial user group as the system has been created to enhance the care and help given to patients' mental health.

Therapist:
Another large user class is that of therapists. They utilize the system to handle patient appointments, keep health records, upload session notes, set therapeutic objectives, check out patients' progress, administration services, and communicate with patients. For therapists, secure access to patient information is crucial, and robust authentication, access controls with different user roles and patient data privacy are critical.

Administrator:
Administrators are in charge of the general functioning of the platform. They approve on allegations for the therapist, doctor/patient relationships and management, subscriptions, stock receipts, and so on., and monitor the action of the device. Admin users can access management features but are not allowed to see private patient medical records, unless it is permitted in system policy. While they are crucial in platform control and maintenance operations, they are not the direct users of the health care services.

Public/Guest User:
General pages like Home, About BENZI are available for public or Guest Users.AI, FAQs, Resources, Contact us and Subscription Plans. They can access information about platforms and opt in as patients or therapists. This user class is not as important as patients and therapists but is essential for user onboarding and awareness.

The two target users are Patients and Therapists, as the focal point of the system is to make it easier for patients and therapists to communicate and for managing the care and mental health needs of patients. The system is handled operationally by the administrators, and the interactions of the Public/Guest users are more geared towards information pages.

1.15 Design and Implementation Constraints

There are several technical, security, regulatory and design requirements that limit the system. Since BENZI.Guardian over sensitive mental health and patient information, robust privacy and security measures are essential with AI. Secure authentication and sessions (using JWT), role based access control, encrypted communication, and database access using security. All patient records, reports, session notes, and AI interactions logs should be kept securely.

The system is realised by developing a MERN stack web application. The web application is built with React.js for the front end, Node.js and Express.js for the back end and MongoDB Atlas as a cloud based database. The AI assistant relies on the Ollama package and a custom fine tuned Llama model, along with a context aware RAG pipeline. The technology choices constrain the developers to only front and backend development in JavaScript/TypeScript, data structures in MongoDB and integration in Python/AI related for the AI engine.

Application was web based and it doesn't have native mobile application. This means that the interface has to be responsive, accessible and friendly for desktop, tablet, and mobile browsers. Additionally, the system relies on internet connectivity, as various features like online consultation, cloud database access, payment processing, notifications, and AI responses can only be made possible with internet.

Constraints also exist that are implemented by third party services. For Payment Processing, Stripe is compulsory, For Video Consultation, Google Meet or Jitsi Meet is compulsory, and for notifications, One has to use email/SMS services. Some features of the system might not function when these services are not available, not configured, or restricted. The AI system should also adhere to ethical limitations of AI: no clinical diagnosis, no medication prescriptions or emergency medical decisions. It should include no more than supportive suggestions and referrals to seek professional assistance, if necessary.

1.16 Assumptions and Dependencies

The system assumes that a stable internet connection and the modern web browser (Chrome, Edge, Firefox or Safari) will be available to the users. It is also assumed that patients and therapists will give correct registration information and therapists will give valid therapist credentials for verification. The system is based on the premise that the administrators conduct research and make wise decisions in approving therapists.

It relies on the following technologies: MongoDB Atlas as its database, Node.js and Express.js for its backend services, React.js for its front end interfaces, and Ollama combined with the fine tuned Llama model for AI capabilities. It also relies on the appropriate setting of environment variables like database URI, JWT secret key, Stripe keys, AI service settings and notification service credentials.

The system also requires connection to other services such as Stripe (payment) and Google Meet or Jitsi Meet (video consultations), as well as email or SMS channels for notifications/remindings. These third party services are subject to change to their API, price, limits and/or availability; this might necessitate updates to the system being used.

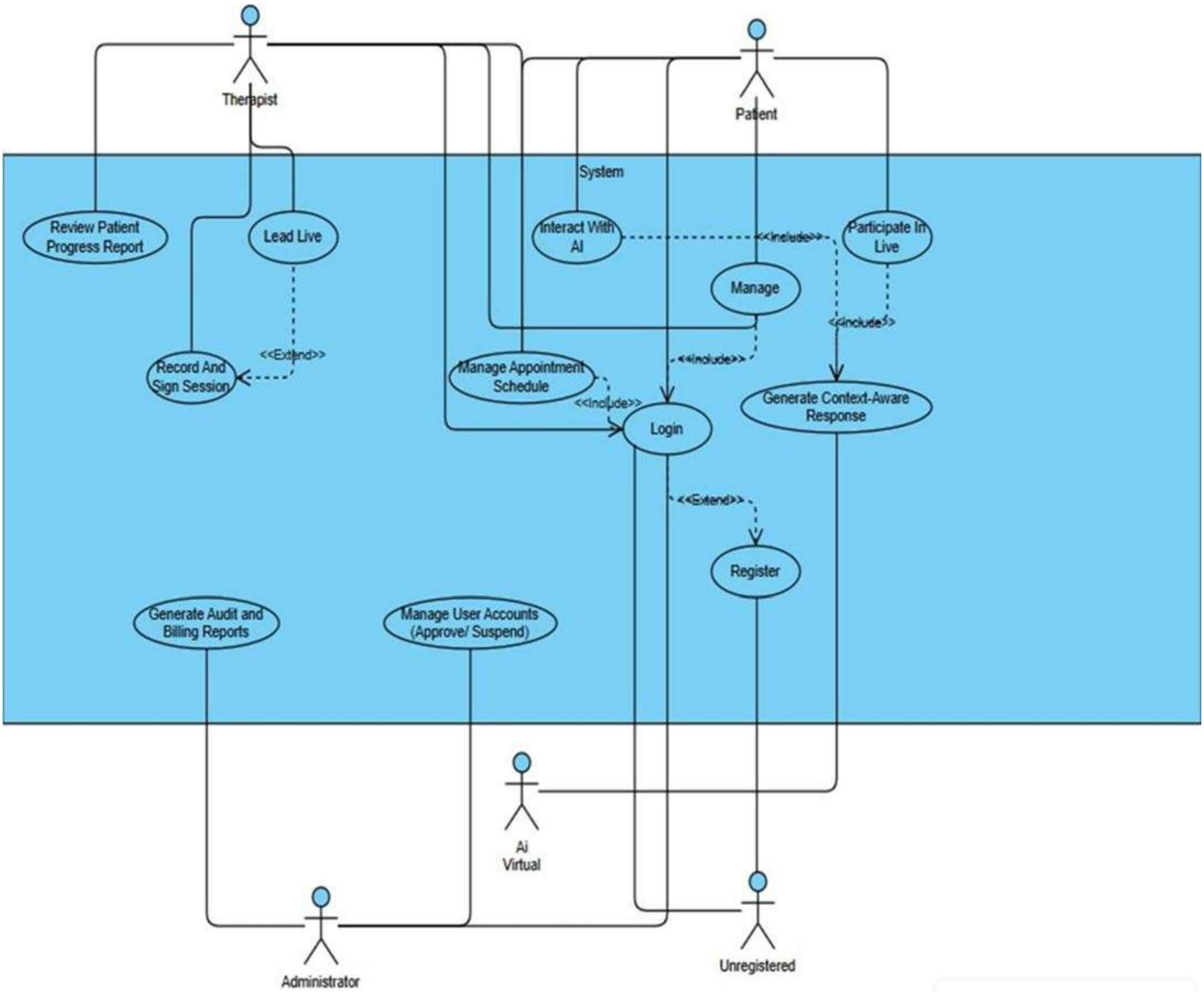
It is additionally presumed that the AI assistant will certainly not change into a licensed specialist therapist and must be made use of just as a supportive tool. The quality of AI responses and their utility is heavily reliant on the quality of the patient records, therapist notes, reports, and knowledge base content provided in the RAG pipeline. Scalability in the future is also reliant on the cloud hosting sources, the speed of the database, safe deployment and correct upkeep from the methods.

Chapter 3. System Requirements

1.17 Functional Requirements

All functional requirements are expressed as use cases. At the heart of the system lies its capacity to handle the interactions between therapists and patients in a secure, efficient and intelligent manner via system automation and AI aids. A use case is a specific task that is carried out by the actors engaged with the system (Admin, Therapist, Patient or AI Assistant).

Figure: Primary Use Case Diagram



1.17.1 Book Appointment with Therapist Use Case 1

Identifier	UC 1
Purpose	To allow a patient to view therapist profiles, check their real time availability, and securely book an appointment through integrated payment and automated confirmation.
Priority	High
Pre conditions	1. Patient must be registered and logged into the system. 2. Therapist profile must be verified and available in the system. 3. Stable internet connection must be active.
Post conditions	1. Appointment details are stored in both patient and therapist dashboards. 2. Automatic confirmation message and reminders are sent via email/SMS.

Typical Course of Action

S#	Actor Action	System Response
1	Patient logs into the portal and opens the "Find Therapist" section.	System displays all verified therapist profiles along with specialization, ratings, session fee, and available slots.
2	Patient selects a therapist and clicks "Book Appointment."	System loads the therapist's real time calendar showing available slots.
3	Patient chooses the desired date and time for the session.	System validates the selected slot and checks for conflicts.
4	Patients choose a payment method (Online or Pay in Person) and proceeds to confirmation.	System either redirects to a secure payment gateway (for online) or marks the slot as "Partially Confirmed" (for pay in person).
5	Patient confirms the booking.	System updates both patient and therapist dashboards, sends confirmation notifications, and schedules automated reminders.

Alternate Course of Action

S#	Actor Action	System Response
1	Patient selects a time slot that has already been taken.	System displays "Slot Unavailable" and prompts the user to choose another time.

2	Payment fails or is cancelled mid process.	System shows "Payment Failed Please Try Again" and logs the event for audit purposes.
3	Therapist cancels or reschedules the session later.	System automatically notifies the patient, offers rescheduling options, and updates both dashboards
4	Patient selects "Pay in Person" instead of online payment.	System reserves the slot and marks it as "Partially Confirmed", notifying both patient and therapist that payment will be completed at the clinic before the session.

Table 1: UC 1

This case describes how a patient schedules a therapy session using the system. The patient can look through therapist profiles after logging in, which contain all the information they need, including availability, reviews, fees, and expertise. After selecting a qualified therapist, the system confirms available timeslots and offers two ways to make a reservation: Pay in Person (for partial confirmation) or Online Payment (for instant confirmation). The chosen time slot is reserved by the system and marked as "Partially Confirmed" until payment is made in person if the patient chooses to pay in person. Both parties receive automated notifications upon successful booking or payment, and the therapist's calendar is updated instantly.
Through automated alerts and rescheduling options, the system gracefully handles exceptions such as failed transactions, double bookings, or therapist cancellations. For both patients and therapists, this procedure maintains openness and dependability while guaranteeing efficient, adaptable, and safe appointment scheduling.

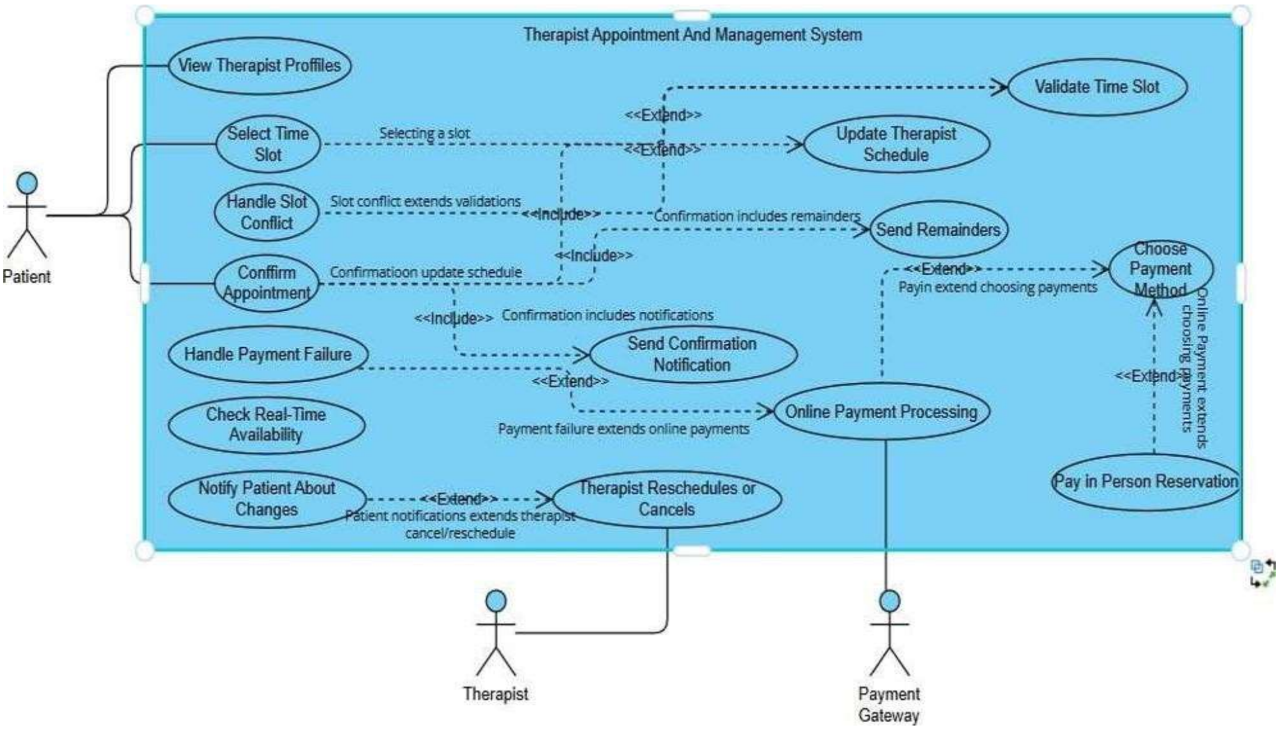


Figure: Use Case 1

1.17.2 Conduct Therapy Session (Encrypted + AI Support) Use Case 2

Purpose	To enable therapists and patients to conduct secure video or chat sessions with encryption and optional AI assistance fo r recommendations during or after sessions.
Priority	High
Pre conditions	Both patient and therapist must be logged in. Appointment must be confirmed. Video session tools (Zoom/Google Meet) integrated.
Post conditions	Session summary recorded. Therapist notes and AI insights stored in database. Notifications sent post session.
Purpose	To enable therapists and patients to conduct secure video or chat sessions with encryption and optional AI assistance for recommendations during or after sessions.

Typical Course of Action

S#	**Actor Action**	**System Response**
S#	Actor Action	System Response
1	Patient and therapist log in and navigate to session dashboard.	System displays "Join Session" button for both.
2	Therapist starts session.	System opens encrypted video/chat room.
3	AI assistant monitors session context (if enabled).	System records key points and suggests follow up recommendations.
4	Therapist ends session and adds notes.	System securely saves session notes and updates patient records.

Alternate Course of Action

S#	**Actor Action**	**System Response**
1	Internet connection drops during session.	System pauses and attempts auto reconnect.
2	Therapist disables AI assistance.	System logs only manual notes without AI suggestions.
3	Patient exits session early.	System marks session incomplete and notifies therapist.

Table: UCS 2

Explanation: This use covers the encrypted online therapy process and AI support, ensuring privacy, personalization, and continuity in therapy.

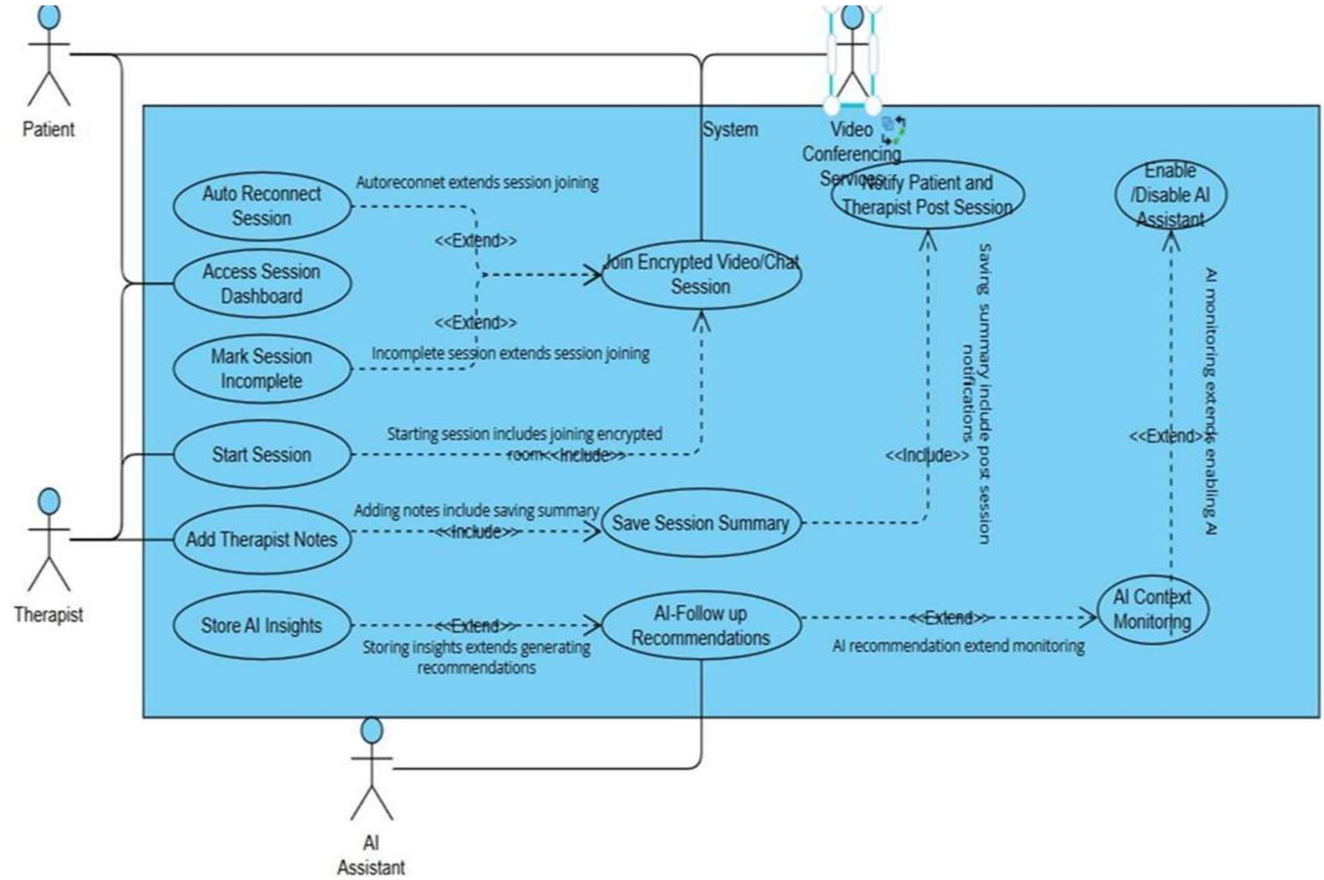


Figure: use case 2

1.17.3 Manage Patient Records and Notes Use Case 3

Identifier	UC 1	
Purpose	To allow therapists to create, update, and securely store patient medical histories, reports, and session notes.	
Priority	High	
Pre conditions	Therapist must be logged in and authorized.	
Post conditions	Updated patient records stored securely in database.	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Therapist logs in and accesses patient profile.	System displays patient history and reports.

2	Therapist uploads new reports or session notes.	System encrypts and stores files securely.
3	Therapist edits or updates medical information.	System validates and updates data.
4	Therapist saves changes.	System confirms successful update.
Alternate Course of Action		
S#	**Actor Action**	**System Response**
1	Therapist uploads unsupported file type.	System displays an error message.
2	Database temporarily unavailable.	System queues update and retry later.

Table: use case 3

Explanation: This use case ensures secure handling of sensitive medical records with proper encryption and validation.

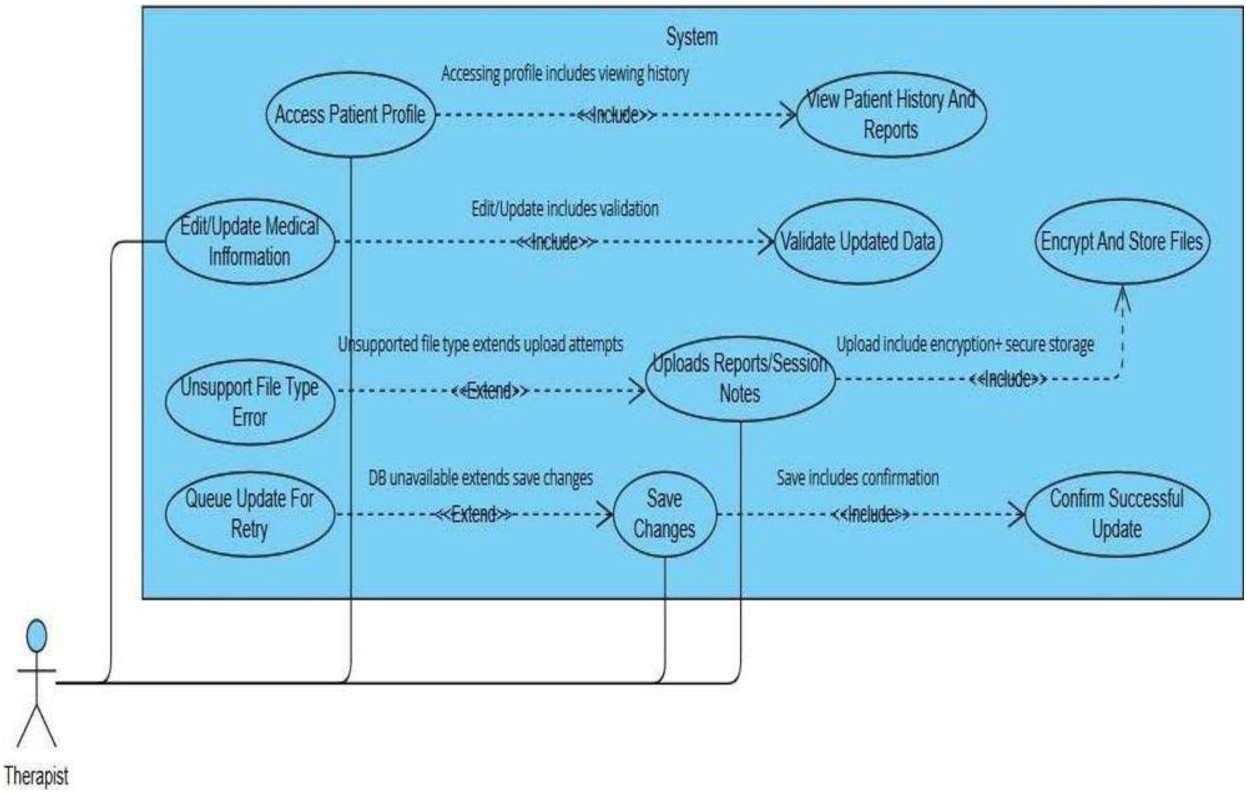


Figure: use case 3

1.17.4 Manage Patient Records and Notes Use Case 4

Identifier	UC 1	
Purpose	To enable therapists to assign wellness or therapeutic goals and allow patients to track progress.	
Priority	Medium	
Pre conditions	Both patient and therapist must be registered and linked.	
Post conditions	Progress data recorded and viewable in dashboards.	
Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Therapist assigns a wellness goal to the	System store's goal details and due date.
	patient.	
2	Patient views assigned goals on dashboard.	System displays goal progress indicators.
3	Patient marks goal as completed.	System updates completion percentage and awards points.
4	Therapist reviews progress.	System shows visual analytics and improvement trends.
Alternate Course of Action		
S#	**Actor Action**	**System Response**
1	Patient misses deadline.	System sends automated reminder notification.
2	Therapist deletes a goal.	System updates both dashboards and logs change.

Table: use case 4

Explanation: This case promotes engagement and motivation through gamified goal tracking and progress monitoring.

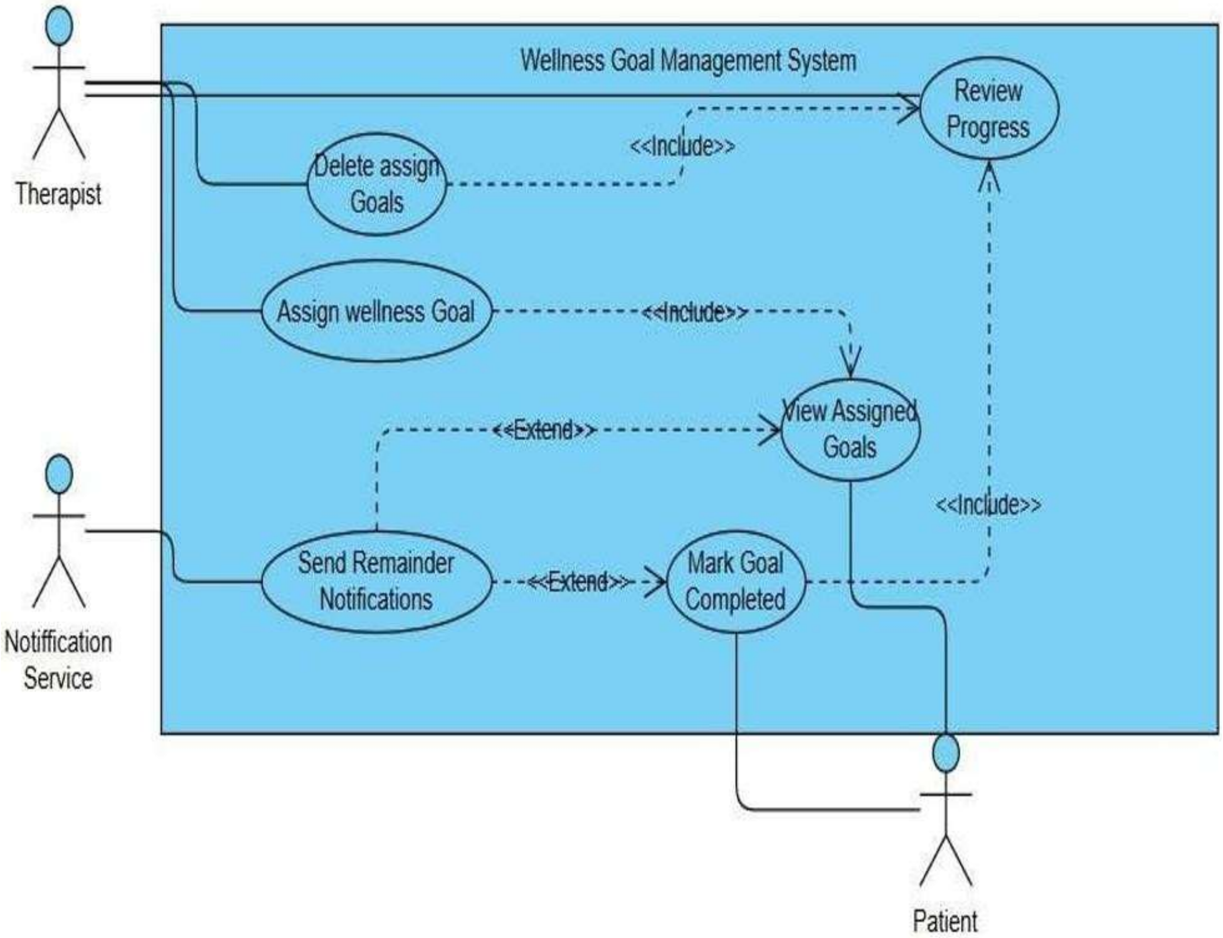


Figure: use case 4

1.17.5 Requirements Analysis and Modeling

Figure ERD:

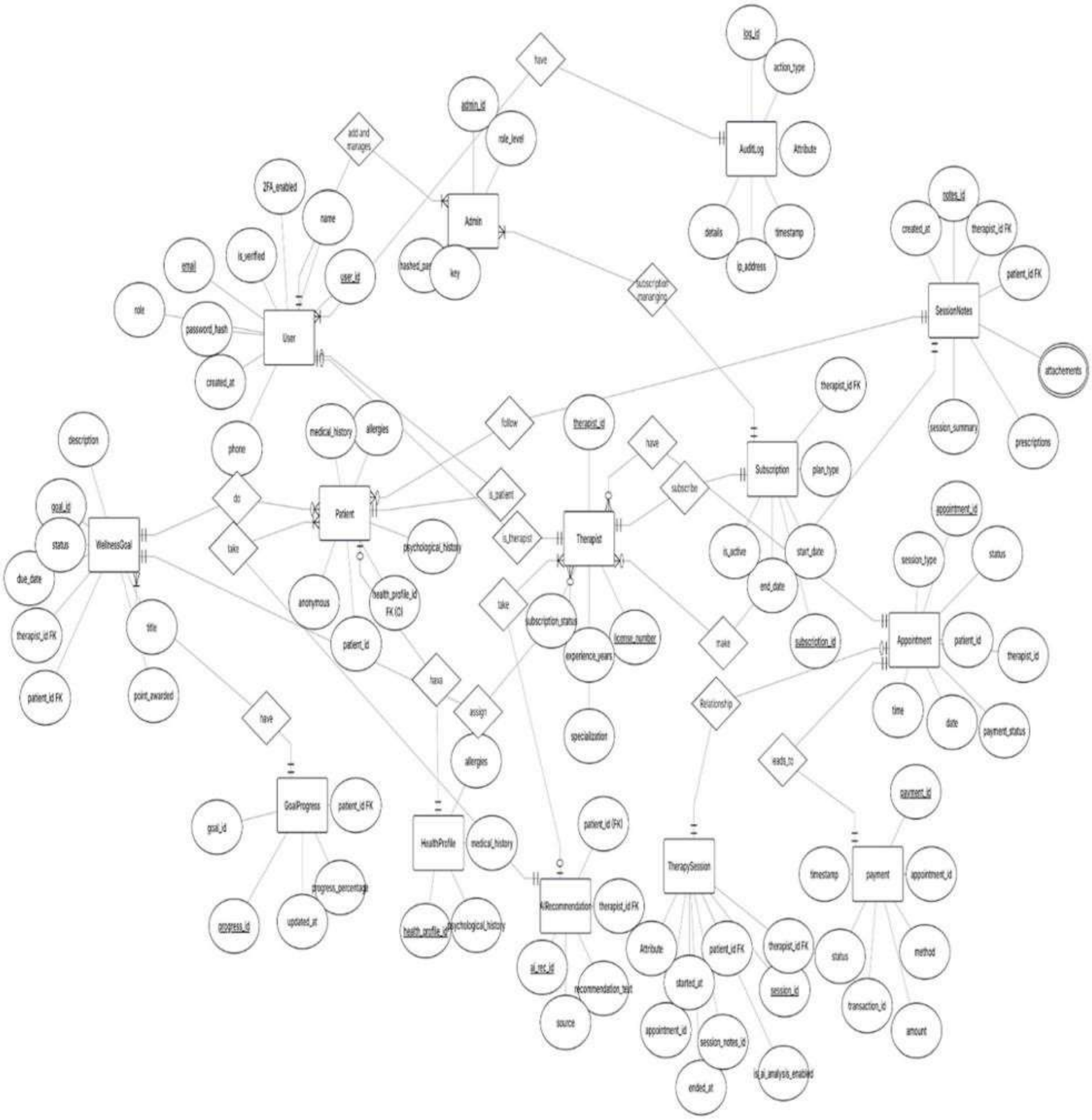


Fig 3.1 ERD

Abstract Class Diagram:

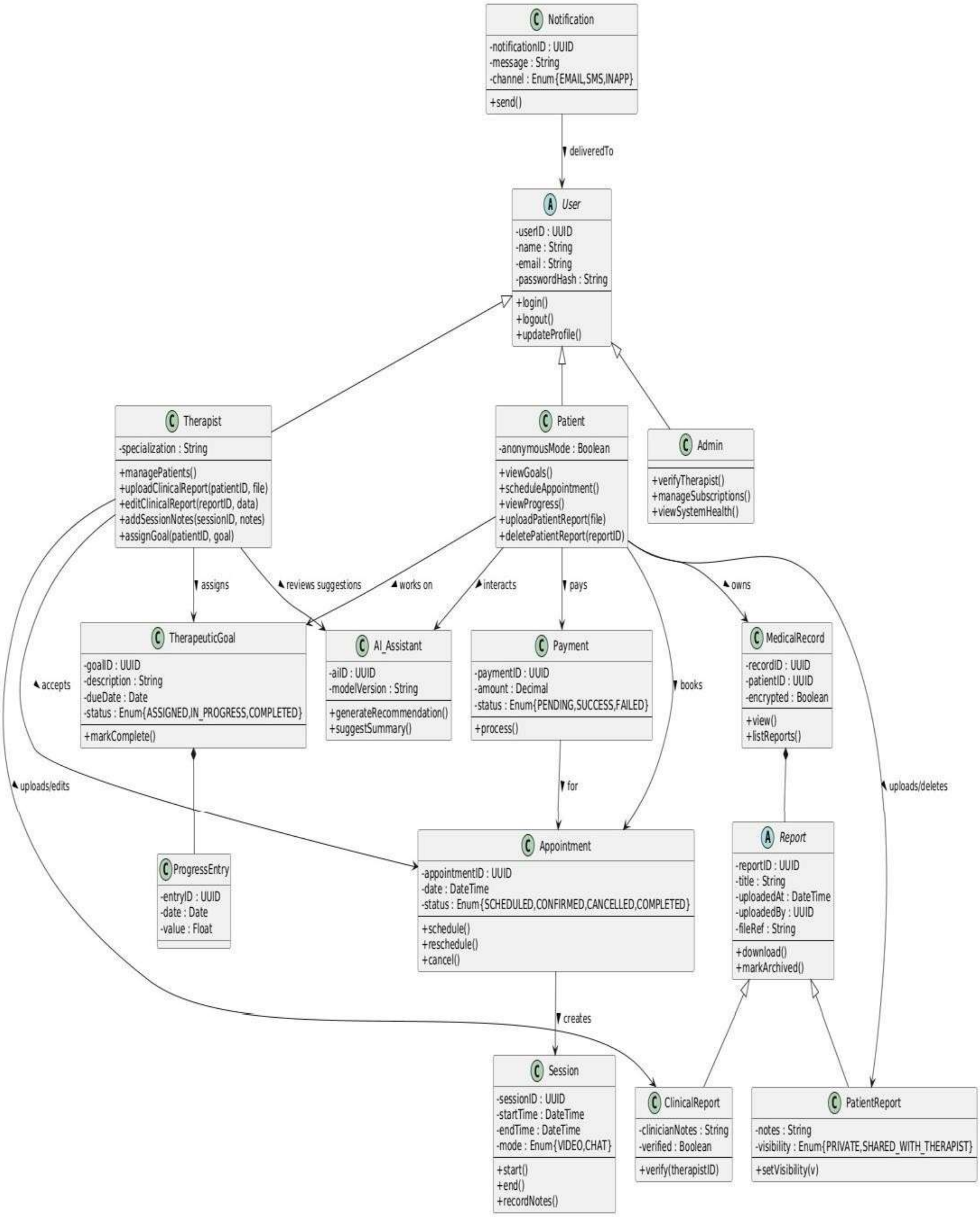


Fig 3.2 Abstract Diagram

DFD:
Level 0:

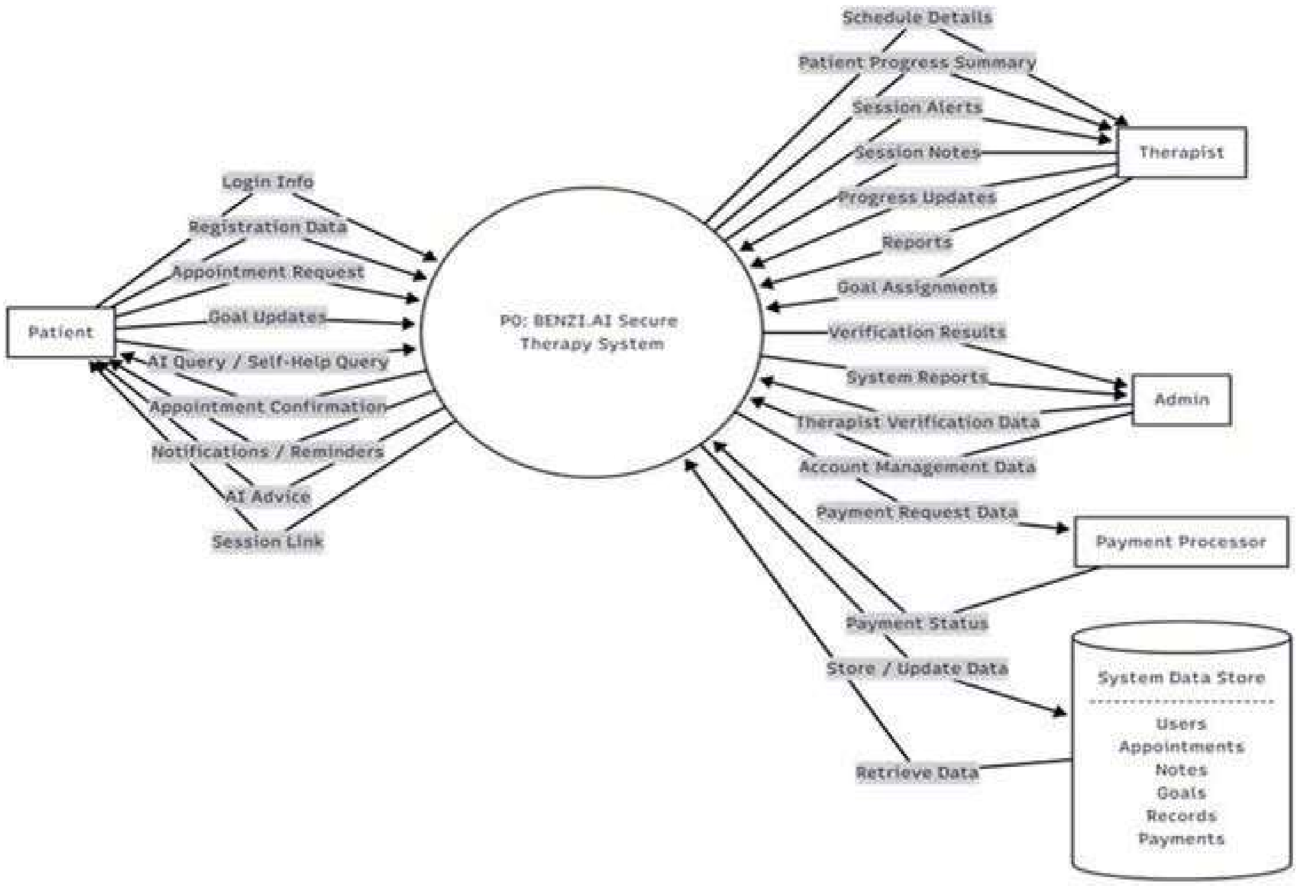


Fig 3.3 Level 0 DFD

DFD Level:1

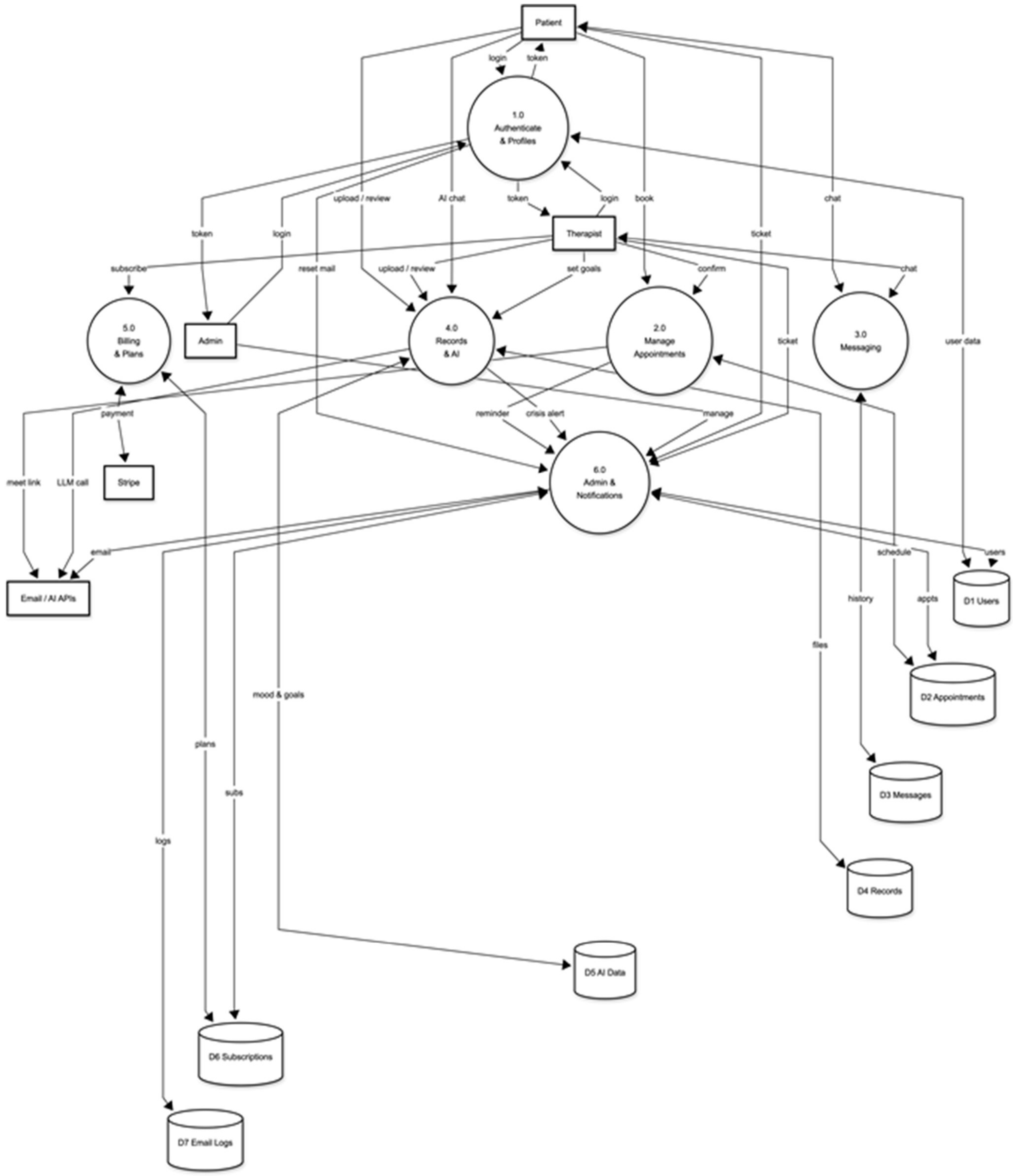


Fig 3.4 Level 1 DFD

DFD Level:2

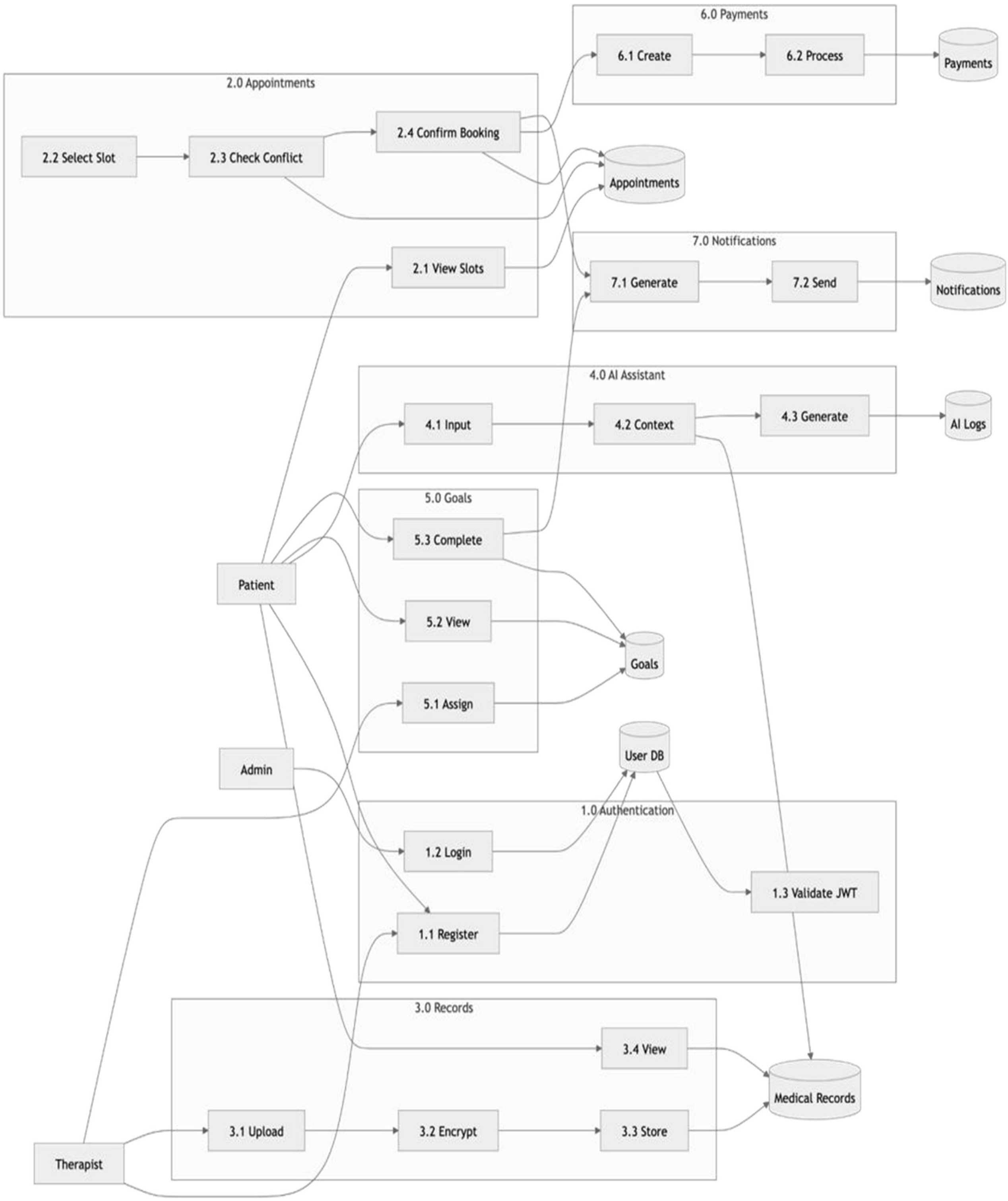


Fig 3.5 Level 2 DFD

Activity Diagram:

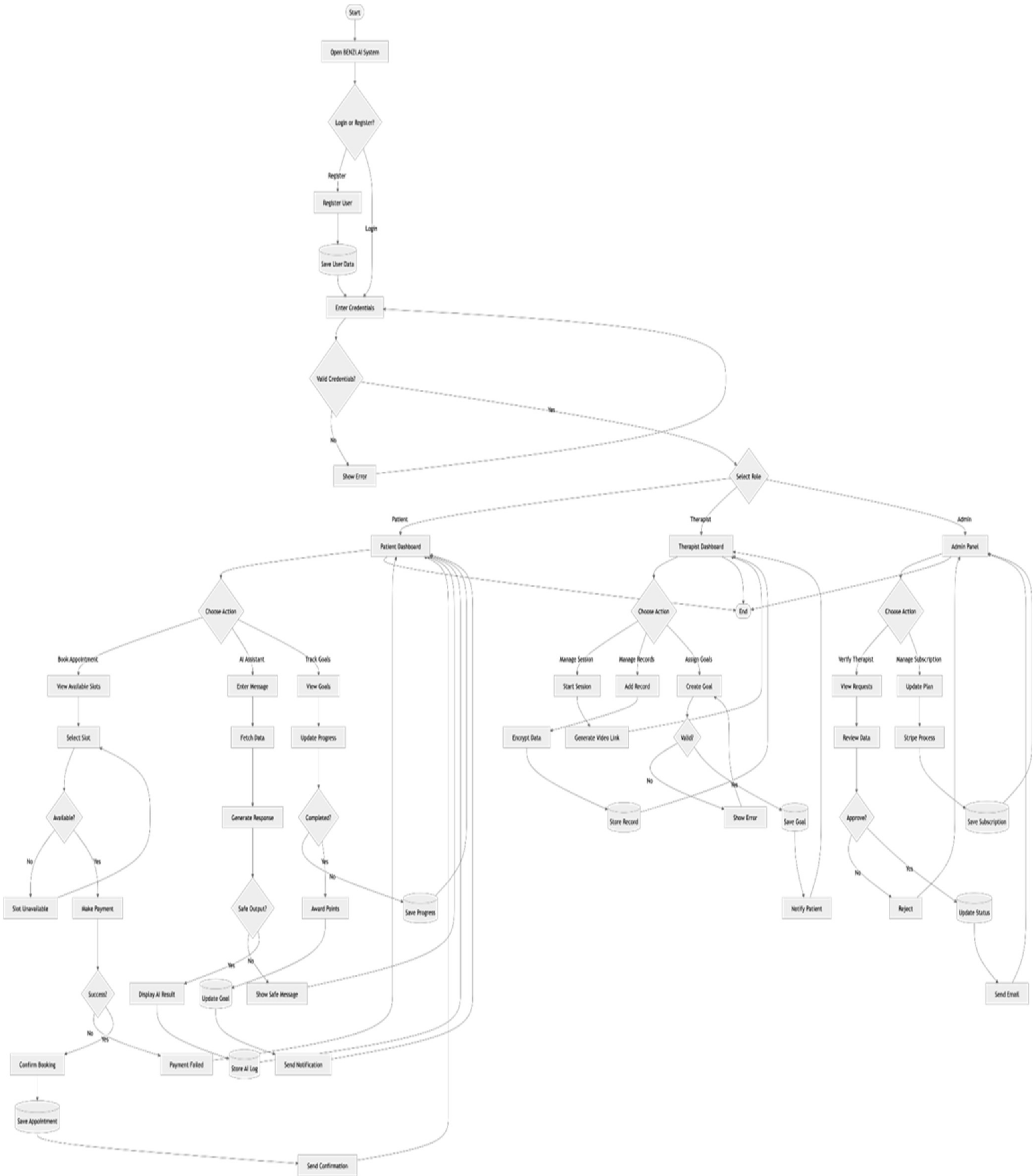


Fig 3.6 Activity Diagram

1.18 Nonfunctional Requirements

1.18.1 Performance Requirements

The AI Inbuilt Secure Therapist Patient Management System should ensure consistent responsiveness in all modules. To guarantee user experience, the system must have the ability to load the main dashboards, login pages and appointment interfaces in three seconds with stable network conditions. All tasks, such as authentication, creating appointments and using the features, must complete within about two seconds to ensure that the user does not notice any limit in responsiveness. The delay in real time chat and messaging functions should be as low as possible, ensuring that patients can communicate with therapist and therapist can communicate with patients within a maximum of 1 second delay. To ensure that there is no disruption in the flow of the conversation, the AI Virtual Assistant should also be able to respond to inquiries or suggestions in less than four seconds.

1.18.2 Concurrent User Handling

Without sacrificing speed or dependability, the platform must be able to accommodate many concurrent users. During normal use, the system should efficiently support up to 500 simultaneous uses by administrators, therapists and patients with consistent performance indices. To accommodate peak use scenarios, such as therapy sessions or extensive testing, the system should support the possibility to scale up to 1,000 concurrent users, by means of a cloud based auto scale feature. The parameters used should be reasonable, ensure the memory usage to be less than 75% and CPU below 70%. These performance levels assure that the responsiveness remains the same and there is no performance degradation for more than one user interaction at a time.

1.18.3 Database Efficiency

The performance of the database is critical for the smooth functioning of the system. Quickly creating, retrieving, updating, and deleting patient information and appointments, and adding or removing therapy logs and others should all take, optimally, less than a second per query, e.g. the lookup of patient information. The system needs to employ the right indexing and query optimization techniques to enhance data access speed in collections with high query frequencies such as appointments, session logs, and AI generated recommendations. Because the database encrypts sensitive data on the client side at the field level, the encryption overhead is minimal and is at most 20% of the query execution latency. The database architecture should be optimized for running the database in parallel without causing performance snags when the system is heavily utilized.

1.18.4 AI Module Processing

The integrated artificial intelligence module, built using Python, needs to be efficient to ensure timely delivery of AI driven insights and recommendations. In the normal load condition, it is expected that the AI assistant will return relevant suggestions to each request in 2 4 seconds. The AI subsystem must complete analysis within ten seconds, even with high data volumes, for example when processing aggregated data for behavioural trend summaries, goal tracking analytics, or performance reports. Automated workflows that schedule tasks, send reminders, and trigger AI powered events should work correctly and reliably, and not delay more than 5 seconds past the scheduled event time. This guarantees that user actions and automated background processes are synchronised in real time.

1.18.5 Real Time Communication Performance

Communication performance is a crucial component in the system for therapist patient interaction to maintain continuous communication. Previously all video sessions created with built in APIs such as Zoom or Google Meet will require at least 720p at 30 fps for visual and aural clarity reasons. In normal broadband conditions, the time elapsed for a Live Video session or chat should not exceed 300 Ms while sessions are live and there must be an auto reconnect attempt within 5 seconds after each of the short few SWIFTY outages to avoid session interruption. The secure communication protocols should not affect the speed of transmission, ensuring that therapy sessions maintain optimal transmission speeds, security, and continuity over time.

1.18.6 System Availability and Throughput

The system has to be "highly available" so that all users are able to always access the system. Uptime is crucial must be at least 99.5 percent, without planned maintenance and upgrades. At least 100 requests per second should be processed by the backend, and average API response times shouldn't be more than 500 Ms. Under a steady internet connection, file uploads, like prescriptions or clinical reports, that are no larger than five megabytes, must finish in about five seconds. The reliable data transfer and minimized server response times ensure uninterrupted system access and user satisfaction.

1.18.7 Scalability and Efficiency

Horizontal scaling should be supported by the cloud infrastructure that houses the platform to handle growing user demand and data growth without experiencing any outages. During periods of increased traffic, additional computational resources must be dynamically provided to ensure the continuous delivery of service. Caching techniques like Redis or in memory coaching should be used to store static data and AI pre processed responses to improve performance efficiency. This approach enhances the reaction time of regularly accessed modules, reduces API latency, and reduces redundant calculations. Together, these coaching and scalability techniques guarantee that the system will continue to be flexible, effective, and able to sustain fast performance even as the number of users rises.

1.18.8 Rationale

These performance requirements were established to ensure a responsive, effective and reliable system that is suitable to the spur and sensitive nature of digital therapy. Low communication latency, quick database access, and optimized AI response times ensure that therapists can carry out sessions without interruption, complementing this, patients benefit from receiving care without latency, thus effectively streamlining administrative processes. Developers can make well informed design and optimization decisions throughout implementation thanks to the established benchmarks, which offer a quantifiable standard for assessing system performance.

1.18.9 Safety Requirements

1.18.9.1 Data Integrity and Loss Prevention

Thanks to an AI Integrated Secure Therapist Patient Management System, the processing, transmission and storage of sensitive data takes place without any data loss or corruption. Ensure all patient and therapist information such as medical history, therapy notes, and session records, is backed up and duplicated and automated recovery procedures are available. Regular encrypted backups will be scheduled at regular intervals to minimize the potential loss of data, in case of a server crash or a system failure. To guarantee that interrupted or partial updates do not jeopardise the accuracy of stored data, the database system will employ atomic operations to preserve transactional integrity. The system should be able to be completely restored from the backup file created at the latest restore point without causing loss of confidentiality or inconsistencies in the consistency of the data structure in the event of hardware and/or software failures.

1.18.9.2 System Fault Tolerance and Failure Recovery

The platform needs to be developed with fault tolerance features to ensure its safe operation and continuous therapy sessions. Load Balancing, Redundant servers and automated failover systems will be implemented to prevent entire systems failure in the event that components fail. Even if the AI module or the video conferencing integration or the automation service goes down, the critical functions such as secure logging in, retrieval of patient information and arranging appointments are all maintained. All failure events shall be recorded in automatic manner with all important characteristics taking into account.

logging of error and exception handling to facilitate diagnostics after incident and to prevent reoccurrence of similar failures. It should also provide safe session termination in case of unforeseen system crashes without causing data corruption or partial writes of therapy records.

1.18.9.3 User and Patient Safety

The data security and the mental health of the system should have the top priority, as it comes to data security and mental health. Given the handling of sensitive mental health data, the platform should have measures in place to protect the privacy and confidentiality of the information. Those who exchange messages with the AI assistant or engage in therapy should be safeguarded against potentially confronting distressing or harmful content. The AI module will operate according to ethical guidelines, ensuring that the suggestions it provides remain non diagnostic, supporting and aligned with accepted therapeutic methods. If the AI detects any statements that could suggest distress or self destructive behaviour, it will trigger an automated alert system that could securely link the user to emergency services or notify the therapist without leaving the private information to remain as plain text on the AI's system.

1.18.9.4 Secure Communication and Session Safety

To safeguard against data misuse or interception, all real time communication such as voice, video and text messages should be complete end to end encrypted. The system is to have secure session verification procedures and encrypted tokens to eliminate eavesdropping, screen recording and unauthorized access to a session. Users shall not be subjected to personally identifiable information or weaknesses that may impair their freedom, in interacting in anonymous mode. With the application, it will be ensured that any modifications in the therapy notes, or medical record, only happen through validated and authenticated workflows and not through risky operations such as data modification during a live video session.

1.18.9.5 Hardware, Network and Operational Safety

The host location must be in monitored, secure data centres and meet international safety and reliability standards. All hardware and networks should be protected from environmental hazards, electrical surges and unauthorized physical access. The cloud provider should assure compliance with known safety practices, such as SOC 2 for operational controls and ISO/IEC 27001 for information security management. Network fireswalls, intrusion detection systems and rate limits need to be deployed to thwart attacks against malicious data injection or denial of service. The system should also track CPU temperature, memory usage, and network bandwidth to ensure safe operating parameters are maintained to prevent any unanticipated downtimes or overloads.

1.18.9.6 Regulatory and Ethical Compliance

The platform must abide by the legal and ethical guidelines that control how medical and psychological data is handled. It needs to be designed keeping with international data protection regulations, including the General Data Protection Regulation (GDPR) and Health Insurance Portability and Accountability Act (HIPAA). The guidelines for compliance ensure that the users are entitled to access their data, correct it, delete it, while the information of patients is collected, stored and treated with the user's consent. Safety audits or vulnerability assessments should be conducted on a regular basis to help maintain compliance with these standards. To avoid human error that could result in safety incidents, all staff members engaged in development or maintenance should receive the proper cybersecurity and data handling training.

1.18.9.7 Rationale

Safety, apart from technical reliability, also entails protecting the users of the system, maintaining proper ethical treatment of data, and ensuring manual retrieval of data in case of emergency. The safeguards are designed to ensure that therapy services can be provided without harming the individual's body, mind or information. Its robust encryption mechanisms, backup plans and compliance frameworks create a secure and regulated environment, maintaining patients' and therapists' trust.

1.18.10 Security Requirements

1.18.10.1 Data Protection and Encryption

The AI Integrated Secure Therapist Patient Management system needs to ensure the highest levels of administrator convenience, patient data availability, confidentiality and integrity through extensive encryption and access controls. To prevent data from being intercepts and/or tampered with during communication, data must be encrypted via Transport Layer Security (TLS 1.3) and secure cypher suites must be utilized between client and server. The Encryption algorithm is AES 256 which is field level encryption [3].

Caution should be taken when storing sensitive data in the database like patient records, therapy notes, prescriptions, session logs, etc., which needs to be protected using the [16] standards. Client side encryption should be considered for sensitive data, so that even the system administrator could not access the plaintext data. Backups and data that has been archived should also be encrypted when it is stored or recovered in case of data exposure.

1.18.10.2 Authentication of an individual by a system (authentication)

User identity management should follow a strong authentication process, otherwise users will have access to information that they are not supposed to have. All the players, the administrators and the therapists should have secure logins with hashed password received from a password generating algorithm such as Argon2 or Bcrypt. Role Based Access Control (RBAC) will be implemented all through the platform to segregate the permissions based on the roles of the users. Therapists can see information about the patients assigned to them, administrators can provide user accounts which can't be accessed by any other patients, and patients can only look at theirs. These accounts, for therapists and administrator, will have Two Factor Authentication (2FA) to enhance security.

This can be through your SMS message or the help of Time based One Time Passwords (TOTP). Session tokens that are used for authentication should be short lived, safe to store in the browser (using HTTP only cookies) and expire on logout or inactivity.

1.18.10.3 Authorization and Session Management:

Safe and traceable users sessions shall be ensured by all the system's modules. To prevent token forgery, there will be a digitally signed JSON Web Token (JWT) provided for each session, which will be checked on every request, and will have encrypted claims. Sessions will automatically end after a pre determined period of inactivity to minimise the likelihood of an unwanted access occurring on any shared or public devices. To stop potential misuse of accounts, the number of concurrent sessions needs to be restricted to avoid multiple user logons with the same user/passwords from different sites. Client and server have a full session termination via logout mechanisms. Session log should be requested for security auditing purposes, and IP addresses, device fingerprint, login times should be used without revealing any private information.

1.18.10.4 Secure Communication (and API Protection)

To prevent man in the middle attacks all system communications, both internal and external, should be secured using secure endpoints with certificate pinning for HTTPS. The backend API will be protected by token based authentication, and IP whitelisting for management endpoints. To prevent DOs and Brute Force attacks, rate limiting and request throttling techniques should be used. APIs need to sanitise all inputs, and ensure that they do not allow any sort of injection attacks, such as SQL Injection or NoSQL Injection, cross site scripting (XSS) and cross site request forgery (CSRF). To lessen browser based threats, security headers such as Content Security Policy (CSP), Strict Transport Security (HSTS), and X Frame Options must be set up. To ensure compliance with the least privilege principle, strict authorisation checks need to be carried out per API call with patient and/or therapist data.

1.18.10.5 Audit Trails and Monitoring

System for continuous monitoring and audit logging of all the security related activity in the platform must be established. All the attempts for successful and unsuccessful login, changes to data, modifications of permissions and administration activities should be recorded. Audit records must be stored in a repository that is tamper proof and encrypted, for future verification. To identify irregularities, send out alerts, and start automated incident responses when suspicious activity is found, the system ought to be integrated with a Security Information and Event Management (SIEM) program. These monitoring features will also enable the quick identification and remedial actions against potential security breaches or misuse. With the need for a proactive security approach, it is essential to regularly feature automated security scans, vulnerability assessments, and dependency checks in the continuous integration and deployment (CI/CD) pipeline.

1.18.10.6 Data Privacy and Regulatory Compliance

The platform's security design and implementation should comply with international standards and laws for the protection of healthcare data privacy. The system needs to comply with the requirements of the Health Insurance Portability and Accountability Act (HIPAA) and General Data Protection Regulation (GDPR). Some of these principles are data minimization, clear user consent for data collecting, safe storage and others. The system should ensure that data is anonymised and pseudonymised to prevent the disclosure of personally identifiable information (PII), especially for training an AI model and for analytics. To ensure proper moral data management, the users are meant to maintain full control over their data which includes the ability to edit or delete accounts which is also upheld.

1.18.10.7 Infrastructure and Physical Security
The application and database servers must be deployed on secure cloud environments such as AWS, Azure or Google Cloud; cloud providers already handle the ISO/IEC 27001 and SOC 2 Type II security certifications. Limit access to server instances with virtual private networks (VPNs) and by using SSH key based authentication. Firewalls, intrusion detection systems (IDS) and intrusion prevention systems (IPS) should be deployed to defend against attacks and unauthorized network access. Multi tren authentication is required for admin actions and admins should adhere the least privilege principle. It is necessary to have penetration testing with certified security experts be performed from time to time in order to detect vulnerability and correct them.
For both layers (infrastructure and application).

1.18.10.8 Rationale

Since the system manages immensely sensitive psychological and medical information, it is highly imperative to have strong security measures. The platform guarantees the confidentiality and protection of all therapist patient interactions by putting in place multi layered encryption, stringent access control, and adherence to international security standards. The ability to guard against outside or internal risks is additionally enhanced on a regular basis monitoring and auditing, compliance with HIPAA and GDPR guaranteeing all information managing actions stay ethically and legally sound. These security features, when combined, make BENZI secure:By ensuring the security of data and online platforms, AI [1] maintains trust, ensures data integrity, and provides a secure environment for receiving mental health care online.

1.18.11 Additional Software Quality Attributes

1.18.11.1 Usability

User friendliness and intuitive interaction for all user roles must be given top priority in the AI Integrated Secure Therapist Patient Management System. Both technical and non technical users, including those with low levels of digital literacy, should be able to utilise the interface's intuitive, easily navigable layout. It's important to align the navigation across the three portals (Patient, Therapist, and Admin), making it easy for users to find the key functions such as bookings, records, and analytics. The system should be designed to meet Web Content Accessibility Guidelines (WCAG) 2.1 standards to accommodate users with physical and/or visual impairment. Simple language to be used in the AI assistant and dashboard interfaces, with simple icons and visual indicators to encourage engagement and reduce the cognitive load. For usability testing, make sure that a new user is able to perform a simple action such as creating a session or browsing through a set of therapy notes in 2 minutes without assistance from another user.

1.18.11.2 Reliability
In environments where mental health services are offered, retaining users' trust is paramount. It should work in all of the State and be reliable, and not drop off when it is called upon. All essential duties including appointment booking, encrypted communication and authentication need to be accurate at least 99% during the outages. Such performance measurement will be actively monitored with automated measurements and if services are down or degraded, an alert will be issued. To guarantee continuous operation even in the event of unplanned hardware or network failures, recovery measures like redundant servers and failover systems must be put in place. Transactional integrity must be warranted by atomic operations and data consistency must be always correctly maintained, to avoid partial updates or losing data in concurrent activities.

1.18.11.3 Availability

With the exception of any planned maintenance periods, users should always be able to access the system and the 'upbringing' should be 99.5% at least. To achieve this level of availability and ensure that hosts are always available for customer access, the hosting infrastructure must provide for load balancing, distributed databases and redundant servers. The user shall be alerted to any planned maintenance and/or update at least 24 hours before it takes place to minimize disruption. In the event of a system component failure, the system will ensure an immediate failover into another instance which will safeguard user sessions and data integrity. To ensure that the trust between therapists and patients is maintained, synchronisation is essential for them to have access to appointments, patient records, and regular contact.

1.18.11.4 Maintainability

Users, data volumes and system functions are constantly increasing, and the platform must be able to handle these increasing demands and size without needing to re configure itself in a significant way. It needs to be horizontally scalable to effectively provide system level scaling, as well as vertically scalable to accommodate increased functionality, data and users within the same system. With only minor adjustments to current components, the architecture should enable the addition of new modules, APIs, or AI models. For instance, wearable device data synchronization, mobile apps, or advanced analytics dashboards could be incorporated in the future without needing to redesign the overall system. Further, the system must adapt to various environments, such as development, staging and production, and allow deployments to be fully automated by using migration scripts. These features provide for flexible operation, effective use and long term sustainability.

1.18.11.5 Interoperability

The system must develop to run efficiently in different environment and platforms with minimal reconfigurations. The web application should be responsive and run across all devices desktop, tablet, mobile, and everything in between, and be fully compatible with the larger browsers Chrome, Firefox, Edge, Safari etc. Enable migration from cloud to cloud (AWS, Azure, Google Cloud) with deployment scripts that do not require much manual intervention. As for differences in hardware or operating systems, containerization (e.g. Docker) will be used to ensure the application will behave in a predictable and consistent fashion over various environments. Portability makes the software product easier to deploy and scale, and enhances its long term sustainability.

1.18.11.6 Testability

The system needs to be built to support lots of levels of comprehensive and repeatable testing, including (but not limited to) unit, integration, system and user acceptance testing. Cypress will be used for frontend testing, Selenium for frontend functionality testing, and Jest/Mocha for testing backend logic. Test cases should include critical scenarios such as data encryption, video session stability, accuracy of AI response, authentication, etc. Each software module should have clear definitions of inputs and outputs to make testing of each module independent. Regression tests should be automatically run by Continuous Integration (CI) pipelines following each code update to make sure that new features don't cause flaws in already existing functionalities. It is essential to regularly review and log test results to assure high levels of software correctness and reliability.

1.18.11.7 Robustness

As a result of incorporating these software quality attributes the system is promised to be dependable, flexible and user oriented throughout its lifecycle. Maintainability and scalability is clearly important for long term sustainability and technological relevance, but so are usability and reliability which directly affects patient trust and adoption. For safe digital healthcare settings, the AI Integrated Secure Therapist Patient Management System strikes a balance between performance, adaptability, and resilience by following industry standards and contemporary software engineering principles.

1.19 Other Requirements

1.19.1 Database Requirements

- The primary database for the system will be MongoDB 6 or above.
- Several collections will be created, such as AI Recommendations, Therapy Notes, Goals, Payments, Users and Appointments to store information.
- Client side FLE, with AES256 keys, will be used to encrypt all private information on the client such as patient names, diagnosis, and session notes.
- Each day encrypted backups will be scheduled and stored for 30 days and periodic automated restoration tests will be performed on every month.
- Indexing: Indexing will be used to ensure query times < 1s, e.g. on the most common fields that are queried like email, appointment date etc.

1.20 External Interface Requirements

1.20.1 Hardware Interfaces

- Cloud servers with a minimum of four virtual CPUs and 8 G of RAM per virtual machine.
- End user devices: desktops/Laptops/Smart Phones with ability to connect to encrypted video sessions, Webcam, and Microphone.

1.20.2 Software Interfaces

- Zoom SDK/Google Meet API: for sessions of therapy that are encrypted.
- To make a secure online payment, you can use Stripe's Sandbox API [12].
- Emails or SMS reminder/notifications (SendGrid or Twilio [13]).
- Automation Server: for orchestrating workflows and scheduling AI tasks.
- Key rotation and storage securely (KMS AWS KMS or Azure Key Vault [7]). Interfaces for communication
- If HSTS is turned on, then network traffic will be encrypted via HTTPS (TLS 1.3).
- This will be done using the WSS protocol to encrypt the WebSocket connections and JWT Tokens securing the API calls.

1.21 Requirements for Localization and Internationalization:

- English will be the first language that the interface of the platform and the answers of the artificial intelligence will support.
- Arabic / Urdu localization files (UTF 8) will be part of future releases.
- Date & Time formats will be changed as per user's location using browser settings (MM/DD/YYYY vs DD/MMM/YYYY).
- All the texts will be externalized to translation files, to easily expand in the future.

1.22 Legal and Regulatory Requirements

- The system design will comply with the principle of processing personal and health information lawfully, in accordance with applicable global privacy principles (i.e. similar to HIPAA and GDPR).
- Prior to creating an account, users will expressly consent to data collection.
- The "Right to Erase" will also be given to patients who may want their personal information deleted.
- Logging and audit trails will be immutable to facilitate legal investigations, if required, while third party APIs will undergo checks for compliance with ethical use of AI guidelines and data protection law.

1.23 Ethical and AI Governance Requirements

- The AI assistant will only provide wellness recommendations and reminders it won't diagnose or treated.
- AI content will undergo filtering to prevent the dissemination of harmful, biased, or inaccurate information, as well as incorrect medical information.
- All datasets used for training AI models, including the PII, will be pseudonymized.
- Annual AI auditor checks will be conducted of reports on AI fairness, bias, and transparency.

1.24 System Reuse and Extensibility Objectives

- The use of the modular architecture of the platform (Node/Express backend, React frontend, n8n automation) enables the reuse in future of parts such as:
- Payment and appointment scheduling modules across other health care sector areas.
- Standalone Chatting Bot MicroService with AI support.
- APIs will follow the REST + OpenAPI 3.0 standards to enable easy integration with other third party wellness or hospital management systems.

Chapter 4. Technical Architecture

BENZI.AI is a fully custom built software system designed and developed from the ground up to address the specialized requirements of secure mental health counselling, therapist patient interaction, and AI assisted meditation guidance. No Commercial Off The Shelf (COTS) solution was adopted, as no existing product sufficiently addresses the combined requirements of healthcare data privacy, AI personalization, and role based multi portal access. The following subsections describe the complete architecture of the system, covering application components, data management, interaction patterns, design patterns, tools and technologies, and infrastructure.

Figure 1.1: BENZI.AI High Level System Architecture Layered Client Server Model

BENZI's high level architecture:The layers of AI architecture are: Presentation Layer: React.js web interfaces, Application Layer: Node.js + Express.js RESTful backend, AI Engine Layer: Python with n8n automation, and Data Layer: MongoDB cloud database. All inter layer communication is made via secure REST APIs and WebSocket connections that we use https for. This system can perform online or batch processing. Online processing refers to real time activities such as user authentication, appointment scheduling, live therapy sessions, AI chatbot queries, and payment processing. Background processing operations for features like automated reminders, generating progress reports, processing AI analytics, doing database backups on schedule, and more, are managed with batch processing. This integrates the transaction processing needed to accept appointments, therapy sessions, and payments with analytical reporting like AI generated progress dashboards and insights for the therapist.

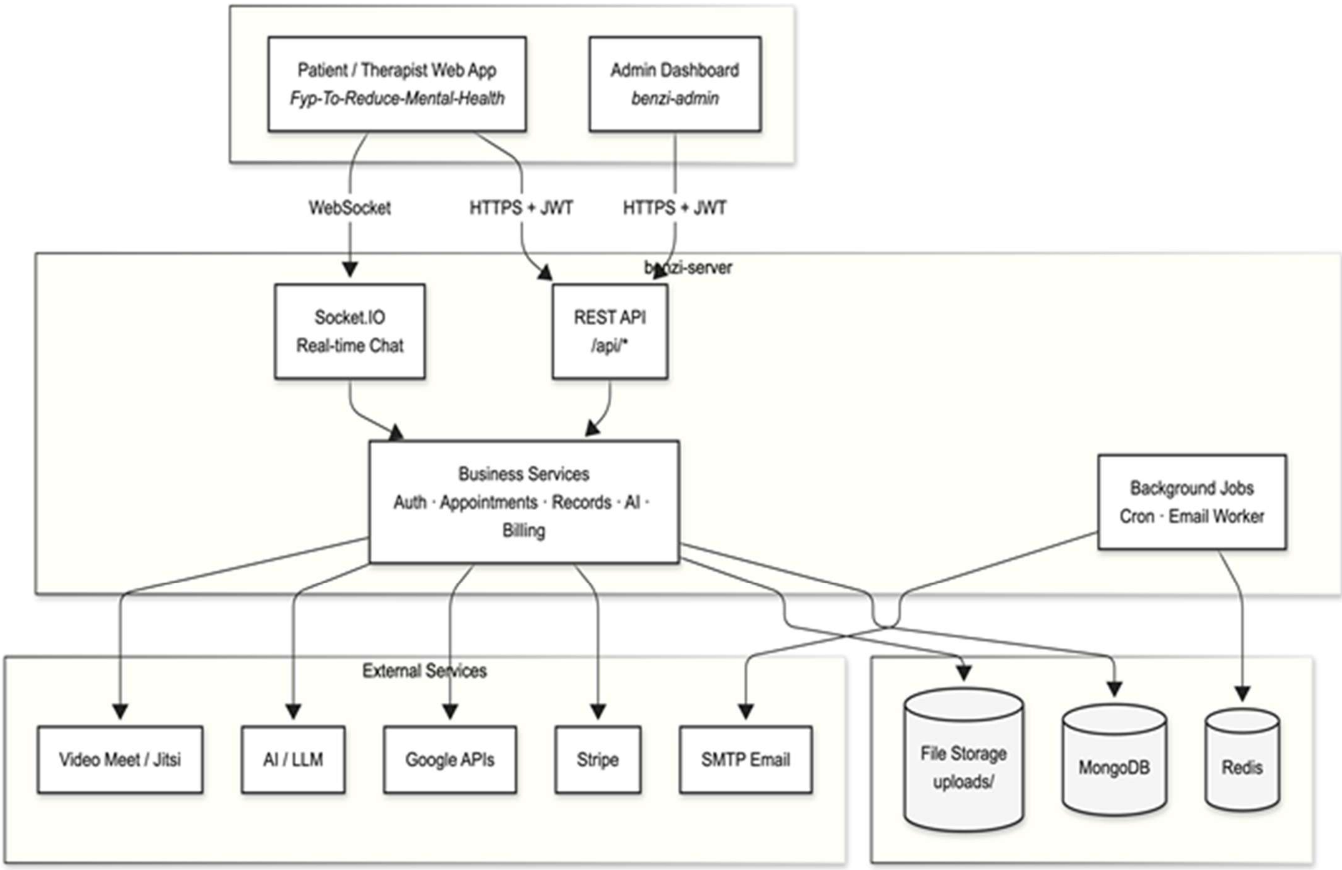


Fig 2.2 Component Diagram

1. Application Components

The frontend web application is implemented in React.js and serves as the browser based interface for all three roles. Three dedicated portals are provided: the Patient Portal, which enables appointment booking, goal tracking, AI assistant interaction, and progress visualization; the Therapist Dashboard, which supports patient registration, session management, medical record maintenance, and analytics; and the Admin Panel, which provides account management, therapist credential verification, subscription control, and platform health monitoring.

The backend application server is implemented using Node.js and Express.js. It provides RESTful API endpoints consumed by all frontend portals and enforces all business logic and security policies. Core backend services include the Authentication and Authorization Service (managing JWT tokens, RBAC, and session control), the Appointment Management Service (real time slot checking, booking, and conflict resolution), the Medical Records Service (encrypted storage and retrieval of patient data, session notes, and clinical reports), and the Notification Service (automated email and SMS dispatching via Stripe, Twilio, and SendGrid).

The AI Engine is developed in Python, using either Flask or FastAPI as the service framework, and is integrated with the backend through secure RESTful APIs. AI workflows and scheduling are orchestrated through n8n, a self hosted workflow automation platform. The AI Engine consists of three sub modules: the AI Virtual Assistant (providing context aware, personalized recommendations based on patient medical history, therapist notes, and session data), the Recommendation Engine (generating progress insights and behavioral trend analysis for therapist facing dashboards), and the Goal Tracking Module (supporting the gamified, points based therapeutic goal system).
The Automation Engine is implemented through n8n workflows and is responsible for all background and scheduled operations, including automated reminder dispatch, report generation, periodic AI analytics execution, and workflow orchestration between the Python AI modules and the Node.js backend.

2. Data Components
The system collects and manages the following categories of data: user registration and authentication credentials, therapist professional profiles and verification documents, patient medical histories (allergies, diagnoses, and treatment plans), therapy session notes and clinical annotations, AI generated recommendations and interaction logs, appointment records and payment transactions, wellness goal assignments and completion histories, system audit logs, and platform usage analytics.

All sensitive healthcare data is stored using MongoDB field level encryption (FLE), with encryption key management handled through a cloud based Key Management Service (KMS) such as AWS KMS. Object storage services on the cloud host are used for file based records (reports, session recordings, uploaded documents). Regular automated backups are configured to ensure data durability and disaster recovery compliance.

1.26 Component Interactions and Collaborations

All components of BENZI.AI interact through a well defined communication architecture. The frontend portals communicate exclusively with the backend application server over HTTPS using RESTful APIs. For features requiring real time bidirectional communication specifically live therapy sessions and chat WebSocket connections are established between the client and the backend. The backend server acts as the central orchestrator, routing requests to the appropriate service module (authentication, records, appointments, or notifications) and invoking the AI Engine through secured API calls when AI processing is required.The AI Engine receives patient profile data and session context from the backend, processes it through th e recommendation and assistant modules, and returns structured outputs that are stored in MongoDB and surfaced to the relevant user dashboards. The n8n automation platform acts as the scheduling and workflow layer, triggering AI pipeline executions, sending notifications through third party APIs, and managing retry logic for failed background operations.Third party integration components connect to the backend through authenticated API clients. Video conferencing is embedded through the Zoom SDK and Google Meet API; payment processing is handled through Stripe; and messaging services (email and SMS) are managed through Twilio and SendGrid. All external API calls are made over HTTPS, with credentials stored securely using environment variables and cloud secret management services.The Sequence Diagram and Activity Diagram for key system workflows are provided in Figures X and X respectively, illustrating the design level interaction flows for appointment booking, therapy session conduct, AI assistant interaction, and goal tracking. These diagrams extend the high level models presented in Phase 2 with full method signatures, return types, and inter service message flows.
The Service Oriented Architecture (SOA) Pattern is applied to core backend functions. Authentication, appointment scheduling, medical records management, AI recommendation delivery, and notification dispatching are each implemented as independent, self contained services that communicate through

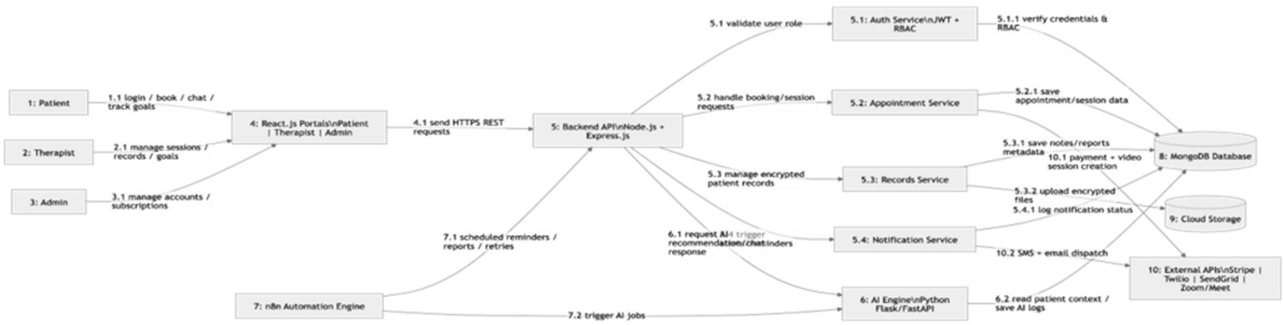


Fig 2.4 Collaboration Diagram

1.27 Design Reuse and Design Patterns

The BENZI.AI system has been architected with a strong emphasis on modularity and reusability, employing the following established software design patterns throughout its implementation.

The Layered Architecture Pattern governs the overall system structure by enforcing strict separation of concerns across the Presentation, Application, AI, and Data layers. No layer communicates with a non adjacent layer directly, ensuring maintainability and independent scalability.

The Model View Controller (MVC) Pattern organizes the backend codebase. Database entity definitions serve as Models, Express.js controllers encapsulate business logic and API request handling, and the React.js frontend components constitute the View layer.

REST APIs. This enables individual services to be scaled, replaced, or extended without affecting other system components.

The Observer Pattern governs the notification subsystem. User facing events including appointment confirmations, goal completion milestones, and session reminders automatically trigger notification dispatch without requiring direct coupling between the business logic and the messaging infrastructure.

The Factory Pattern is applied to user role instantiation and AI response handler creation. The system dynamically generates the appropriate user context object (Patient, Therapist, or Admin) and AI response type at runtime based on authenticated session data.

The Singleton Pattern ensures that system wide resources including the MongoDB database connection, configuration managers, and logging services are instantiated once and shared consistently across all modules, preventing resource duplication and ensuring uniform state management.

Reusable Components: A centralized Authentication and RBAC module is reused across all three portals. React.js UI components including navigation bars, appointment cards, modal dialogs, notification banners, and form elements are developed once and reused across all screens. RESTful API endpoints for user profiles, appointments, and analytics are designed as shared services consumed by multiple modules. The AI recommendation logic is implemented as a standalone, reusable service invokable from therapy sessions, goal tracking, and progress dashboards. The n8n automation workflows for reminders and report generation are designed as reusable templates applicable across multiple system functions.

1.28 Technology Architecture

BENZI.AI is designed to operate as a cloud based, scalable, and highly available platform. The complete technology stack is summarized in Table X below.

Table 1: Technology Stack Summary

Layer	Component	Technology / Tool
Presentation	Web Application	React.js
Application	Backend Server	Node.js, Express.js
Application	Workflow Automation	n8n
AI Engine	AI Module	Python (Flask / FastAPI)
Data	Primary Database	MongoDB
Data	Key Management	AWS KMS / Azure Key Vault
Integration	Video Conferencing	Zoom SDK, Google Meet API
Integration	Payment Processing	Stripe API
Integration	Messaging (SMS)	Twilio API
Integration	Messaging (Email)	SendGrid API
Infrastructure	Cloud Hosting	AWS / Microsoft Azure / GCP
Security	Data Encryption	AES 256, TLS/HTTPS, E2EE
Security	Authentication	JWT, 2FA, RBAC
DevOps	Version Control	Git / GitHub

System Hosting: The system is hosted on an external cloud data center Amazon Web Services (AWS), Microsoft Azure, or Google Cloud Platform (GCP) providing elastic scalability, high availability, automated backups, and disaster recovery. Containerized or virtualized deployment of all application services, AI modules, databases, and automation workflows is supported.

Hardware Infrastructure: The system operates on cloud based virtual machines providing scalable CPU and memory resources for concurrent user sessions, optional GPU support for AI model inference, secure encrypted storage volumes, and load balancers to distribute incoming traffic across application instances. No dedicated on premises hardware is required.

Network Architecture: BENZI.AI operates entirely over the Internet using secure HTTPS communication. All client server interactions are encrypted using TLS 1.2 or higher. Real time features (live chat, video sessions) require stable broadband connectivity with low latency and are supported through WebSocket connections. The system is not available on a LAN or private WAN it is a public internet, cloud hosted platform accessible from any modern web browser without client side software installation.

End User Interface: The system provides a browser based end user interface accessible through all major modern browsers (Chrome, Firefox, Edge, Safari). No thick client installation is required. The interface is responsive and supports desktops, tablets, and mobile browsers.

Modes of Operation: The system supports three operational modes. Online Mode enables real time interactions including user authentication, appointment booking, therapy sessions, AI chatbot communication, and dashboard updates. Background Mode manages automated workflows for reminders, notifications, analytical report generation, and scheduled AI tasks. Maintenance Mode supports system updates, security patches, and performance monitoring with minimal service interruption.

1.29 Architecture Evaluation

The architectural decisions made for BENZI.AI were guided by the specific requirements of a secure, scalable, AI integrated mental health platform. This subsection justifies the key technology choices and evaluates each against alternative options.

React.js vs Angular / Vue.js: React.js was selected for the frontend due to its component based architecture, which directly supports the creation of reusable UI elements across the three portals, its large ecosystem, and its flexibility for integration with third party libraries. Angular, while offering a more opinionated structure, introduces higher complexity and steeper learning curve unsuitable for the project timeline. Vue.js, although lightweight, has a comparatively smaller ecosystem and less community support for enterprise scale healthcare UI patterns.

Node.js + Express.js vs Django / Spring Boot: Node.js was selected as the backend framework for its non blocking, event driven architecture, which is well suited to real time operations such as WebSocket based therapy sessions and concurrent API requests. Django (Python) was considered but excluded because it would require maintaining two separate Python environments one for Django and one for the AI modules adding unnecessary operational complexity. Spring Boot (Java) was excluded due to its greater infrastructure overhead and mismatch with the team's primary skill set.

MongoDB vs PostgreSQL / MySQL: MongoDB was selected as the primary database due to its schema flexibility, which is critical for managing semi structured healthcare data (session notes, AI generated recommendations, and variable length medical histories), and its native support for field level encryption. PostgreSQL, while more mature for relational data, would require more complex schema migrations when the data model evolves. The system's data is predominantly document oriented, making a NoSQL solution structurally appropriate.

Python (Flask / FastAPI) for AI vs Node.js native AI: Python was selected for all AI components due to its unmatched ecosystem of machine learning libraries (TensorFlow, PyTorch, scikit learn, HuggingFace Transformers). Implementing AI logic natively in Node.js would significantly limit the availability of trained model architectures and community support. The separation of the AI module as an independent Python service, integrated through REST APIs, also supports independent scaling and model replacement without affecting the main application backend.

n8n vs AWS Step Functions / Apache Airflow: n8n was selected for workflow automation due to its self hosted nature (supporting HIPAA/GDPR compliance by keeping data processing within the controlled environment), its visual workflow designer that simplifies maintenance, and its native REST

API integration: AWS Step Functions would introduce vendor lock in and higher operational cost. Apache Airflow, while powerful, is over engineered for the scheduling complexity required by this project and would require significant DevOps investment to maintain.

Cloud Hosting (AWS / Azure / GCP) vs On Premises: Cloud hosting was selected to provide elastic scalability, high availability, and built in disaster recovery without the capital investment of on premises infrastructure. HIPAA and GDPR compliance is achievable on all three major cloud providers through their respective Business Associate Agreements (BAA) and compliance certification programs (e.g., AWS HIPAA eligible services, Azure Healthcare APIs).

Chapter 5. Detailed Design and Implementation

Deployment Diagram:

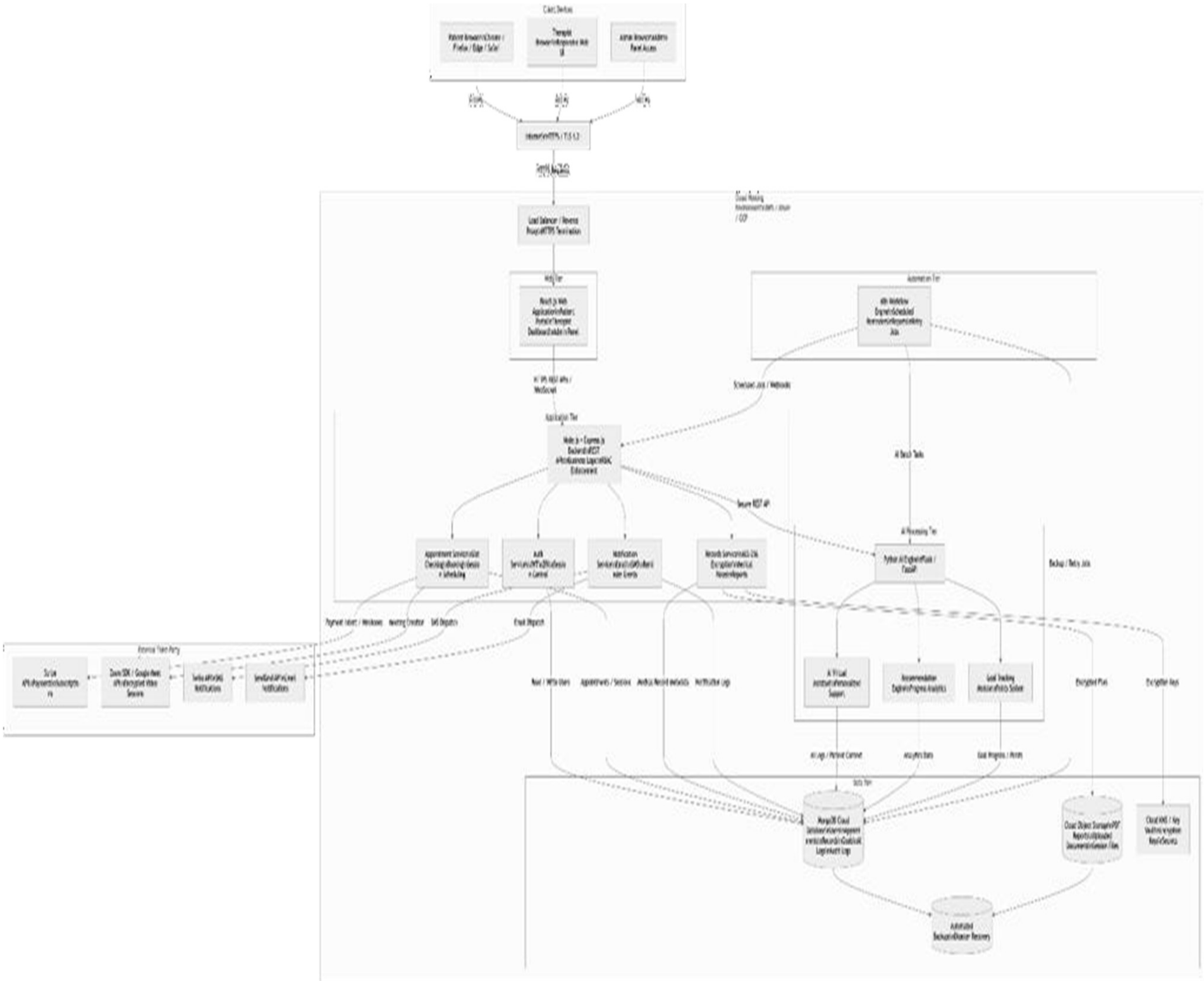


Fig 5.1 Deployment Diagram

DataBase Diagram:

Fig 5.2 Database Diagram

State Diagram:

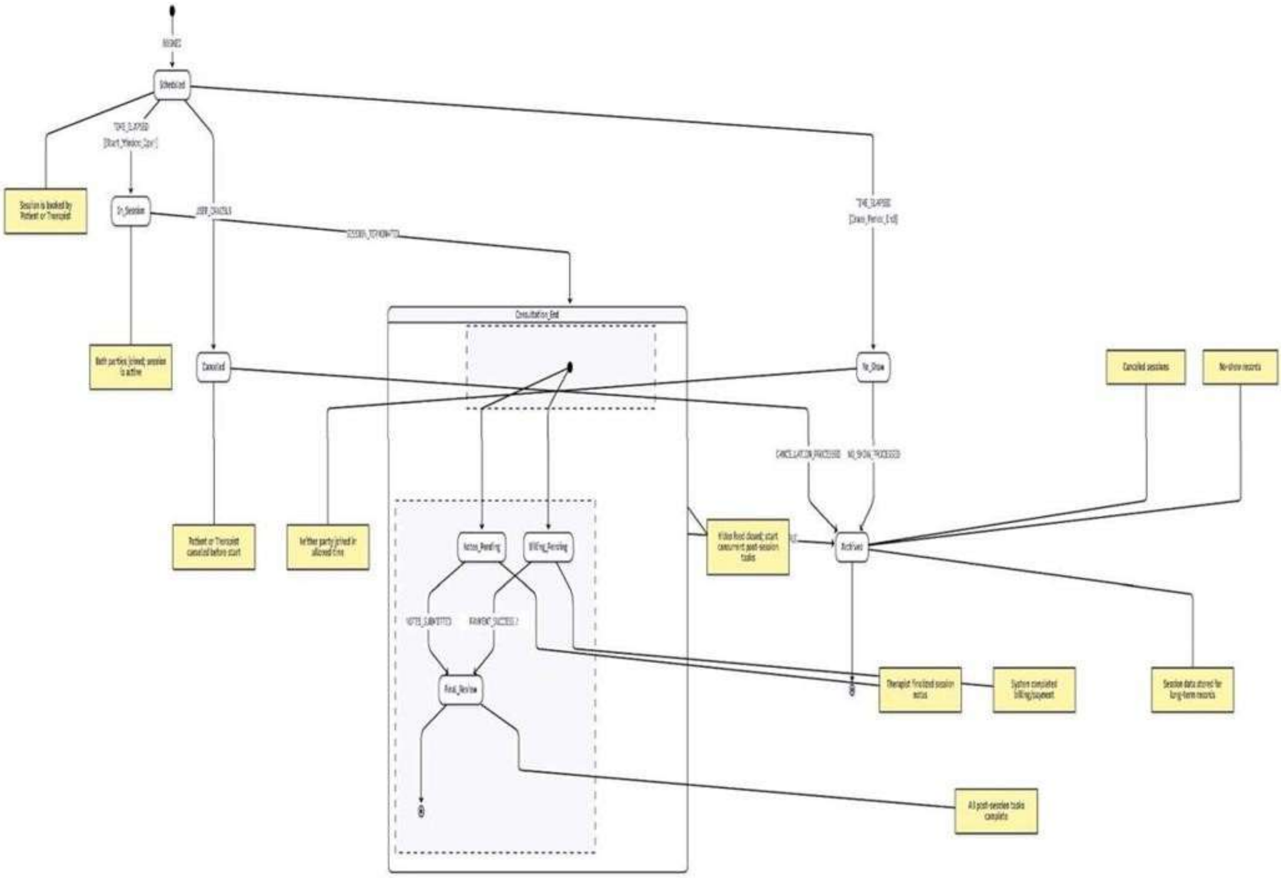


Fig 5.3 State Diagram

Petrinet Diagram:

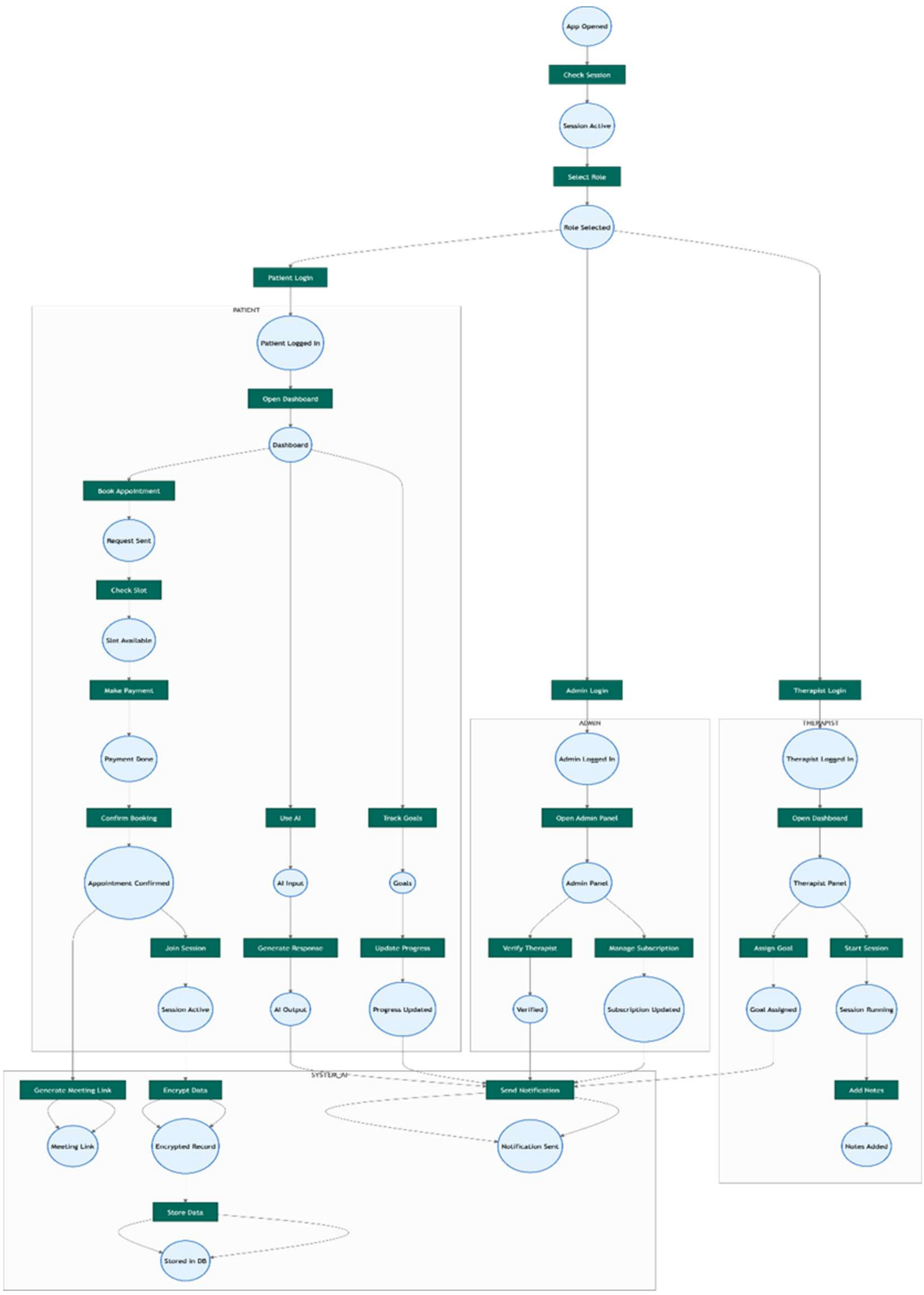


Fig 5.4 Petrinet Diagram

1.30 Component Component Interface

1.30.1 UC 1: Book Appointment with Therapist

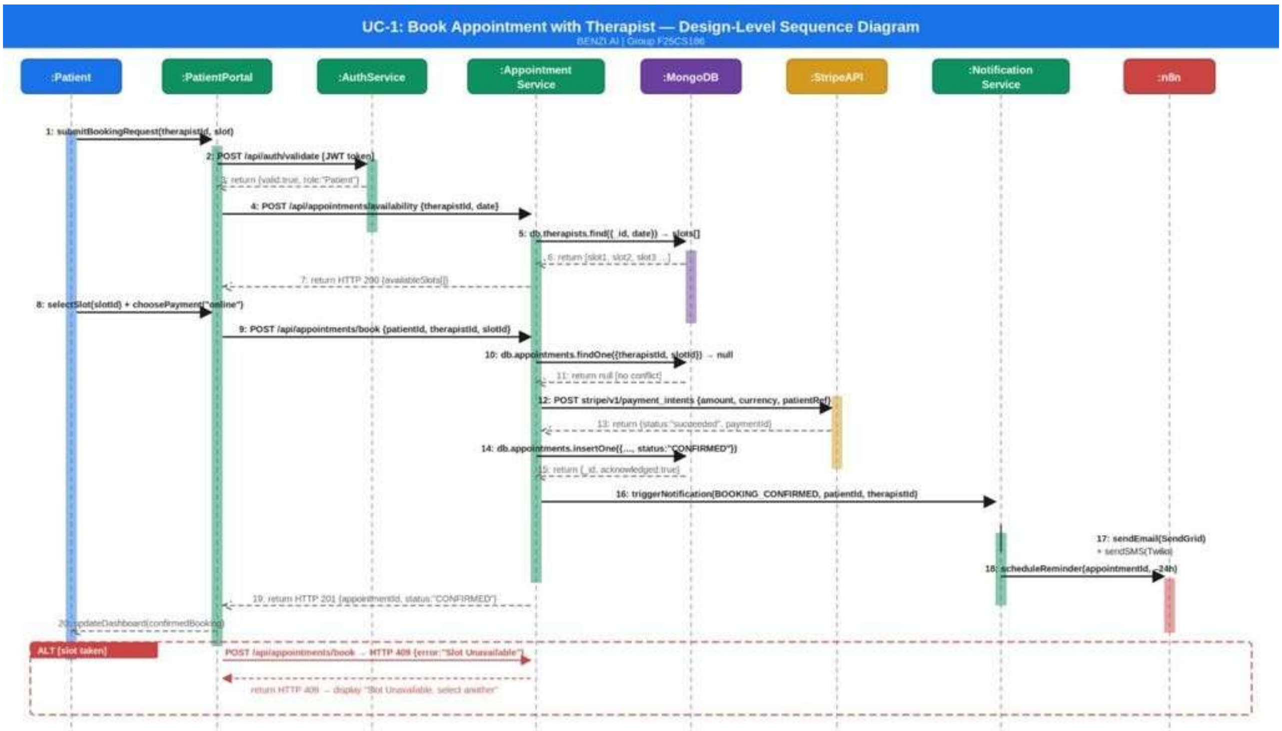


Figure 1: UC 1 Book Appointment Design Level Sequence Diagram

1.30.2 UC 2: Conduct Therapy Session (Encrypted + AI Support)

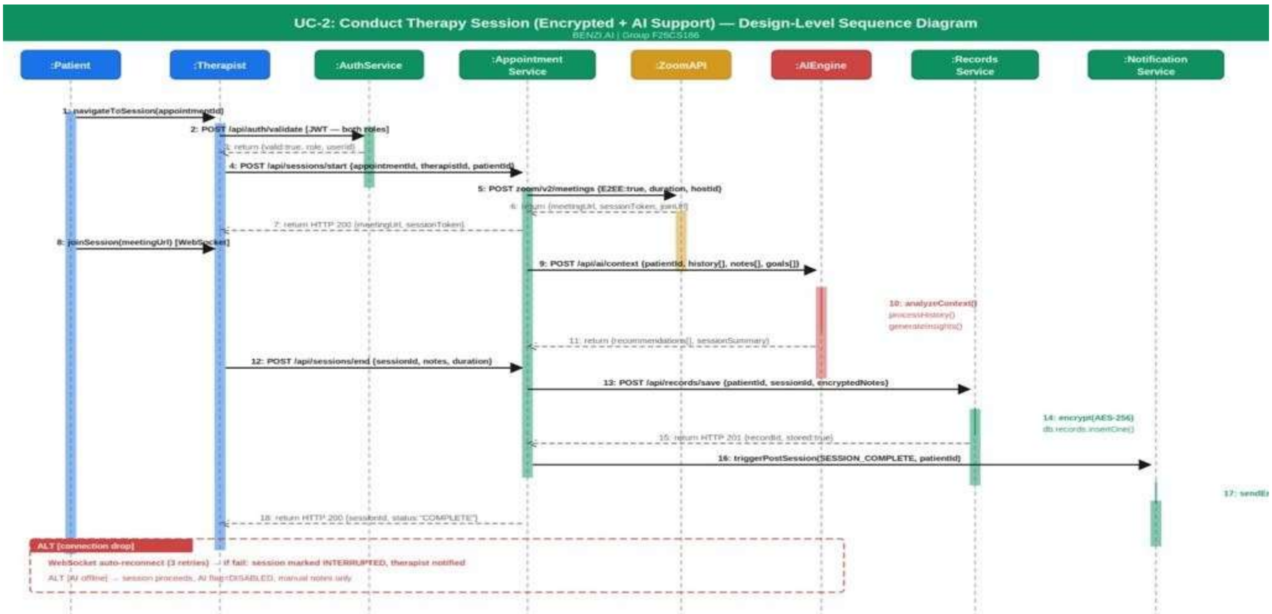


Figure 2: UC 2 Conduct Therapy Session   Design Level Sequence Diagram

1.30.3 UC 3: Manage Patient Records and Notes

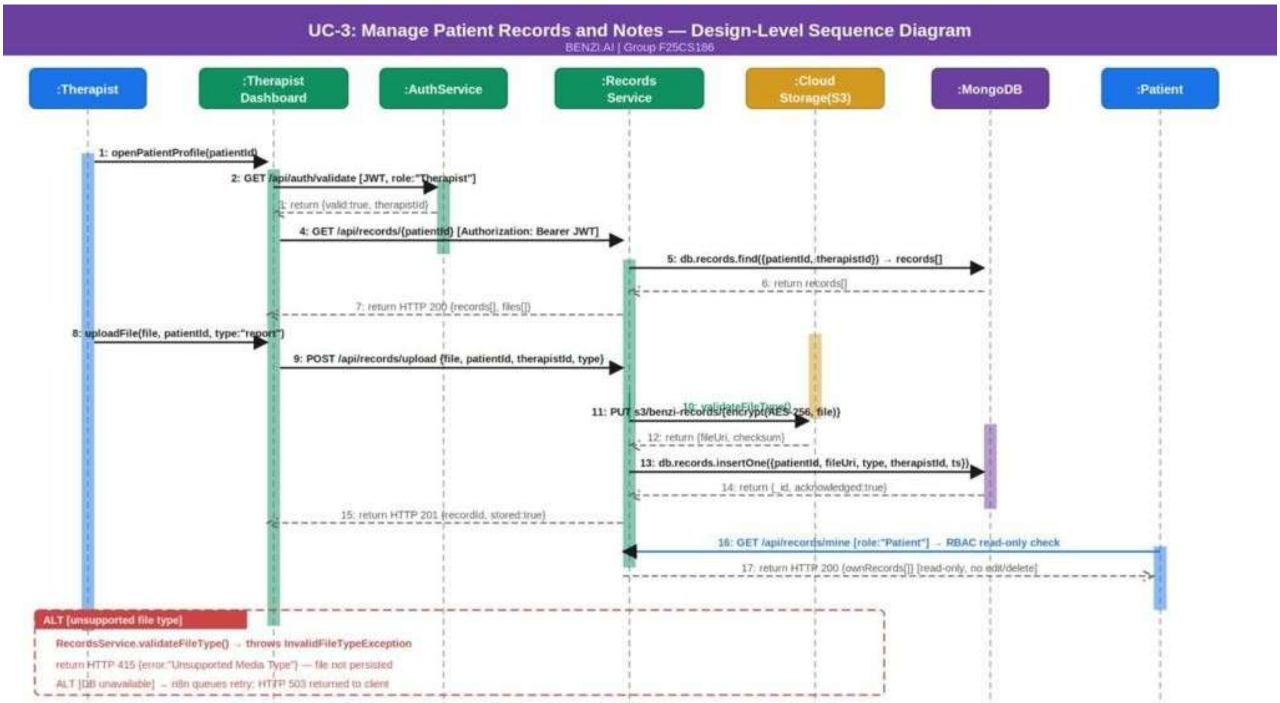


Figure 3: UC 3 Manage Patient Records   Design Level Sequence Diagram

1.30.4 UC 4: Assign and Track Wellness Goals

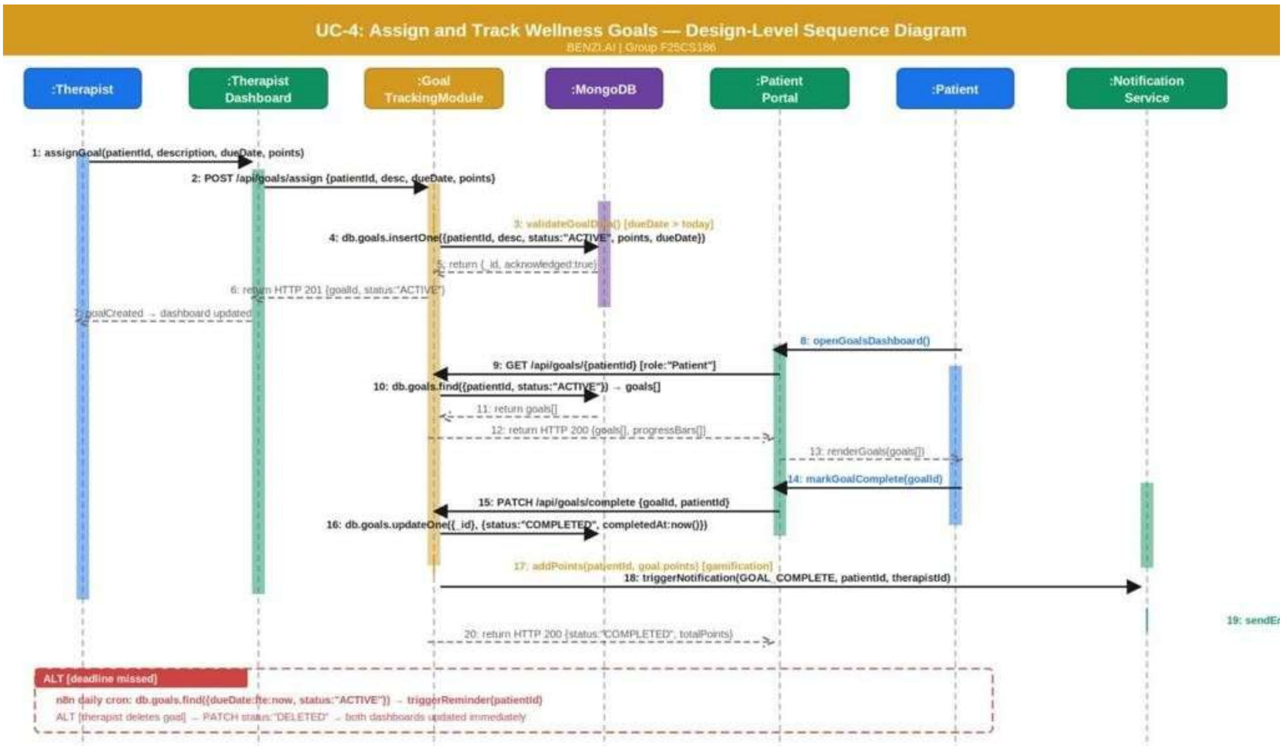


Figure 4: UC 4 Assign and Track Wellness Goals Design Level Sequence Diagram

1.30.5 UC 5: Interact with AI Virtual Assistant

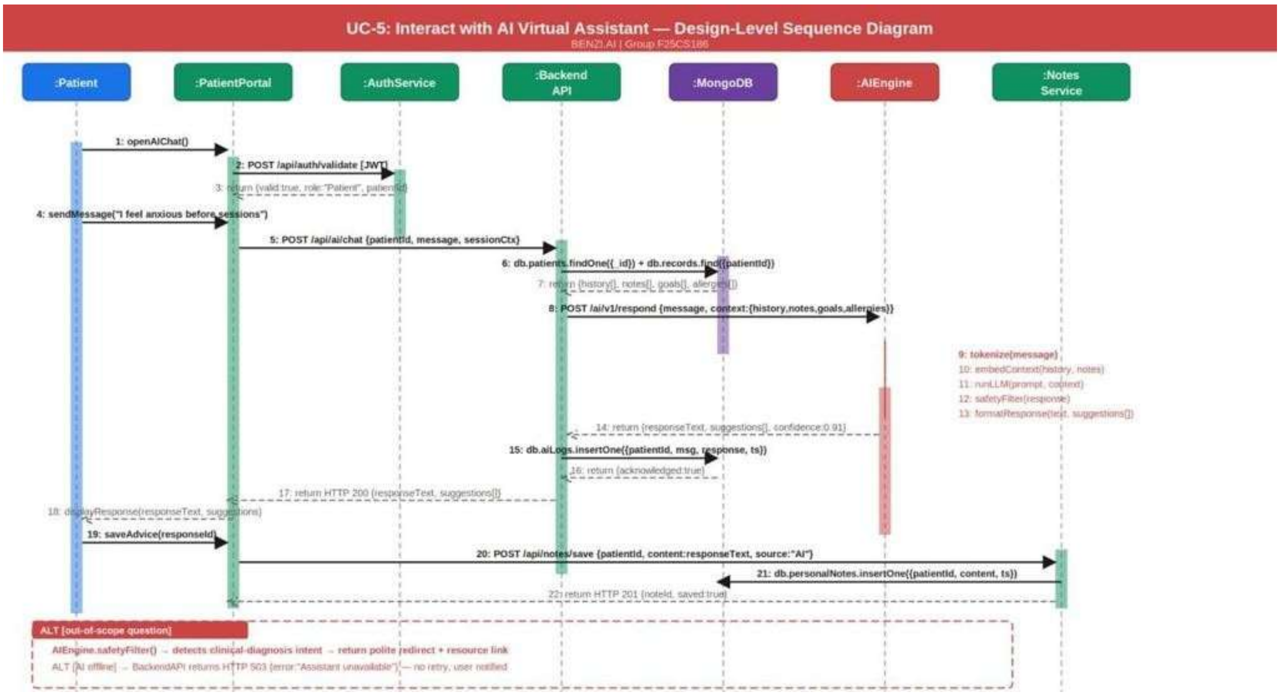


Figure 5: UC 5 AI Virtual Assistant Interaction Design Level Sequence Diagram

1.30.6 UC 6: Manage User Accounts and Subscriptions

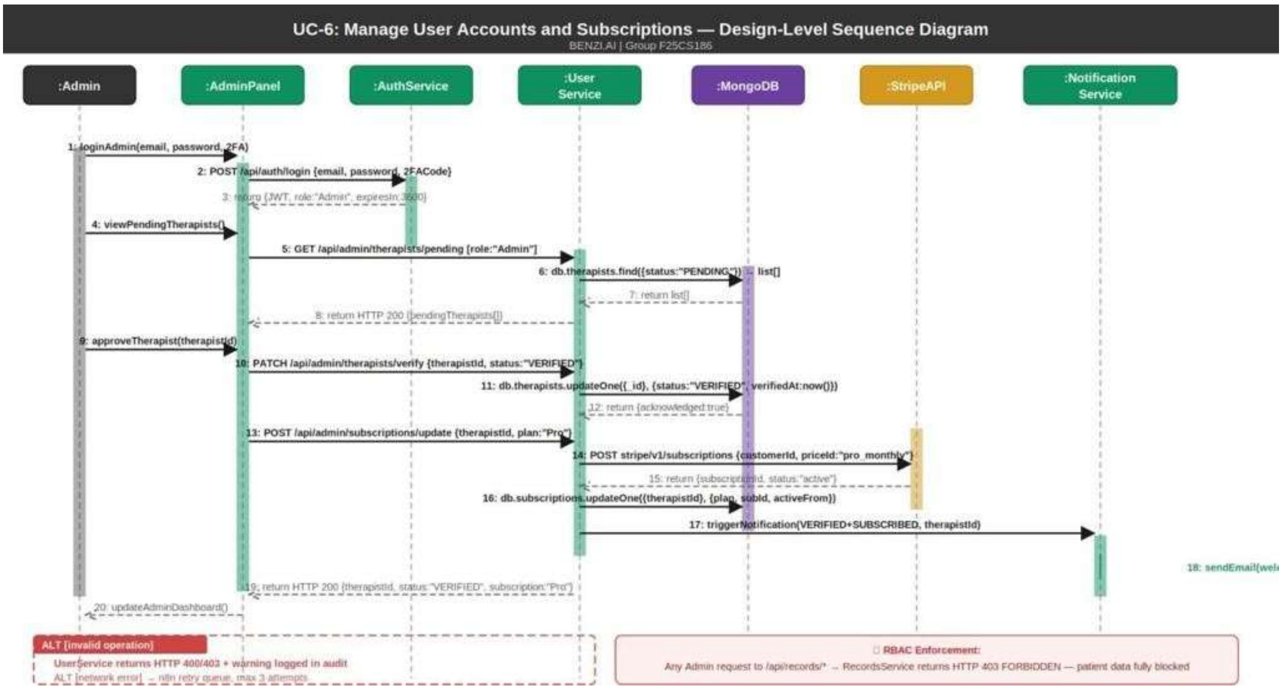


Figure 6: UC 6 Manage Accounts and Subscriptions Design Level Sequence Diagram

Component Interaction Summary Table

Insert the following table after the six sequence diagram subsections.

Table 2: Internal Component Interaction Summary

Calling Component	Called Component	Method / Endpoint	Protocol	Returns
PatientPortal	AuthService	POST /api/auth/validate	HTTPS REST	{valid, role, userId}
PatientPortal	AppointmentService	POST /api/appointments/book	HTTPS REST	HTTP 201 {appointmentId}
AppointmentService	MongoDB	db.appointments.insertOne()	MongoDB Driver	{_id, acknowledged}
AppointmentService	StripeAPI	POST stripe/v1/payment_intents	HTTPS REST	{status, paymentId}
AppointmentService	NotificationService	triggerNotification()	Internal Event	void

NotificationService	n8n	scheduleReminder()	HTTP Webhoo k	{jobId}
TherapistDashboa rd	AIEngine	POST /api/ai/context	HTTPS REST	{recommendations []}
BackendAPI	AIEngine	POST /ai/v1/respond	HTTPS REST	{responseText, suggestions[]}
RecordsService	CloudStorage	PUT s3/benzi records/	AWS SDK	{fileUri, checksum}
RecordsService	MongoDB	db.records.insertOne()	MongoD B Driver	{_id, acknowledged}
GoalTrackingModu le	MongoDB	db.goals.updateOne()	MongoD B Driver	{acknowledged}
GoalTrackingModu le	NotificationServic e	triggerNotification()	Internal Event	void
UserService	StripeAPI	POST stripe/v1/subscriptions	HTTPS REST	{subscriptionId, status}
AdminPanel	RecordsService	GET /api/records/*	HTTPS REST	HTTP 403 FORBIDDEN

1.31 Component External Entities Interface

BENZI.AI communicates with four categories of external systems. All external communications are performed over HTTPS using authenticated API clients with credentials stored securely in environment variables and cloud secret managers (AWS Secrets Manager / Azure Key Vault). No external system has direct access to the BENZI.AI database. All external API calls are made exclusively from the backend Application Layer the frontend never communicates directly with any third party service.

External Entity 1: Zoom SDK / Google Meet API (Video Conferencing)
Direction: BENZI.AI Backend → Zoom / Google Meet
Communication: The Appointment Service calls the Zoom REST API or Google Meet API at session start time to dynamically generate an encrypted meeting room. The returned session URL is distributed to the Patient and Therapist through their respective portals. End to end encryption of the video session is enforced at the provider level (Zoom E2EE / Google Meet encryption).
Data Exchanged: Session creation request (therapist ID, patient ID, scheduled time, duration) → Encrypted meeting URL and session token returned.
Error Handling: If the API returns an error or timeout, the backend retries once. On second failure, both parties are notified via the Notification Service and the session is flagged for manual rescheduling.

External Entity 2: Stripe API (Payment Processing)
Direction: BENZI.AI Backend → Stripe
Communication: The Appointment Service calls the Stripe API during the online payment path of appointment booking. The backend creates a payment intent, which is processed by Stripe. BENZI.AI never stores raw card data all payment data is handled exclusively by Stripe's PCI DSS compliant infrastructure. Webhook callbacks from Stripe notify the backend of payment confirmation or failure.

Data Exchanged: Payment intent creation request (amount, currency, patient reference) → Payment confirmation or failure event returned via webhook.
Error Handling: Payment failure triggers appointment cancellation, patient notified via Notification Service. Network timeout triggers retry; on persistent failure, session flagged as PAYMENT_PENDING.

External Entity 3: Twilio API (SMS Notifications)
Direction: BENZI.AI Notification Service → Twilio
Communication: The Notification Service calls the Twilio REST API to dispatch SMS messages. Triggered events include appointment confirmations, 24 hour session reminders, goal deadline reminders, and system alerts. n8n workflows schedule and manage the timing of all SMS dispatches.
Data Exchanged: SMS dispatch request (recipient phone number, message body) → Delivery status response (DELIVERED / FAILED).
Error Handling: Failed delivery logged in MongoDB audit log. n8n workflow retries failed SMS up to three times with exponential backoff.

External Entity 4: SendGrid API (Email Notifications)
Direction: BENZI.AI Notification Service → SendGrid
Communication: The Notification Service calls the SendGrid API to dispatch HTML formatted email notifications. Triggered events mirror Twilio triggers and additionally include post session summaries, AI advice logs (when saved by patient), and subscription receipts.
Data Exchanged: Email dispatch request (recipient address, subject, HTML body, template ID) → Delivery acknowledgement.
Error Handling: Undelivered emails logged in audit system. Bounced emails flagged in patient/therapist profile for admin review.

External Entity Interaction Summary

Table 3.2.1: Component External Entity Interaction Summary

External System	BENZI.AI Component	Protocol	Purpose	Data Direction
Zoom SDK	Appointment Service	HTTPS REST	Encrypted video session creation	Outbound→Inbound URL
Google Meet API	Appointment Service	HTTPS REST	Encrypted video session creation	Outbound→ Inbound URL
Stripe API	Appointment Service	HTTPS REST+ Webhook	Payment processing	Outbound→ Inbound event
Twilio API	Notification Service	HTTPS REST	SMS dispatch	Outbound→ Status

SendGrid API	Notification Service	HTTPS REST	Email dispatch	Outbound→ Status
AWS KMS / Azure Key Vault	Records Service	SDK	Encryption key management	Bidirectional
AWS S3 / Azure Blob	Records Service	SDK	Encrypted file storage	Outbound→ Inbound URI

1.32 Component Human Interface

This subsection identifies all screens and interaction points through which users provide input to or receive output from the BENZI.AI system. The interface is entirely browser based (no thick client required) and is accessible via Chrome, Firefox, Edge, and Safari. The design follows Human Computer Interaction (HCI) principles including clarity, minimal cognitive load, role appropriate information presentation, and accessibility conscious design consistent with WCAG guidelines.

HCI Principles Followed

Role Based Interface Segregation: Each user role (Patient, Therapist, Admin) accesses a completely distinct portal with only the functions relevant to their role visible. No role sees interface elements belonging to another role, reducing confusion and enforcing the principle of least privilege at the UI level.

Minimalist and Calm Design: Given the mental health context of the application, the UI employs a minimalistic design language with soft color palettes, generous whitespace, and simple navigation. This is intentional to reduce visual fatigue and anxiety for users who may already be in emotionally vulnerable states.

Progressive Disclosure: Complex features (AI assistant configuration, subscription management, analytics) are not surfaced immediately. Users are progressively guided to advanced features only after completing primary actions, preventing overwhelm on first use.

Feedback and Confirmation: All destructive or significant actions (appointment cancellation, goal deletion, account suspension) require explicit confirmation dialogs before execution. Success and error states are communicated with clear, non technical messages.

Accessibility: Font sizes meet minimum readability standards. Contrast ratios between text and background are maintained at WCAG AA level. All interactive elements are keyboard navigable.

The following table lists all screens identified from the actual implemented prototype.
Table 3.1: Screen Inventory Input/Output and Component Mapping

#	Screen Name	User Role	Input Received	Output Provided	Backend Component
1	Home	Public / Guest	Navigation clicks, newsletter email subscription	Platform overview, care offerings (NLP, Emotion Recognition, Personalized Recommendations, Risk Assessment, 24/7 Availability, Continuous Learning), subscription plan tiers, CTA buttons	None (static + SendGrid)
2	Auth Role Selection	Public / Guest	Role card selection (Register As Therapist / Register As User)	Redirects to role specific Register or Login screen	AuthService
3	Login	All Roles	Email / username, password, Remember Me checkbox	JWT session token, role based dashboard redirect, Forgot Password link	AuthService
4	Register	Patient / Therapist	Full name, email, password, role selection; Therapist additionally submits credentials	Account creation confirmation, email verification link dispatched	AuthService, SendGrid

5	**Patient Dashboard**	**Patient**	**Mood selection (Happy / Good / Normal / Bad / Awful), Submit button**	**Task score counter, mood log confirmation, Current Progress rings (Mental Health / Self Care / Therapy), weekly Task Progress bar chart, Overall Report annual line chart**	**RecordsService, GoalTrackingModule**
6	**Patient Conversations (AI Chat)**	**Patient**	**Free text message, voice input button**	**BENZI AI response bubbles, chat history panel (Today / Yesterday / Previous), New Chat button, search, save option**	**AIEngine, NotesService**
7	**Patient Goals**	**Patient**	**Goal preset selection (Enhance Sleep Quality / Improve Stress Management / Improve Communication Skills),current status sliders (0 100), custom goal text input, Submit**	**Active goals with progress percentage, Completed / In Progress / Pending pie chart (70% / 20% / 10%), Community Reviews sentiment (Negative / Neutral / Positive), AI Insights & Recommendations feed**	**GoalTrackingModule, AIEngine**

8	**Patient Progress**	**Patient**	**Date range filter, Monthly / Annual toggle**	**Individual goal progress bars (Stress / Anxiety / Depression / Improve Sleep at 100% / 50% / 30% / 10%), Overall Goal Progress area chart (Jan May), Benzi Usage hours bar chart, Chatbot Usage weekly bar chart, Overall Report line chart**	**RecommendationEngine, MongoDB**
9	**Patient Appointment**	**Patient**	**+ Appointment button, keyword search, per row action icons**	**Appointments table (ID, Therapist, Date & Time, Duration, Location, Status: Pending / Completed), Video Call link generation panel, Clinic / Hospital address generation panel, paginated list**	**AppointmentService, ZoomAPI**
10	**Patient Reports**	**Patient**	**Keyword search, Download action, Feedback action, per row menu**	**Reports table (Report ID, Therapist, Date & Time, PDF File, Location, Status: Reviewed / Not review / Half review), Task generation section, Notes / Guidance generation section**	**RecordsService, MongoDB**

11	**Patient Profile**	**Patient**	**Profilephoto upload, Full Name, Email, Password change (old / new / confirm)**	**Updated profile confirmation, password change confirmation**	**AuthService, MongoDB**
12	**Patient Help & Support**	**Patient**	**Support query input, topic selection**	**Help articles, support ticket submission, response confirmation**	**NotificationService, SendGrid**
13	**Subscription (Public)**	**Public / Guest**	**Plan selection (Basic $100 / Standard $200 / Premium $300), Monthly / Annual billing toggle, Get Started button**	**Plan comparison cards with feature lists, redirect to payment / registration**	**StripeAPI**
14	**Therapist Dashboard**	**Therapi st**	**Navigation clicks, calendar date selection**	**KPI cards: Active Services (12, +4.2%), New Services (2, −1.2%), Avg Reviews (20, +3.4%), Avg Reply Time (15 min, +3.4%); Today's Appointments card with patient name and session topic; monthly calendar view; Most Bought Package donut charts (Stress Management 60%, Career Counseling 75%, Couples**	**AppointmentService, MongoDB**

				Counseling); Generated Revenue horizontal bar chart (Aug $1000 / Sep $700 / Oct $800)	
15	**Therapist Appointment**	**Therapi st**	**Keyword search, + Calendar button, per row action icons**	**Appointments table (Appointment ID, Patient, Date & Time, Duration, Location, Status: Confirmed / Pending / Cancelled), paginated list**	**AppointmentService, MongoDB**
16	**Therapist Clients**	**Therapi st**	**+ New Client button, keyword search, per row messaging and action icons**	**Clientstable (Appointment ID, Patient name, Date & Time, Duration, Location, Status: Confirmed / Pending /**	**RecordsService, MongoDB**
				Cancelled), paginated list	

17	**Therapist Services**	**Therapi st**	**+ New Service button, Edit / Delete per service card**	**Service cards displaying: service name, type, price per 60 minute session, description (e.g. Mindful Counseling Individual Therapy $100/60min; Career Counseling $120/60min; Couple Counseling $120/60min)**	**MongoDB**
18	**Therapist Subscription**	**Therapi st**	**Plan selection, upgrade / downgrade action**	**Active subscription plan details, plan comparison, Stripe payment redirect**	**StripeAPI, MongoDB**
19	**Therapist Payments**	**Therapi st**	**Date range filter**	**Payment history list, revenue summary by period**	**StripeAPI, MongoDB**
20	**Therapist Profile**	**Therapi st**	**Full Name, WhatsApp, Email (Personal Information section); Specialization, Qualification, Practice, Experience, Session count, Client count, Bio / Message (Professional Information**	**Updated profile confirmation, password change confirmation**	**AuthService, MongoDB**

			section); Old Password, New Password, Confirm Password (Change Password section); Change Photo		
21	Therapist Profile Setup	Therapi st	Initial onboarding fields (name, credentials, specialization)	Profileactivation, onboarding completion confirmation	AuthService, MongoDB
22	Meditation Counselor (Public)	Public / Guest	Navigation clicks, See Our Benefits / See more buttons	AI approach explanation ("Your journey begins with an initial consultation"), How it Works 6 step process, feature highlights (Risk Assessment, Emotion Recognition, 24/7 Availability), Interactive Therapy Modules section, AI feedback & learning description	None (static)

23	**Resources**	**Public / Guest**	**Navigation / category filter clicks**	**Mental health resource articles and materials library**	**None (static / CMS)**
24	**About Us**	**Public / Guest**	**Navigation click**	**Team information, mission statement, company background**	**None (static)**
25	**About BENZI.AI**	**Public / Guest**	**Navigation click**	**Product overview, brand values, platform description**	**None (static)**
26	**FAQs**	**Public / Guest**	**Navigation click, question expansion**	**Frequently asked questions list with expandable answers**	**None (static)**

27	**Contact Us**	**Public / Guest**	**Full name, email address, message text, Submit**	**Support ticket creation, submission confirmation message dispatched**	**NotificationService, SendGrid**
28	**Admin Dashboard**	**Admin**	**Navigation clicks, date range filter (7 days / 30 days / 12 months)**	**KPI cards: Total Doctors (48, +3.2%), Total Patients (1,240, +5.1%), Monthly Revenue ($24,800, +4.2%), Active Subscriptions (36, +1.8%); Most Sold Packages donut charts (Standard Plan 60%, Pro Plan 75%, Enterprise Plan 45%); Revenue Overview 12 month line chart; Today's Doctor Appointments card with calendar**	**UserService, MongoDB, StripeAPI**
29	**Admin Doctors**	**Admin**	**Keyword search, + Add Doctor button, per row actions (Edit / View Profile / Send Credentials / Delete)**	**Doctors table: Doctor ID, Name, Specialization, Patients count, Subscription plan (Pro / Standard / Enterprise), Status (Active / Pending / Inactive), action menu; paginated list of 6 doctors per page**	**UserService, MongoDB**

30	**Admin Patients**	**Admin**	**Keyword search, view (eye icon) and menu actions per row**	**Patients table grouped by doctor (Doctor Name, Specialization, Total Patients, Active count, Inactive count, Last Session date); Patient Distribution donut chart (Mental Health 40% / Self Care 35% / Therapy 25%); New Patients This Month daily bar chart**	**UserService, MongoDB**
31	**Admin Subscription s**	**Admin**	**+ New Plan button, Edit / Delete per plan card, Manage per assignment row**	**Plan cards: Standard ($299/mo, 36 active doctors), Pro ($499/mo, Most Popular, 12 active), Enterprise ($799/mo, 8 active) with full feature lists; Subscription Assignments table (Doctor ID, Name, Plan, Start Date, Expiry Date, Status: Active / Pending / Inactive)**	**StripeAPI, MongoDB**
32	**Admin Revenue**	**Admin**	**Date range filter, per payment actions (Edit / Delete)**	**Revenue KPI cards: Total Revenue ($124,800, +8.3%), This Month ($24,800, +4.2%), Pending Payouts ($3,200, −1.1%); Payments table (Payment ID, Doctor, Patient, Date, Service, Amount, Status: Completed / Pending); Monthly**	**StripeAPI, MongoDB**

				Revenue horizontal bar chart; Revenue by Plan donut chart (Standard / Pro / Enterprise)	
33	**Admin Send Credentials**	**Admin**	**First Name, Last Name, Email, Phone No, Specialization, Subscription Plan, Temporary Password (manual or Auto generate), Welcome Message text, Send Credentials button**	**Credential dispatch confirmation; Recently Sent log showing doctor avatar, name, email, timestamp and delivery status (Sent ✓ / Failed ✗)**	**AuthService, NotificationService, SendGrid**
34	**Admin Send Credentials (Nav Active State)**	**Admin**	**Same as #33 Send Credentials nav item highlighted**	**Same output as #33 shows active navigation state for Send Credentials section**	**AuthService, NotificationService, SendGrid**
35	**Admin Profile**	**Admin**	**(In development no input fields active)**	**Placeholder message: "Admin profile settings coming soon."**	**AuthService**

36	**Admin About Benzi**	**Admin**	**Navigation click**	**Platform description: "Benzi is a mental wellness platform connecting patients and therapists."**	**None (static)**

1.33 Screenshots/Prototype

1.33.1 Workflow

The BENZI.AI system operates across three parallel user journeys that converge at the AI Engine and database layer. The Public/Guest user lands on the Home screen and navigates to the Auth screen to select their role. A Patient registers, logs in, and accesses the Patient Portal where they can track mood, book appointments, chat with the BENZI AI assistant, view goals assigned by their therapist, monitor progress analytics, and download session reports. A Therapist registers with credentials, awaits admin verification, then accesses the Therapist Portal where they manage their service listings, view and confirm appointments, monitor their client list, and track their generated revenue. All authenticated interactions mood submissions, AI chat messages, appointment bookings, report downloads pass through the Node.js backend to MongoDB and, where applicable, to the AI Engine or third party APIs. The n8n automation engine runs in the background managing reminders, notifications, and analytics workflows.

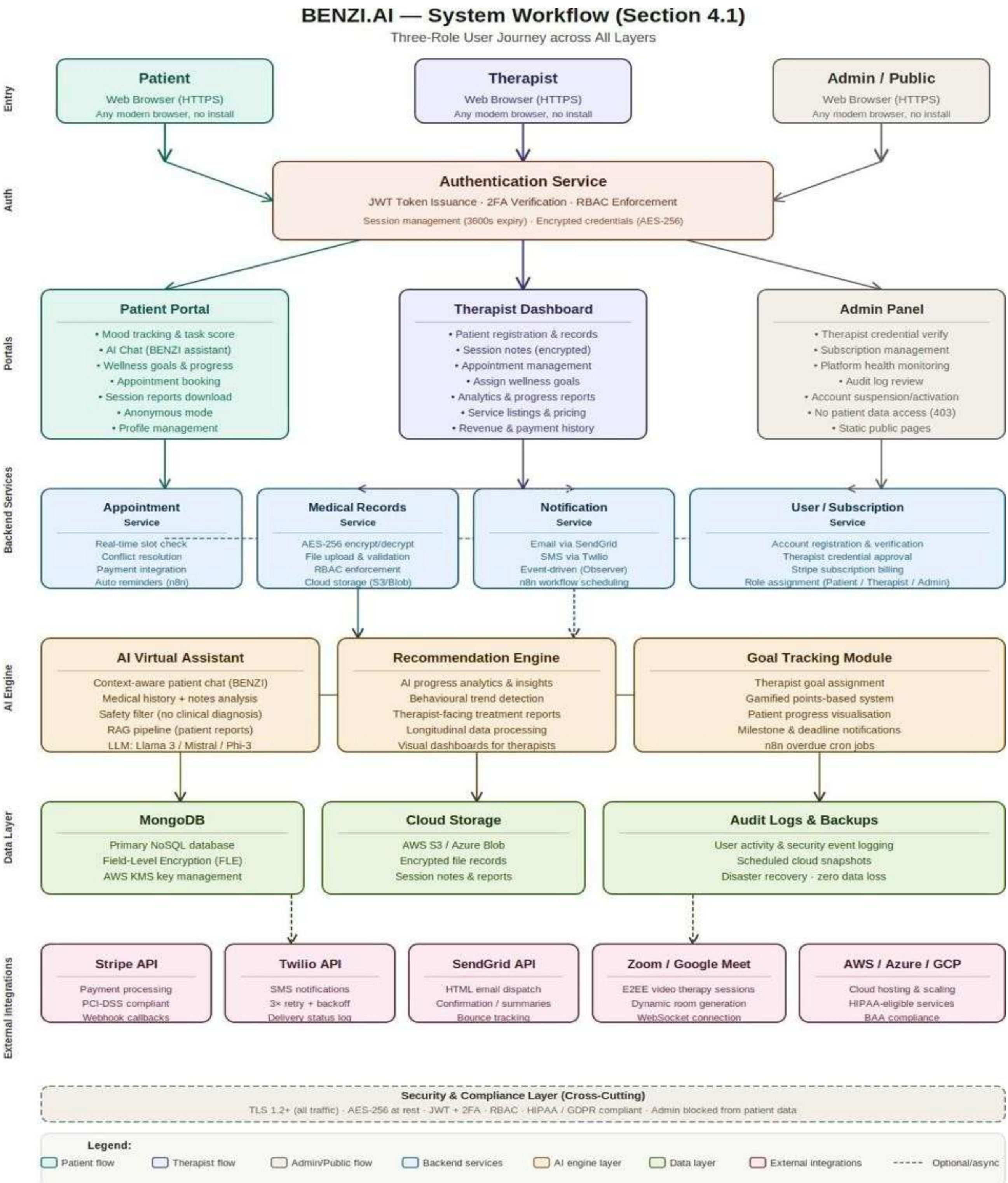
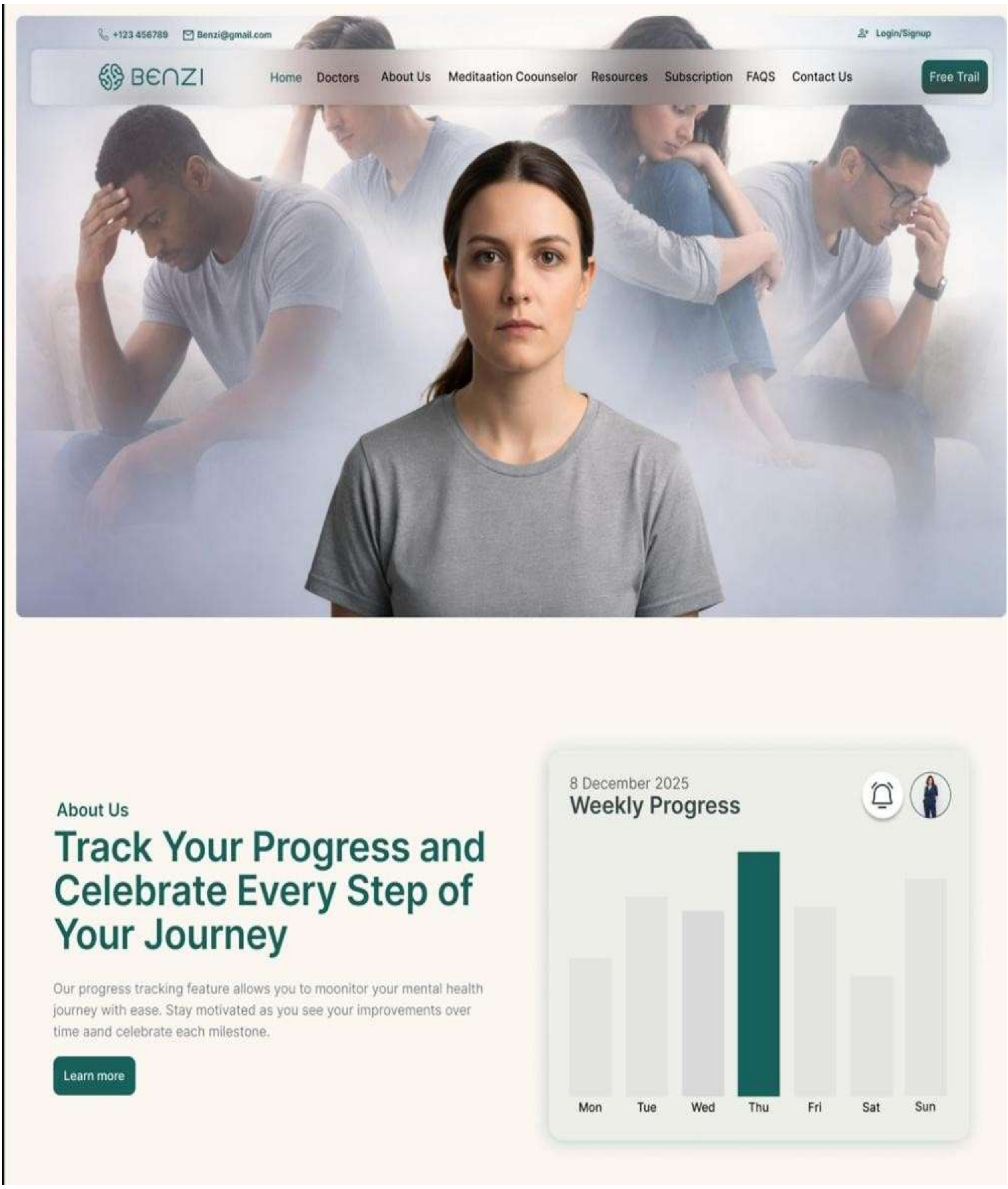
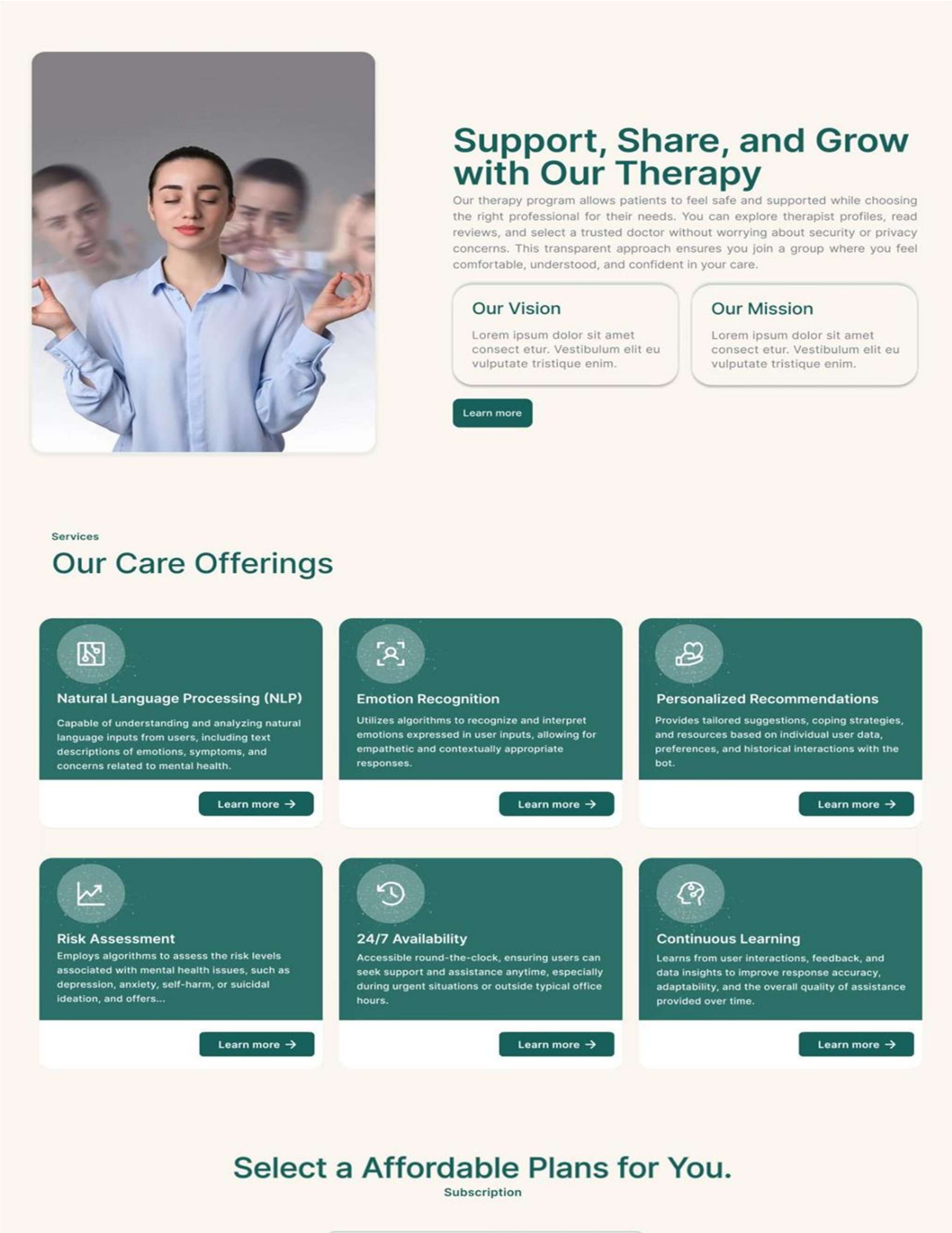


Figure 4.1: BENZI.AI System Workflow — Three-Role User Journey across All Layers | Group F25CS186

1.33.2 Screens





Support, Share, and Grow with Our Therapy
Our therapy program allows patients to feel safe and supported while choosing the right professional for their needs. You can explore therapist profiles, read reviews, and select a trusted doctor without worrying about security or privacy concerns. This transparent approach ensures you join a group where you feel comfortable, understood, and confident in your care.
Our Vision
Lorem ipsum dolor sit amet consect etur. Vestibulum elit eu vulputate tristique enim.
Our Mission
Lorem ipsum dolor sit amet consect etur. Vestibulum elit eu vulputate tristique enim.
Learn more
Services
Our Care Offerings
Natural Language Processing (NLP)
Capable of understanding and analyzing natural language inputs from users, including text descriptions of emotions, symptoms, and concerns related to mental health.
Learn more →
Emotion Recognition
Utilizes algorithms to recognize and interpret emotions expressed in user inputs, allowing for empathetic and contextually appropriate responses.
Learn more →
Personalized Recommendations
Provides tailored suggestions, coping strategies, and resources based on individual user data, preferences, and historical interactions with the bot.
Learn more →
Risk Assessment
Employs algorithms to assess the risk levels associated with mental health issues, such as depression, anxiety, self-harm, or suicidal ideation, and offers...
Learn more →
24/7 Availability
Accessible round-the-clock, ensuring users can seek support and assistance anytime, especially during urgent situations or outside typical office hours.
Learn more →
Continuous Learning
Learns from user interactions, feedback, and data insights to improve response accuracy, adaptability, and the overall quality of assistance provided over time.
Learn more →
Select a Affordable Plans for You.
Subscription

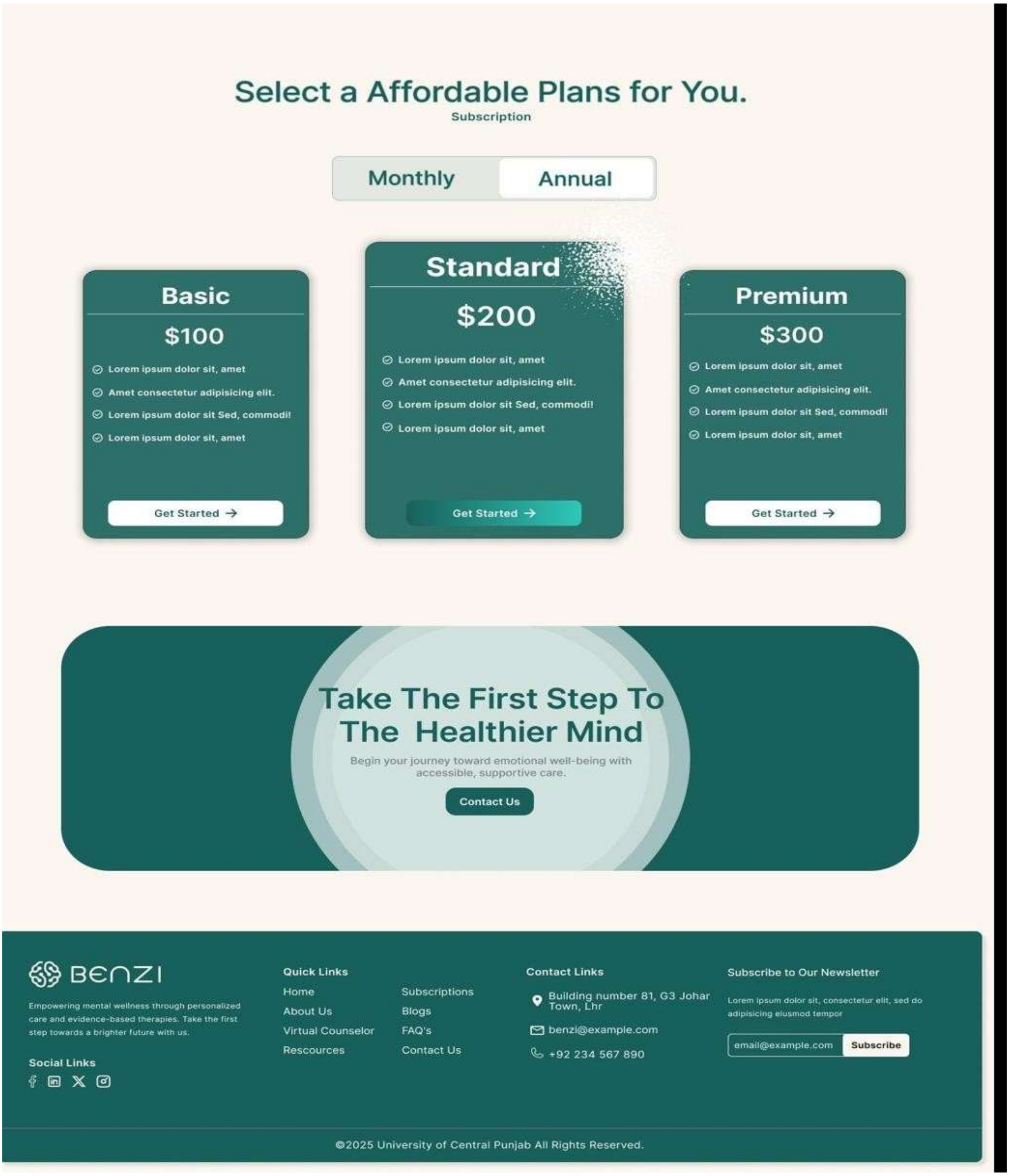


Figure 4.1: Home Page Platform landing page with care offerings and subscription plans

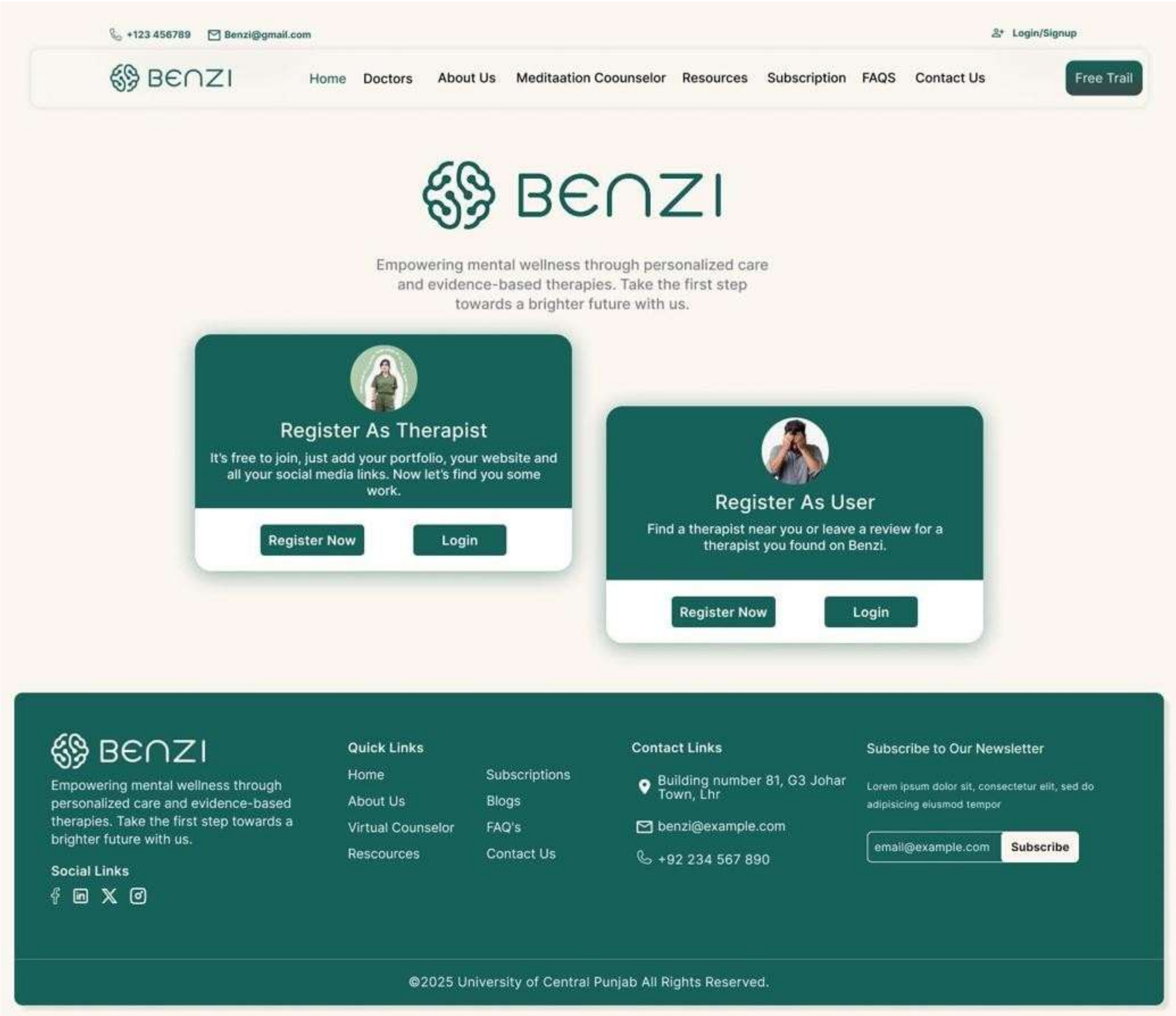


Figure 4.2: Auth Screen Role selection for Therapist or Patient registration/login

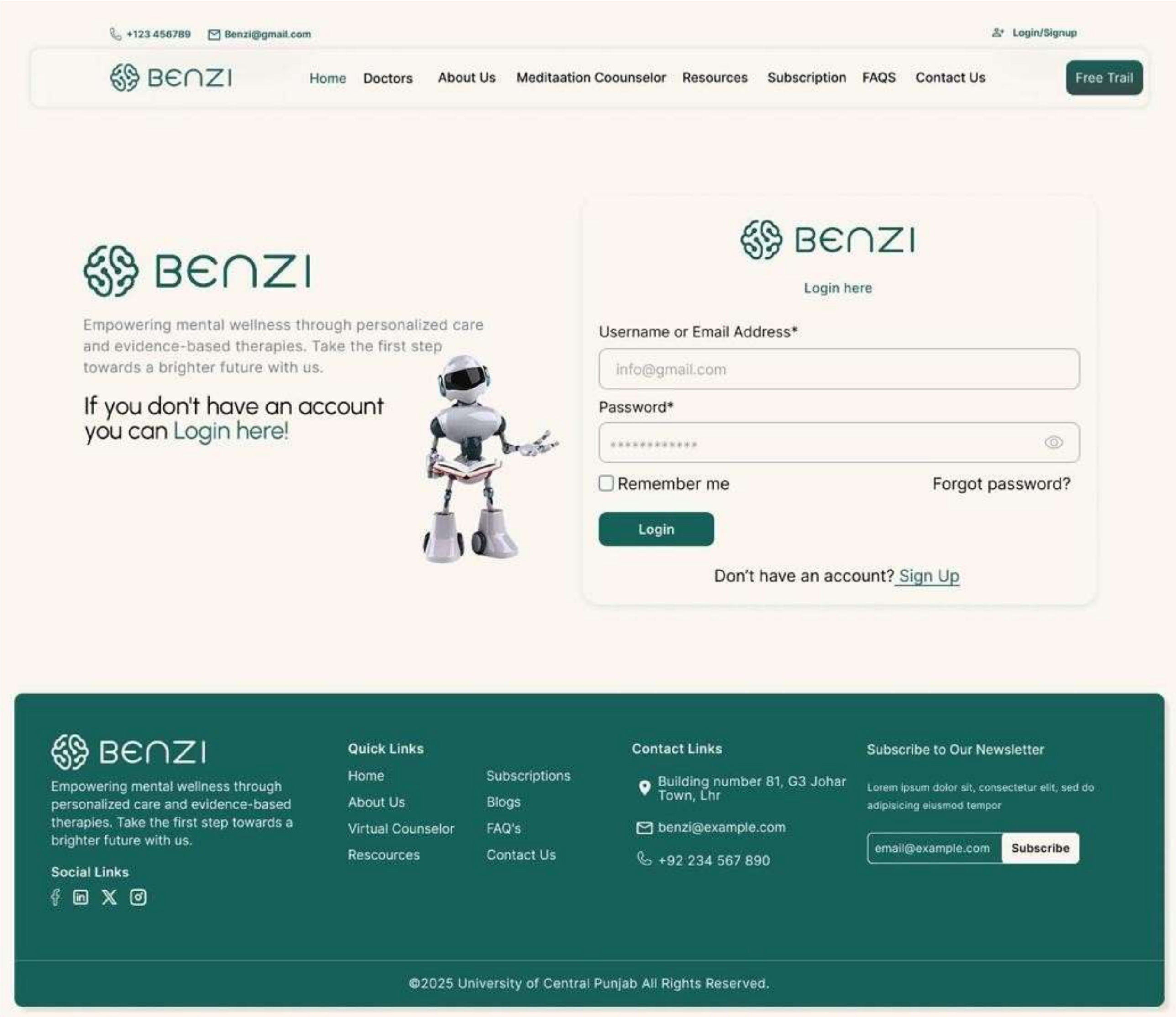


Figure 4.3: Login Screen Credential entry with Remember Me and Forgot Password

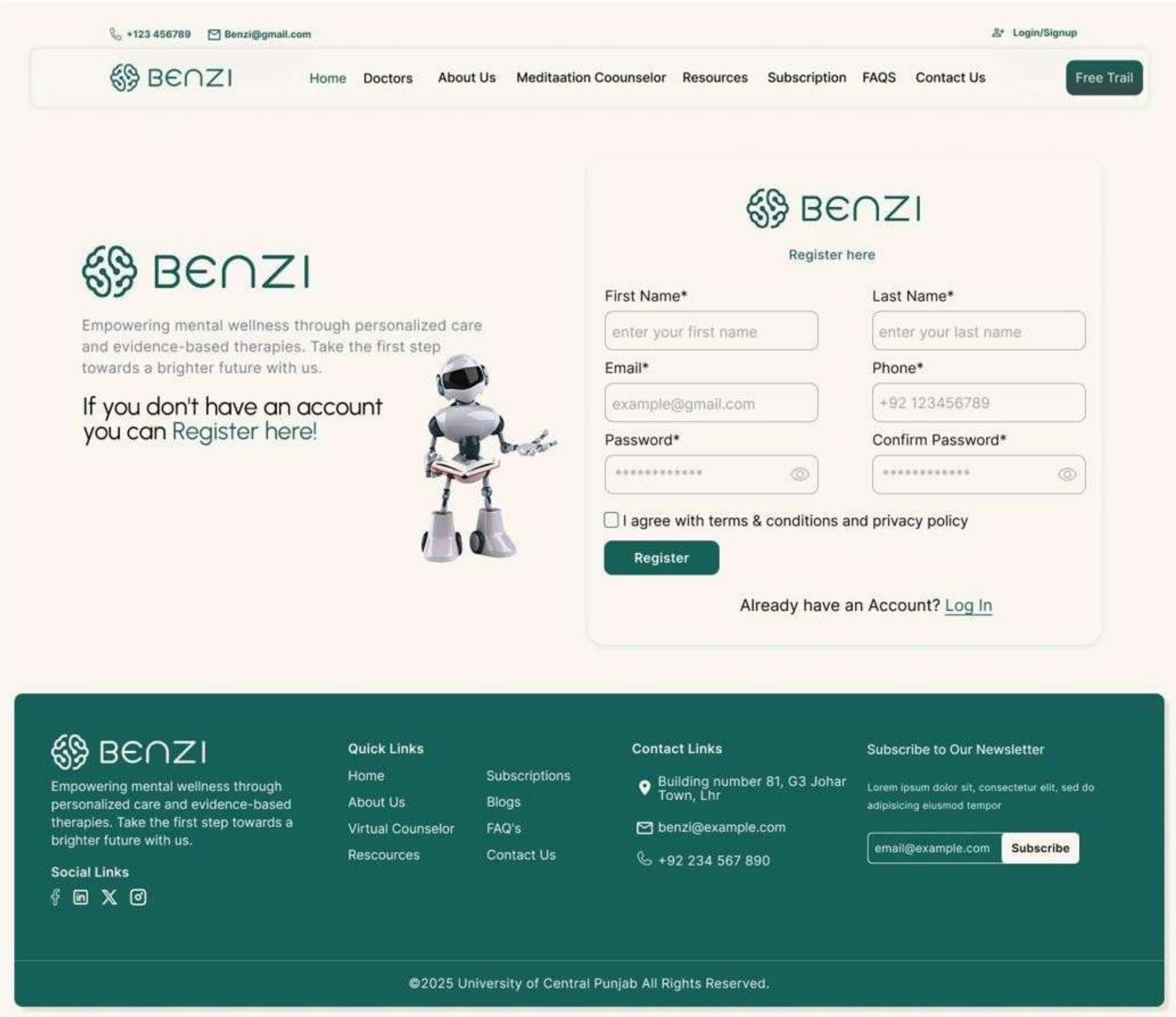


Figure 4.4: Register Screen New user account creation

Figure 4.5: Patient Dashboard, Mood tracker, task score, progress rings and weekly chart

Figure 4.6: Patient Conversations BENZI AI chatbot interface with history panel

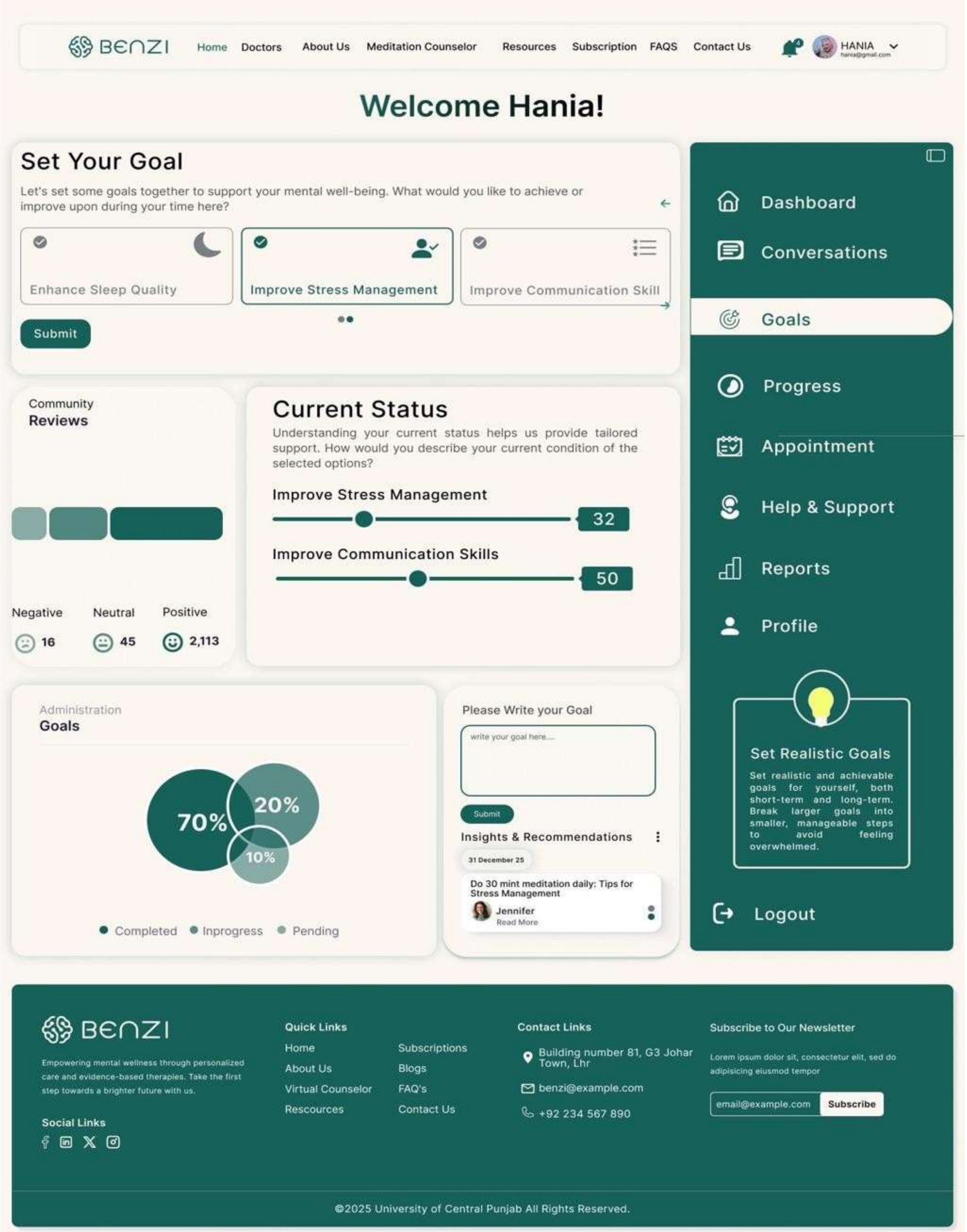


Figure 4.7: Patient Goals Goal selection, status sliders, completion pie chart and AI insights

Figure 4.8: Patient Progress   Individual goal bars, Benzi usage, chatbot activity charts

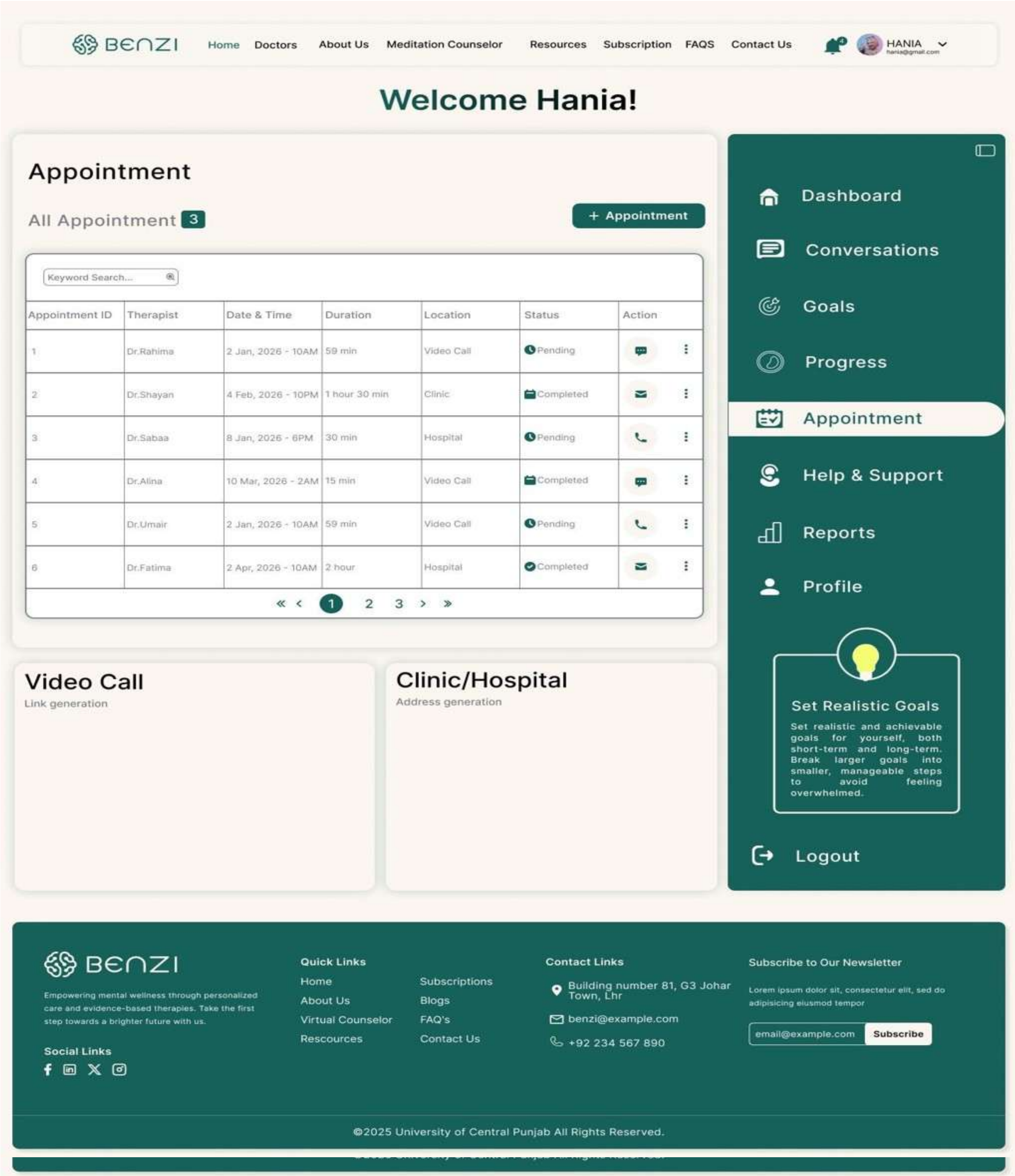


Figure 4.9: Patient Appointments Appointment table with video call and clinic link generation

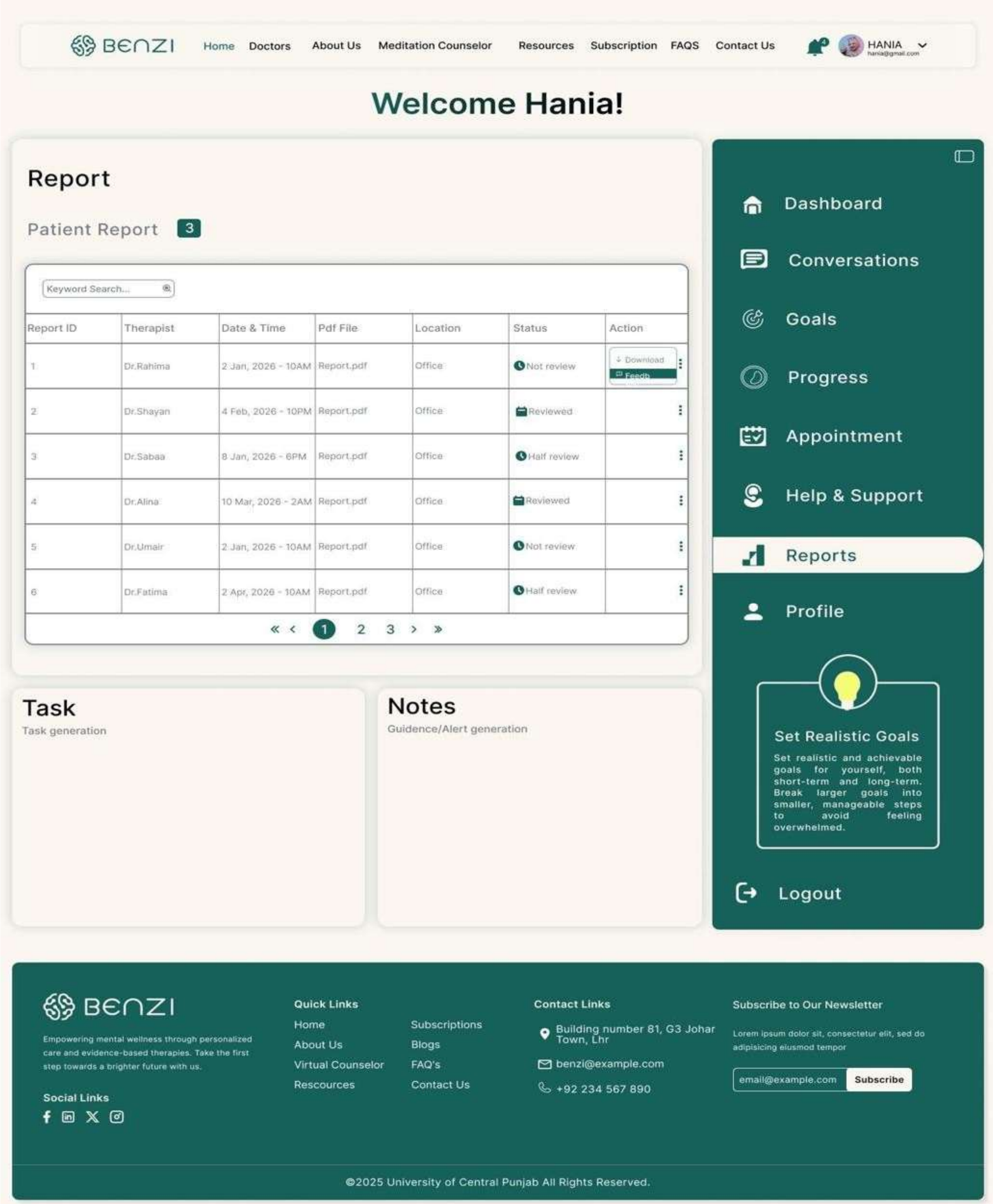


Figure 4.10: Patient Reports   Report table with download, feedback, task and notes sections

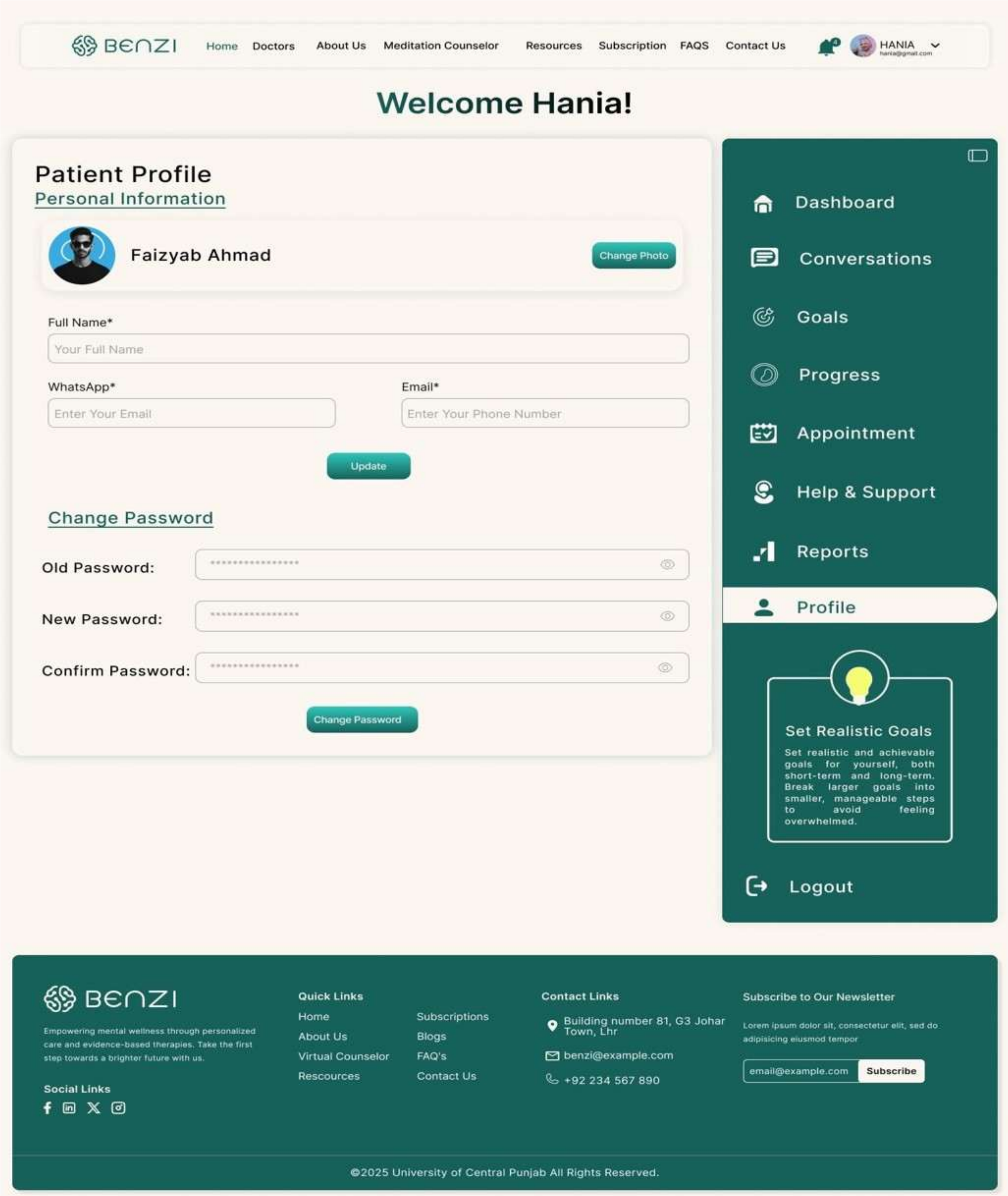


Figure 4.11: Patient Profile   Personal info and password management

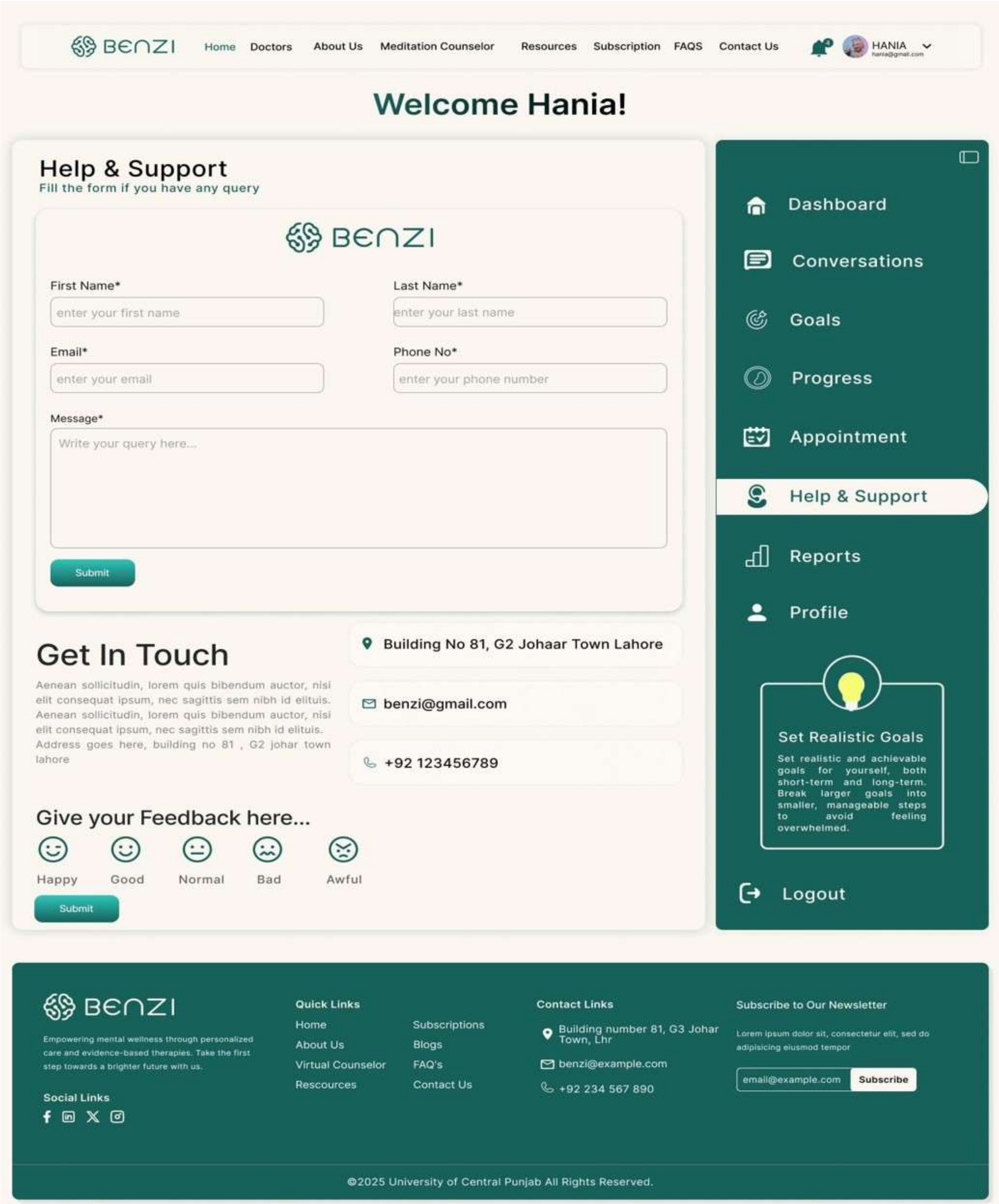


Figure 4.12: Patient Help & Support Support access screen

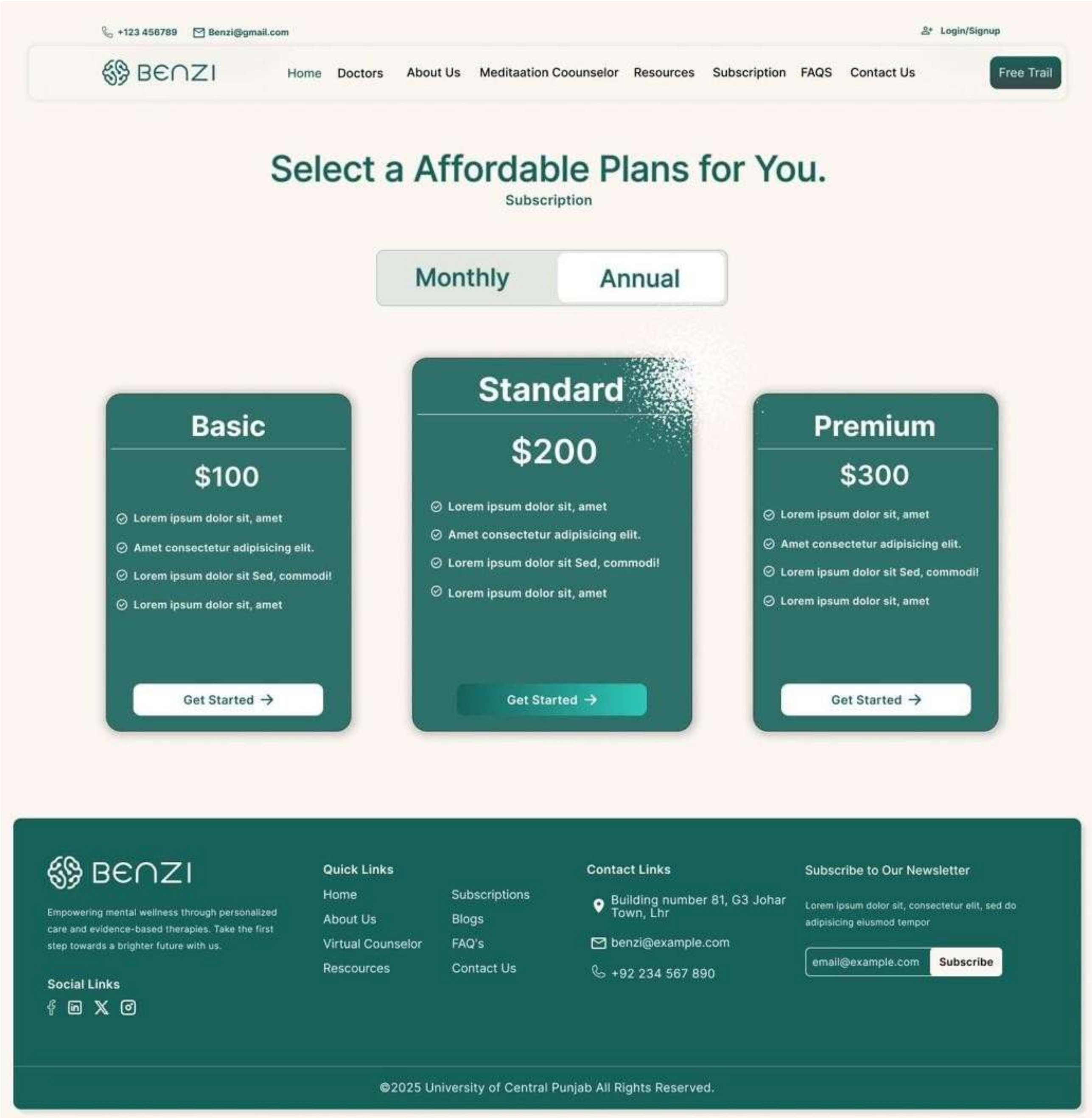


Figure 4.13: Subscription Plans Basic/Standard/Premium with Monthly/Annual toggle

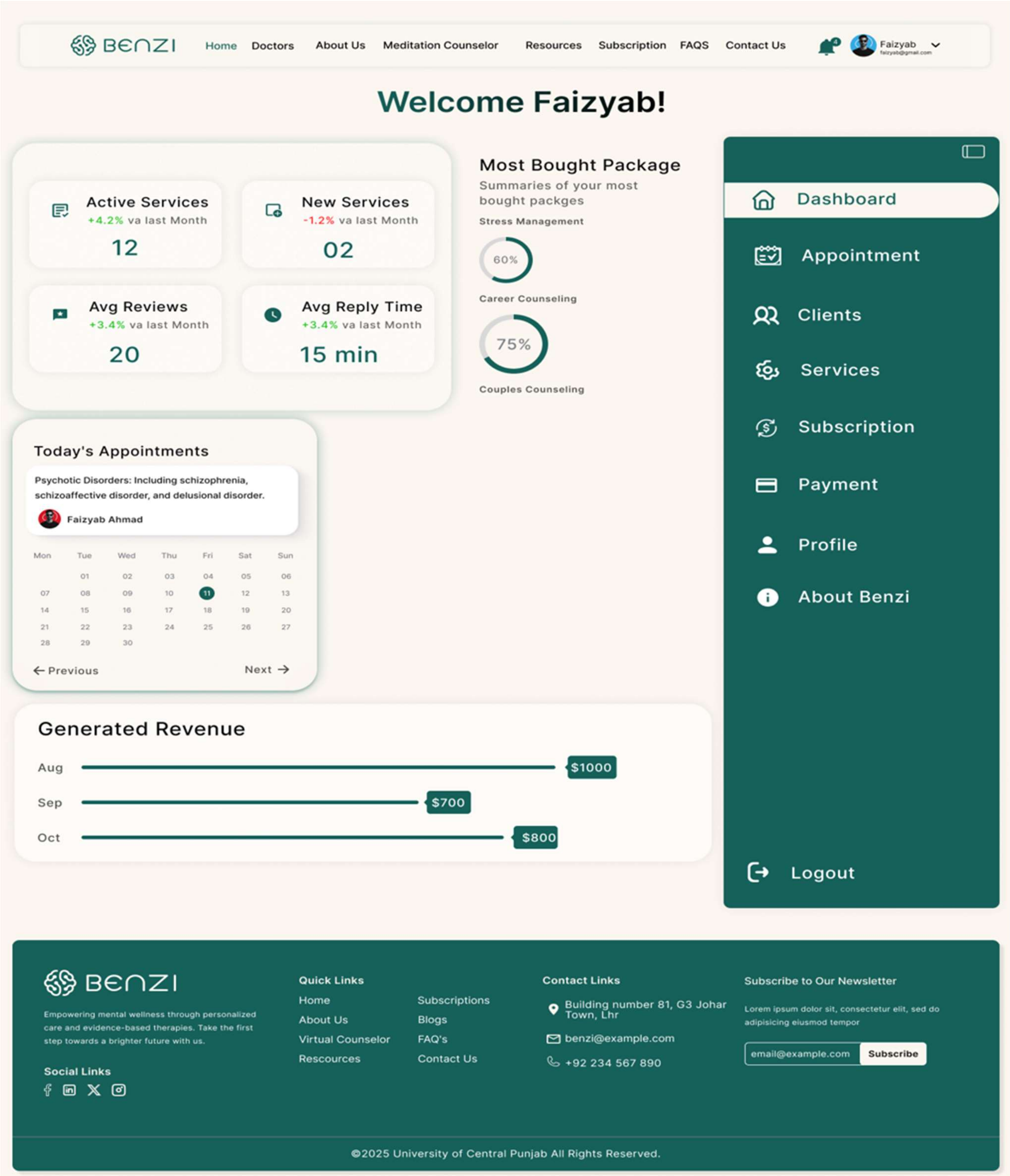


Figure 4.14: Therapist Dashboard KPI cards, today's appointments, revenue chart

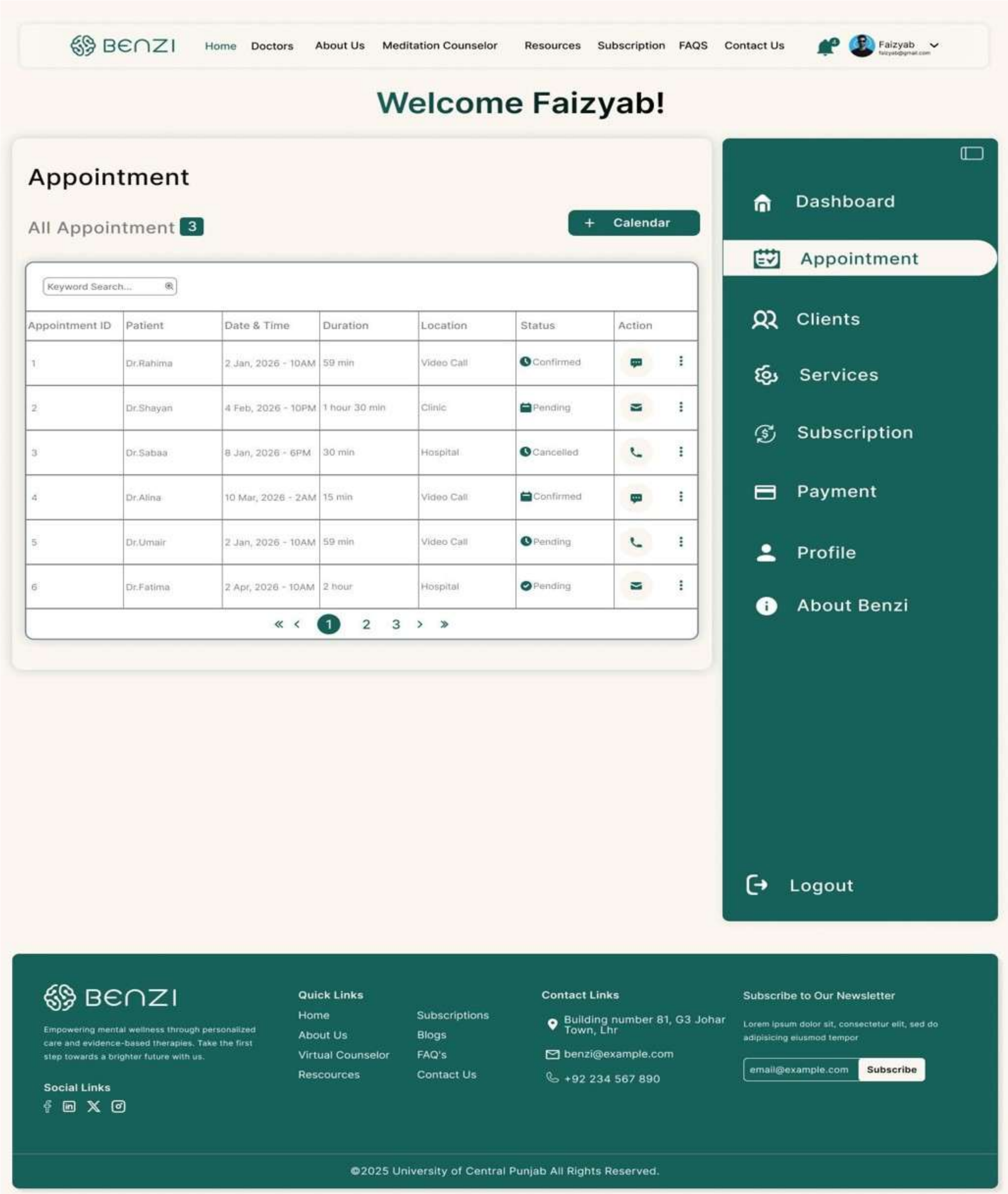


Figure 4.15: Therapist Appointments Full appointment list with calendar integration

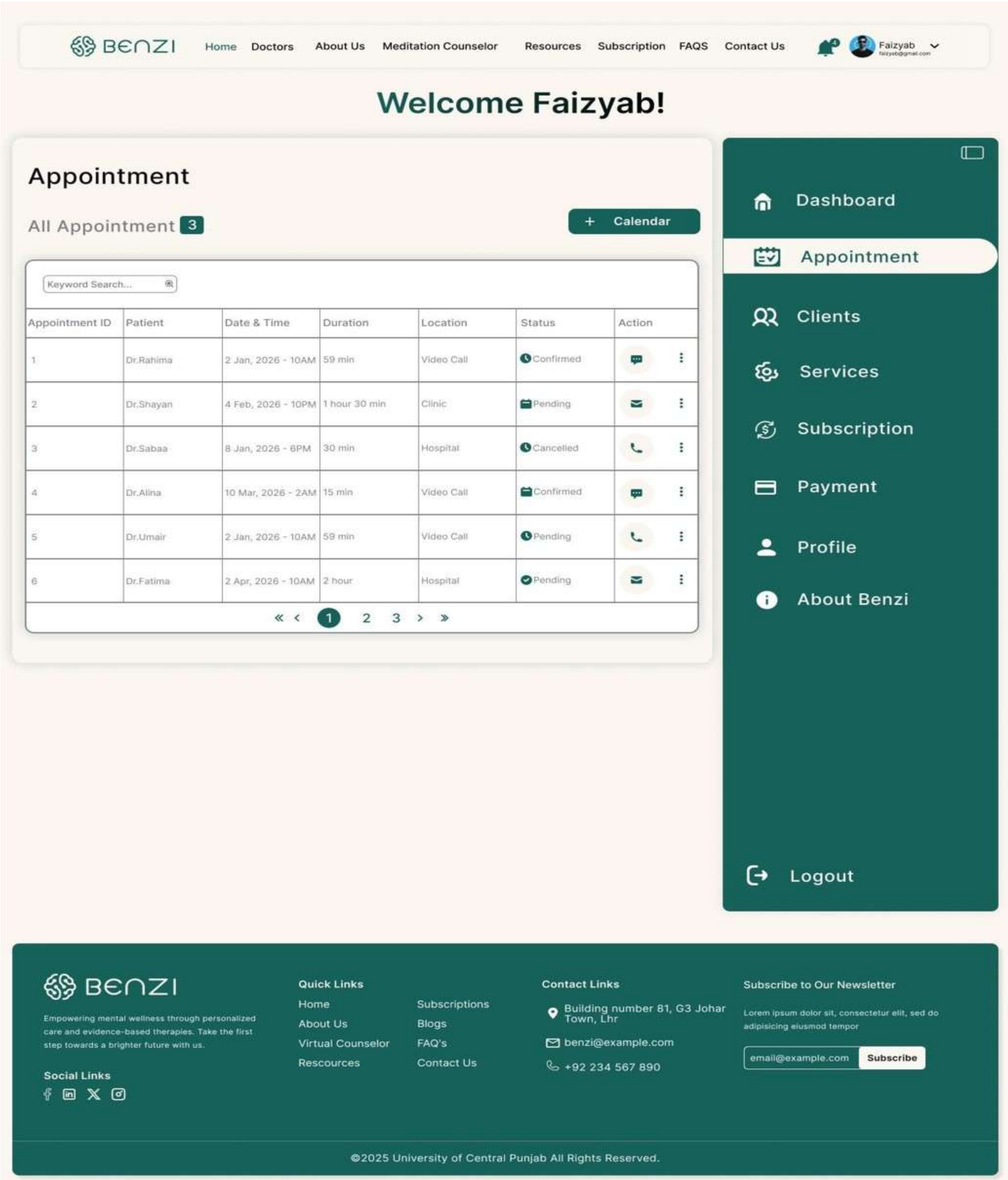


Figure 4.16: Therapist Clients Client management table with status tracking

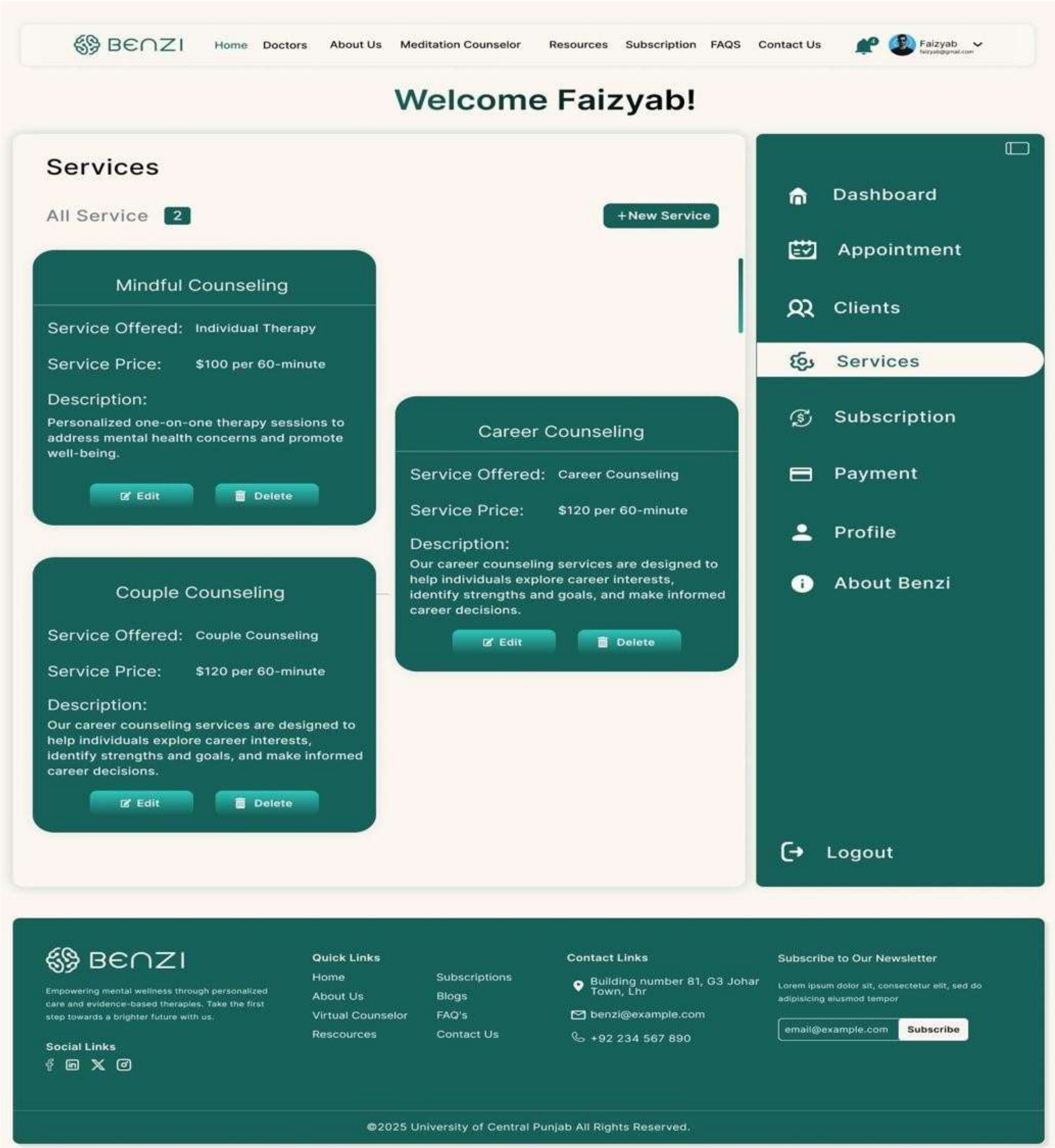


Figure 4.17: Therapist Services Service listing with pricing, edit and delete controls

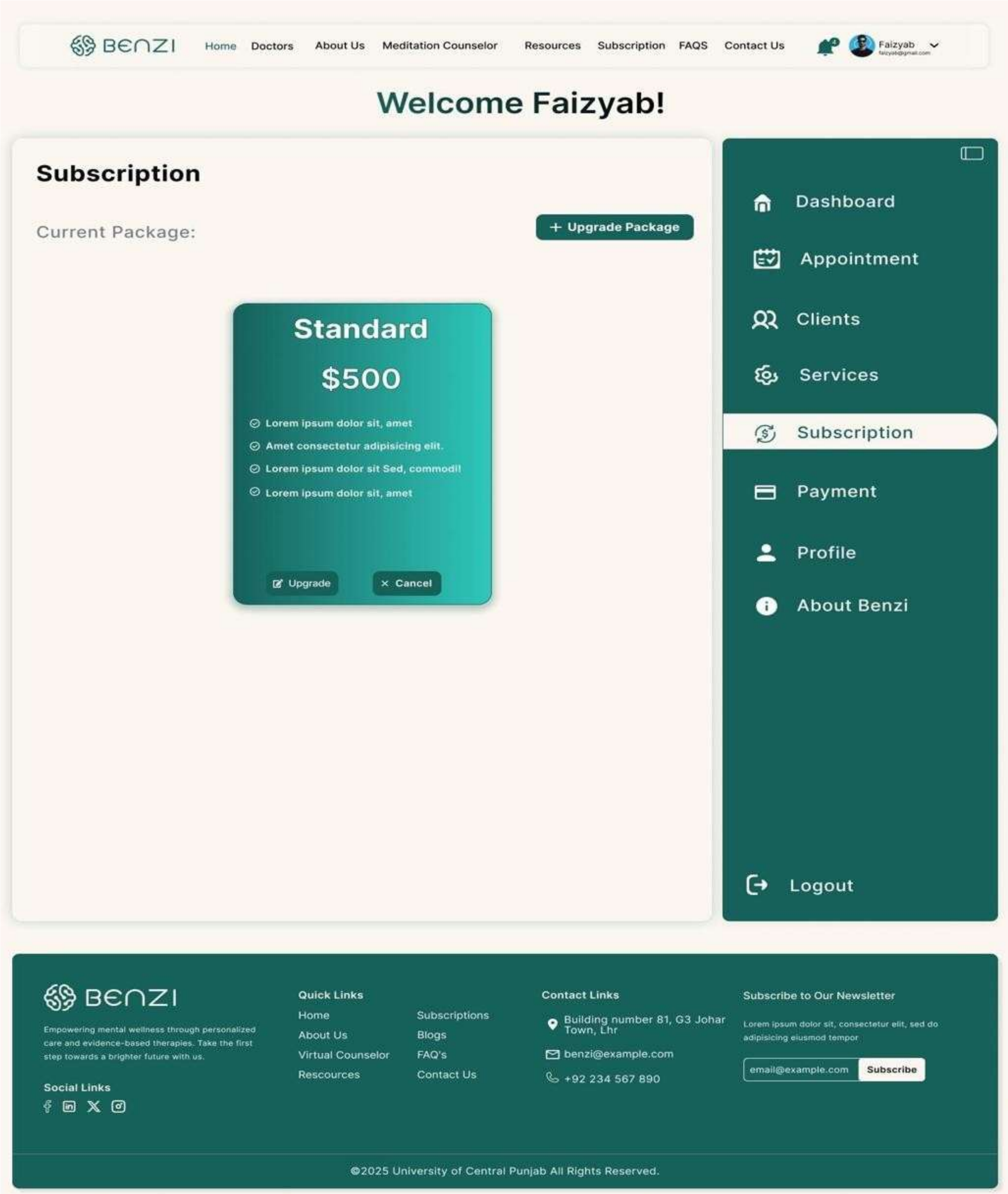


Figure 4.18: Therapist Subscription Plan management for therapist accounts

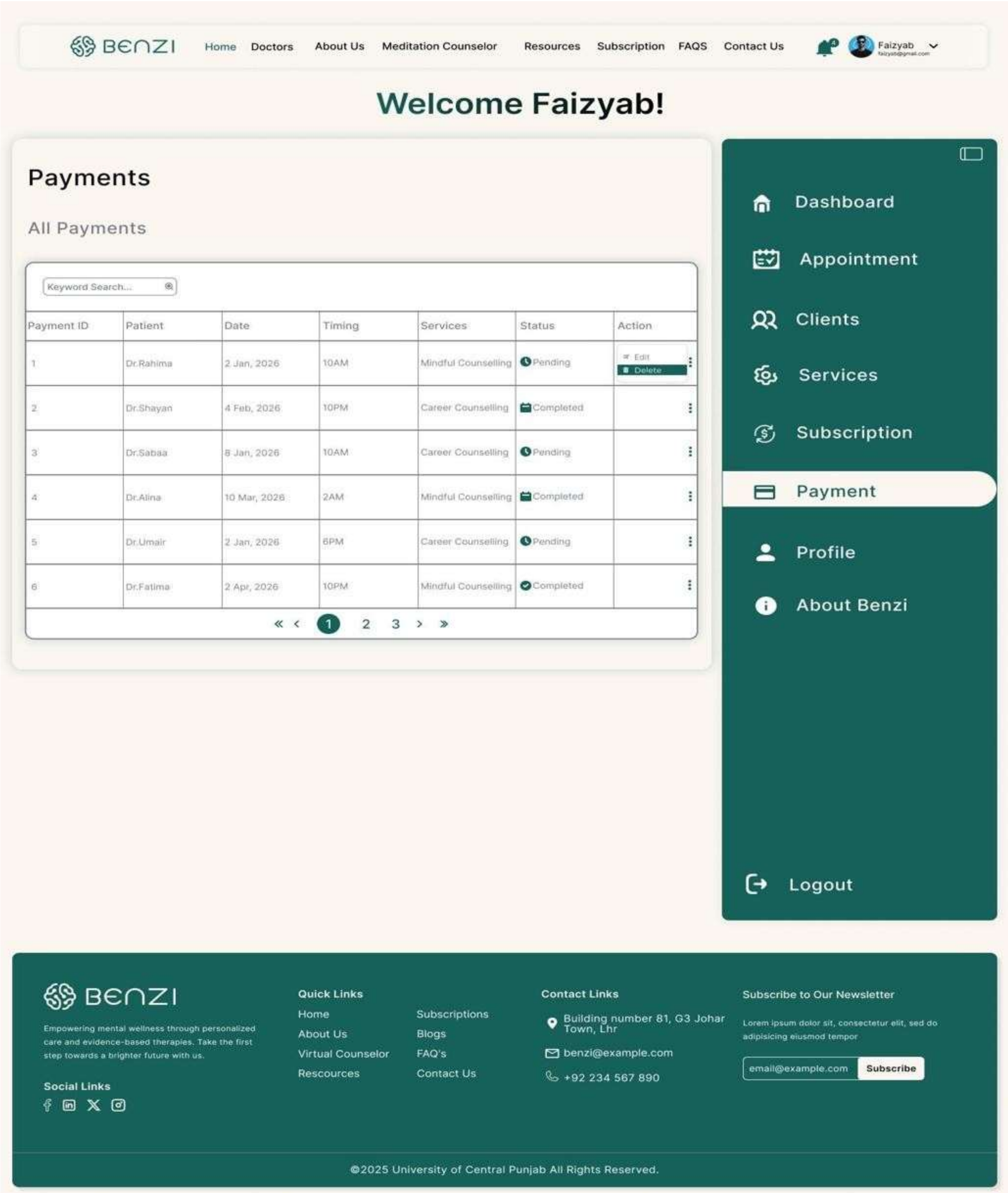


Figure 4.19: Therapist Payments Revenue and payment history

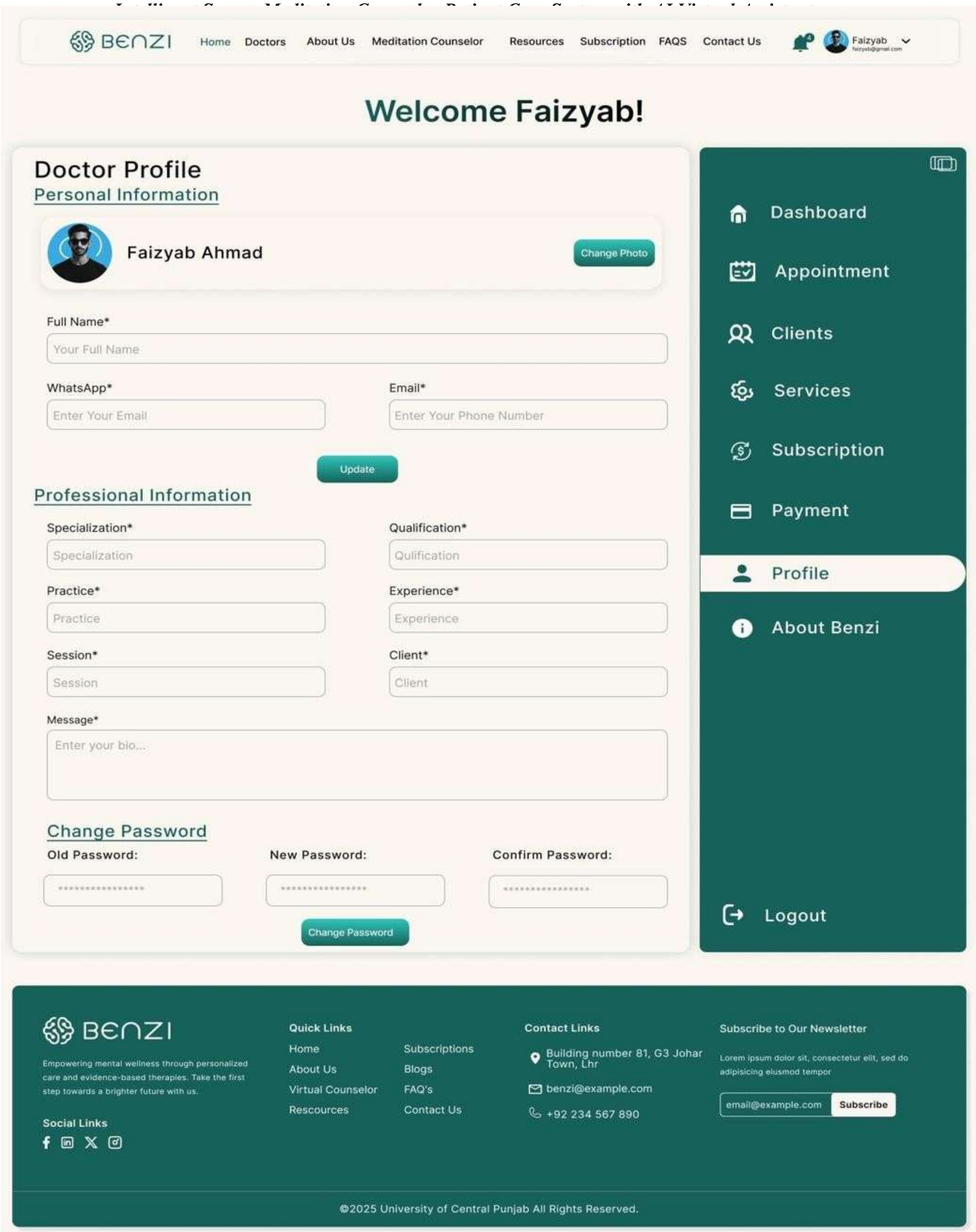


Figure 4.20: Therapist Profile   Personal and professional info with credential fields

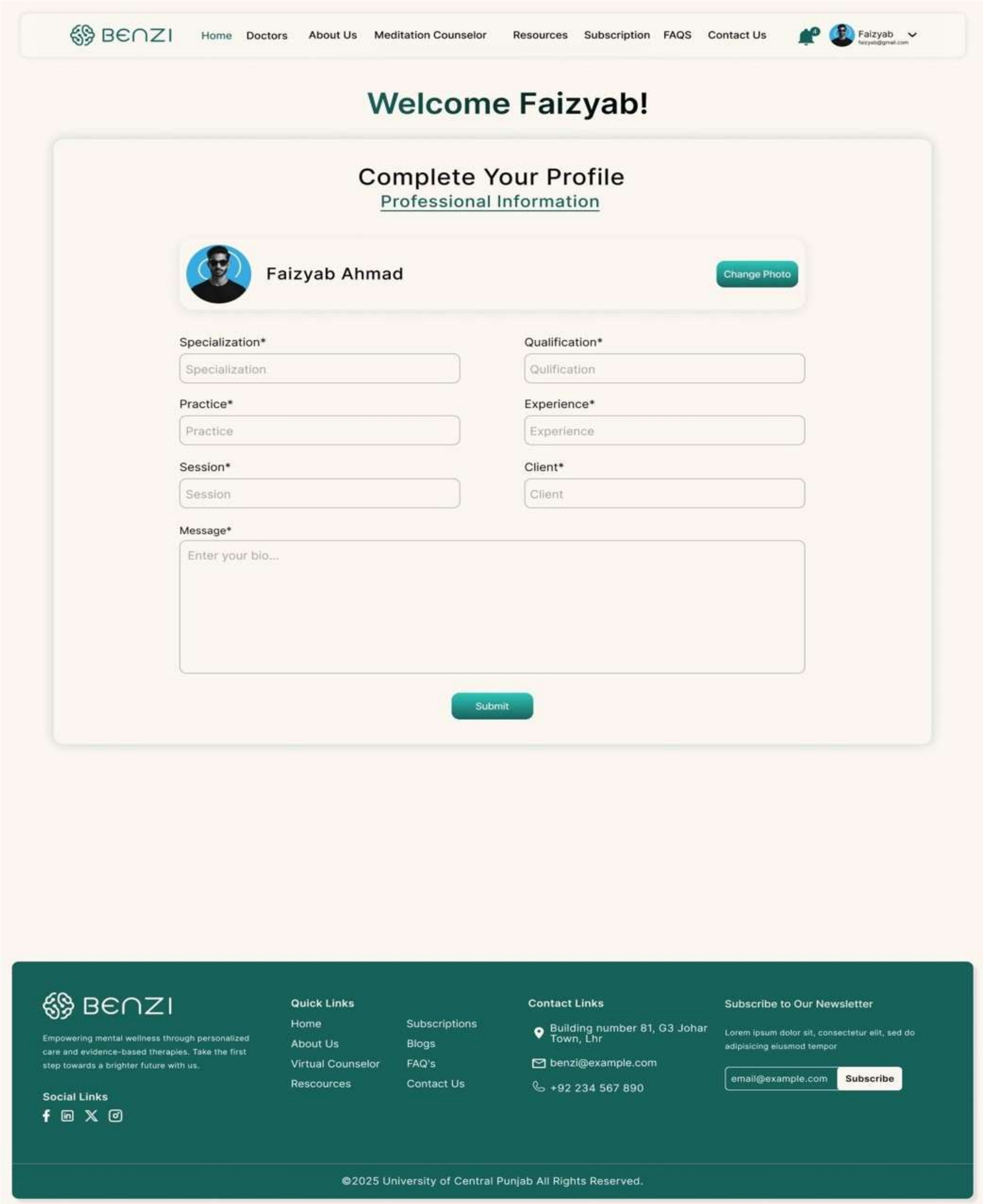
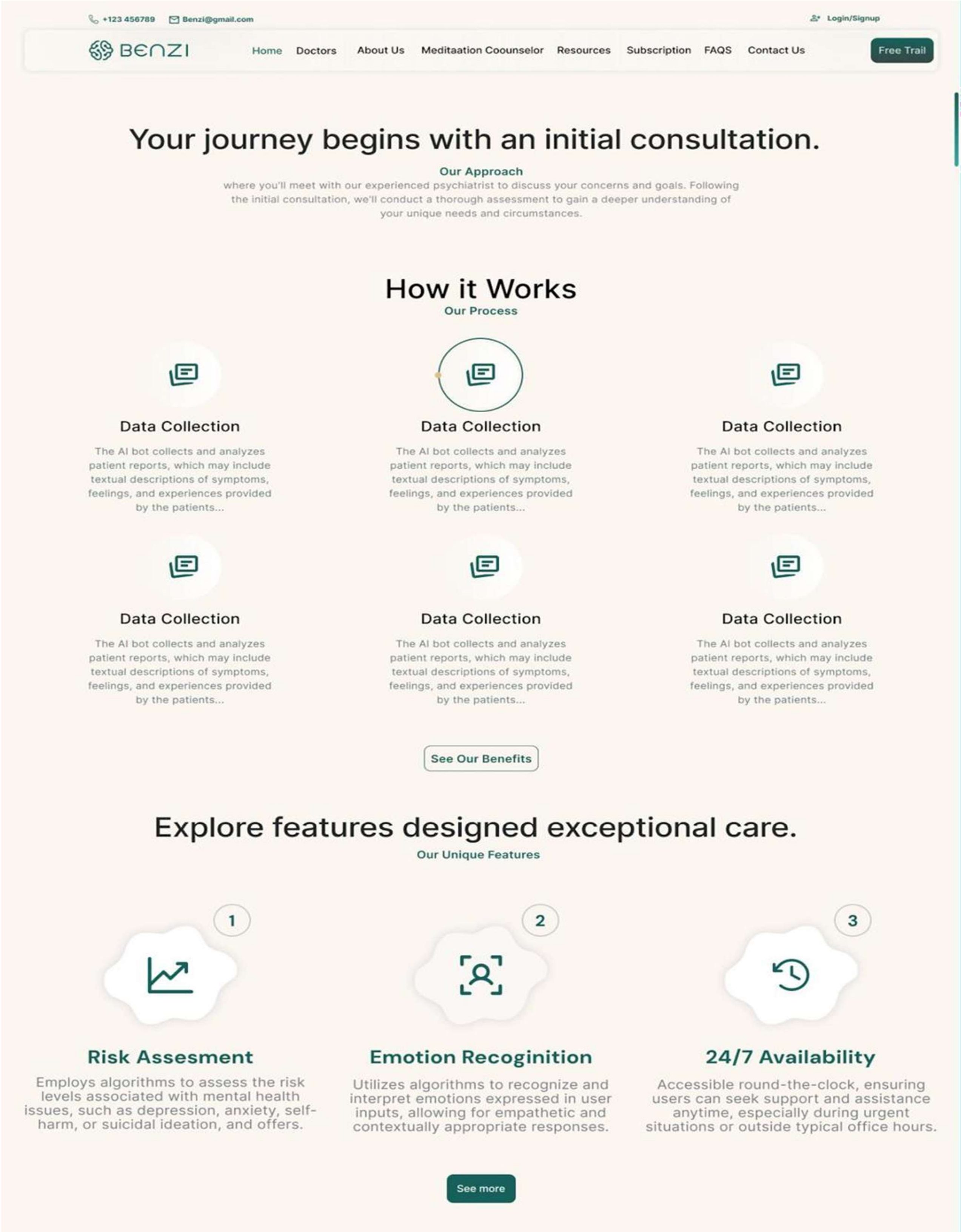


Figure 4.21: Therapist Profile Setup   Initial onboarding form


+123 456789
Benzi@gmail.com
Login/Signup
BENZI
Home
Doctors
About Us
Meditaation Coounselor
Resources
Subscription
FAQS
Contact Us
Free Trail
Your journey begins with an initial consultation.
Our Approach
where you'll meet with our experienced psychiatrist to discuss your concerns and goals. Following the initial consultation, we'll conduct a thorough assessment to gain a deeper understanding of your unique needs and circumstances.
How it Works
Our Process
Data Collection
The AI bot collects and analyzes patient reports, which may include textual descriptions of symptoms, feelings, and experiences provided by the patients...
Data Collection
The AI bot collects and analyzes patient reports, which may include textual descriptions of symptoms, feelings, and experiences provided by the patients...
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Data Collection
The AI bot collects and analyzes patient reports, which may include textual descriptions of symptoms, feelings, and experiences provided by the patients...
See Our Benefits
Explore features designed exceptional care.
Our Unique Features
1
2
3
Risk Assesment
Employs algorithms to assess the risk levels associated with mental health issues, such as depression, anxiety, self-harm, or suicidal ideation, and offers.
Emotion Recoginition
Utilizes algorithms to recognize and interpret emotions expressed in user inputs, allowing for empathetic and contextually appropriate responses.
24/7 Availability
Accessible round-the-clock, ensuring users can seek support and assistance anytime, especially during urgent situations or outside typical office hours.
See more

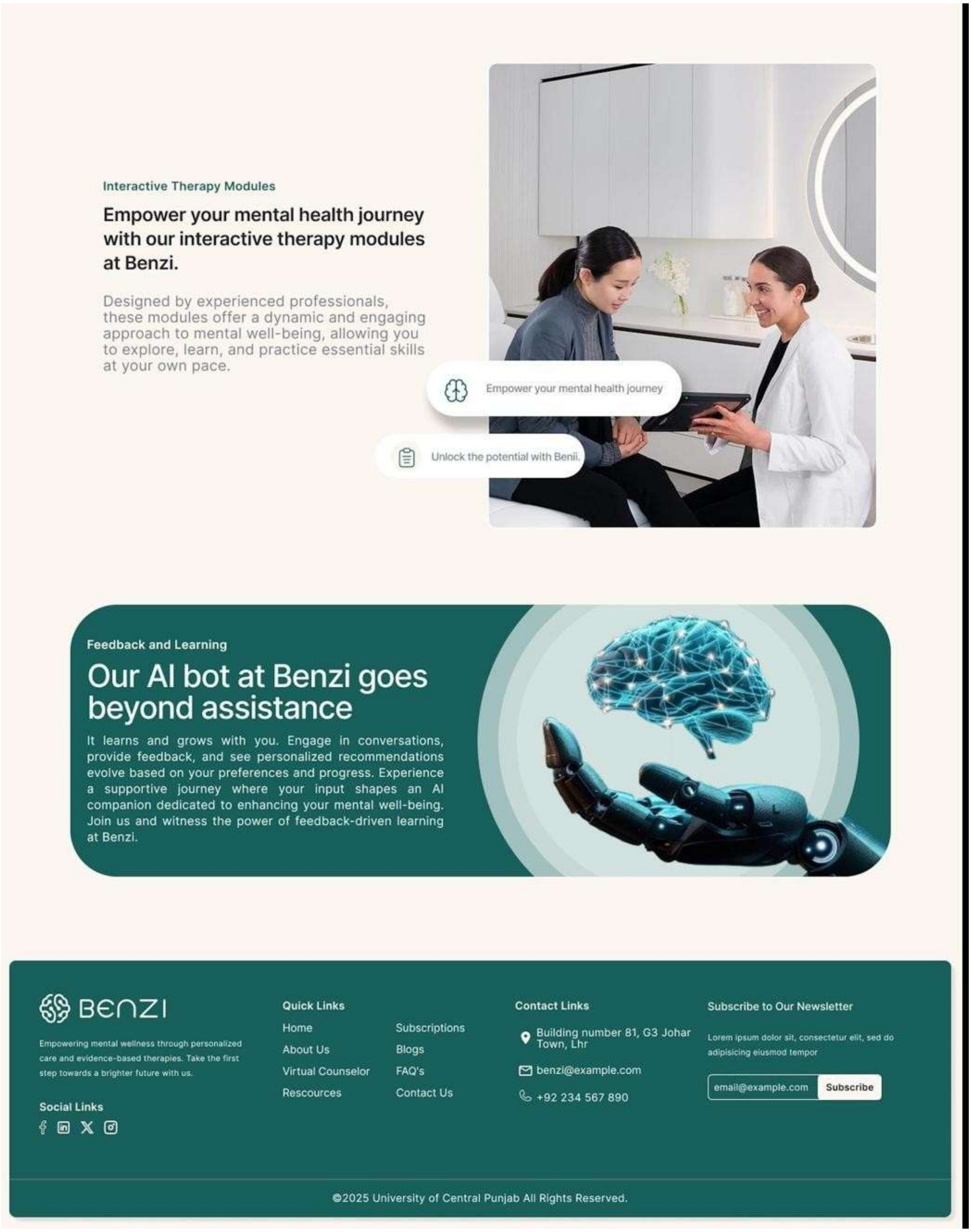


Figure 4.22: Meditation Counselor Page AI approach, How it Works, feature highlights

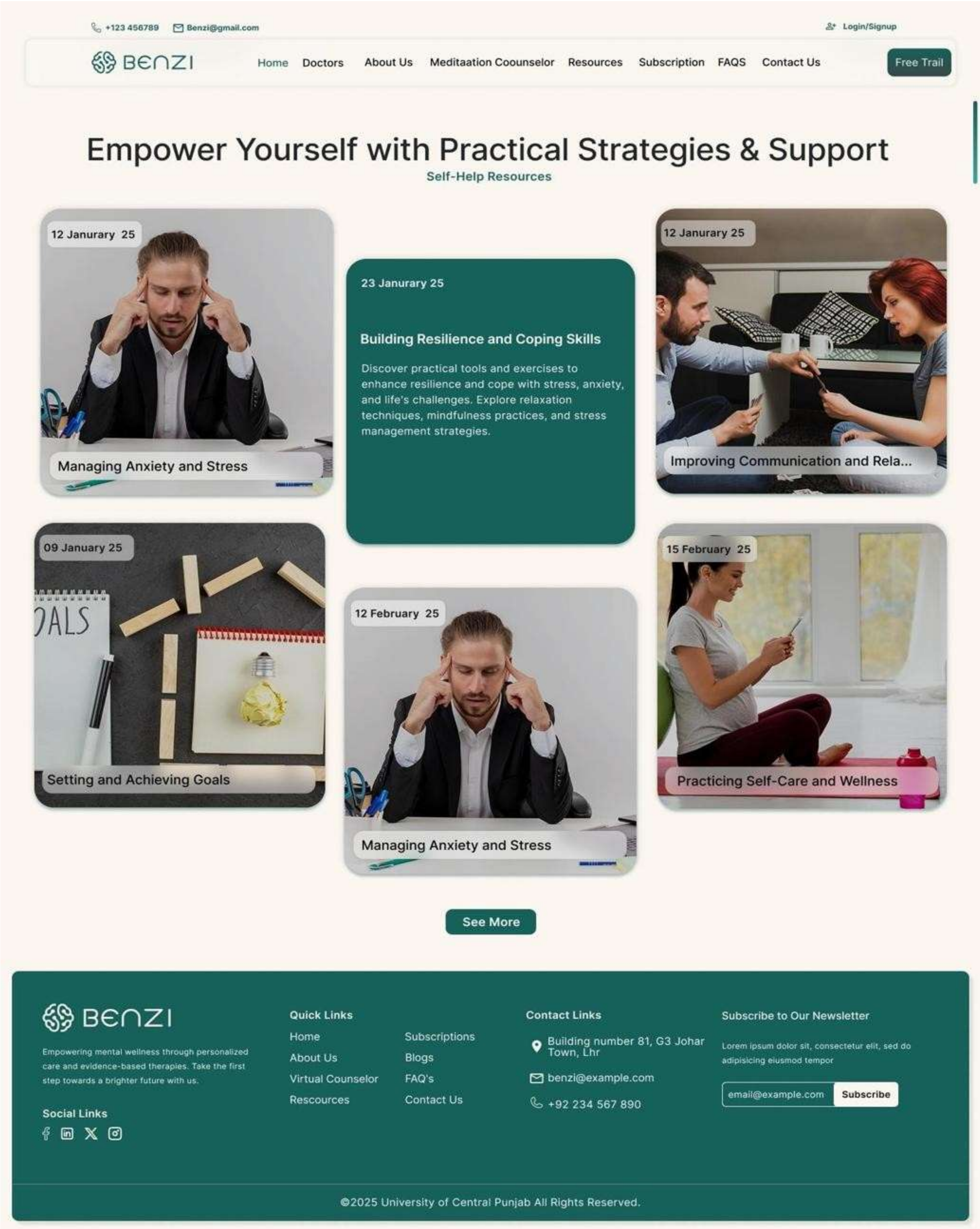
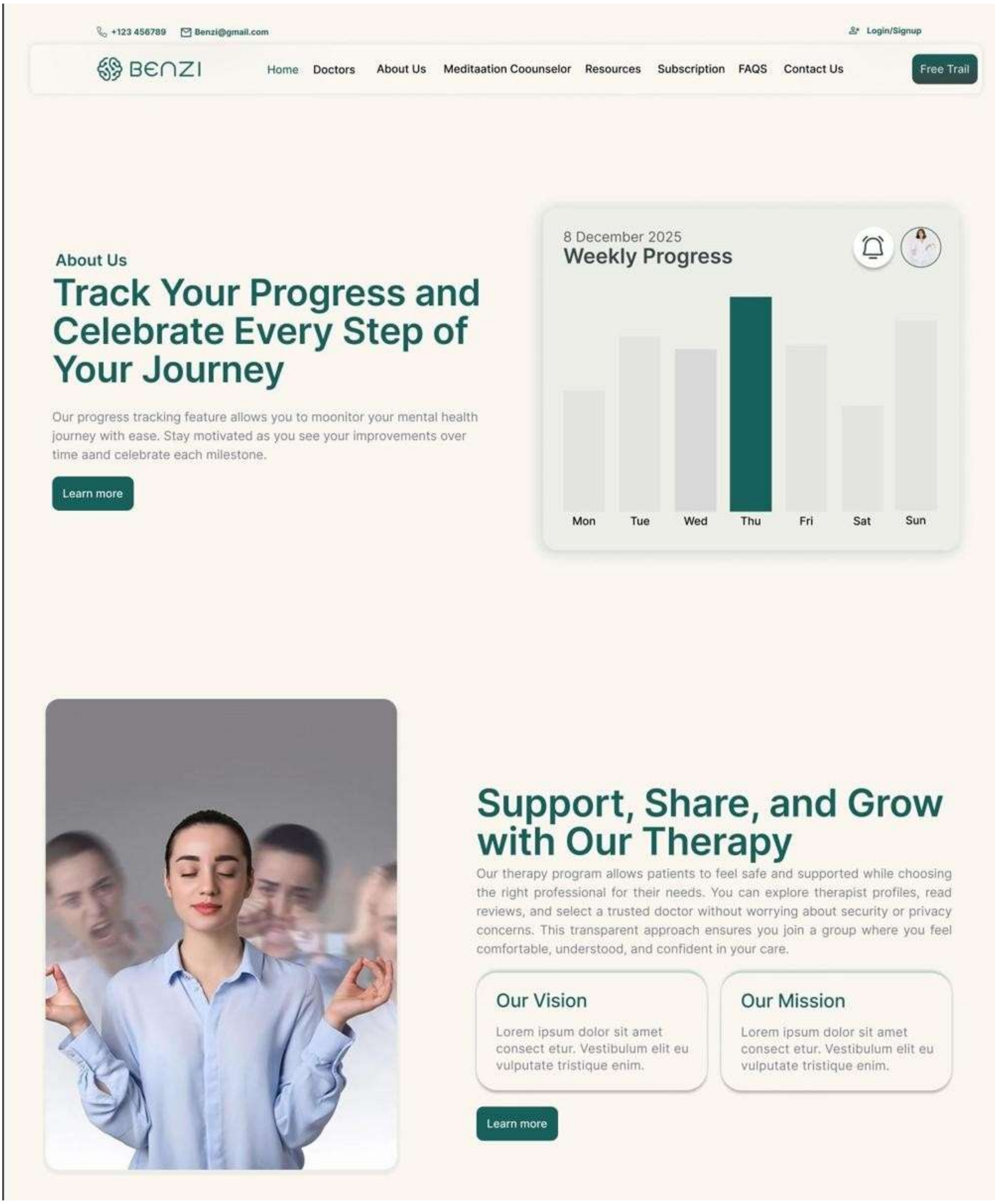


Figure 4.23: Resources Page   Mental health resource library


+123 456789
Benzi@gmail.com
Login/Signup
BENZI
Home
Doctors
About Us
Meditaation Coounselor
Resources
Subscription
FAQS
Contact Us
Free Trail
About Us
Track Your Progress and Celebrate Every Step of Your Journey
Our progress tracking feature allows you to moonitor your mental health journey with ease. Stay motivated as you see your improvements over time aand celebrate each milestone.
Learn more
8 December 2025
Weekly Progress
Mon
Tue
Wed
Thu
Fri
Sat
Sun
Support, Share, and Grow with Our Therapy
Our therapy program allows patients to feel safe and supported while choosing the right professional for their needs. You can explore therapist profiles, read reviews, and select a trusted doctor without worrying about security or privacy concerns. This transparent approach ensures you join a group where you feel comfortable, understood, and confident in your care.
Our Vision
Lorem ipsum dolor sit amet consect etur. Vestibulum elit eu vulputate tristique enim.
Our Mission
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Learn more

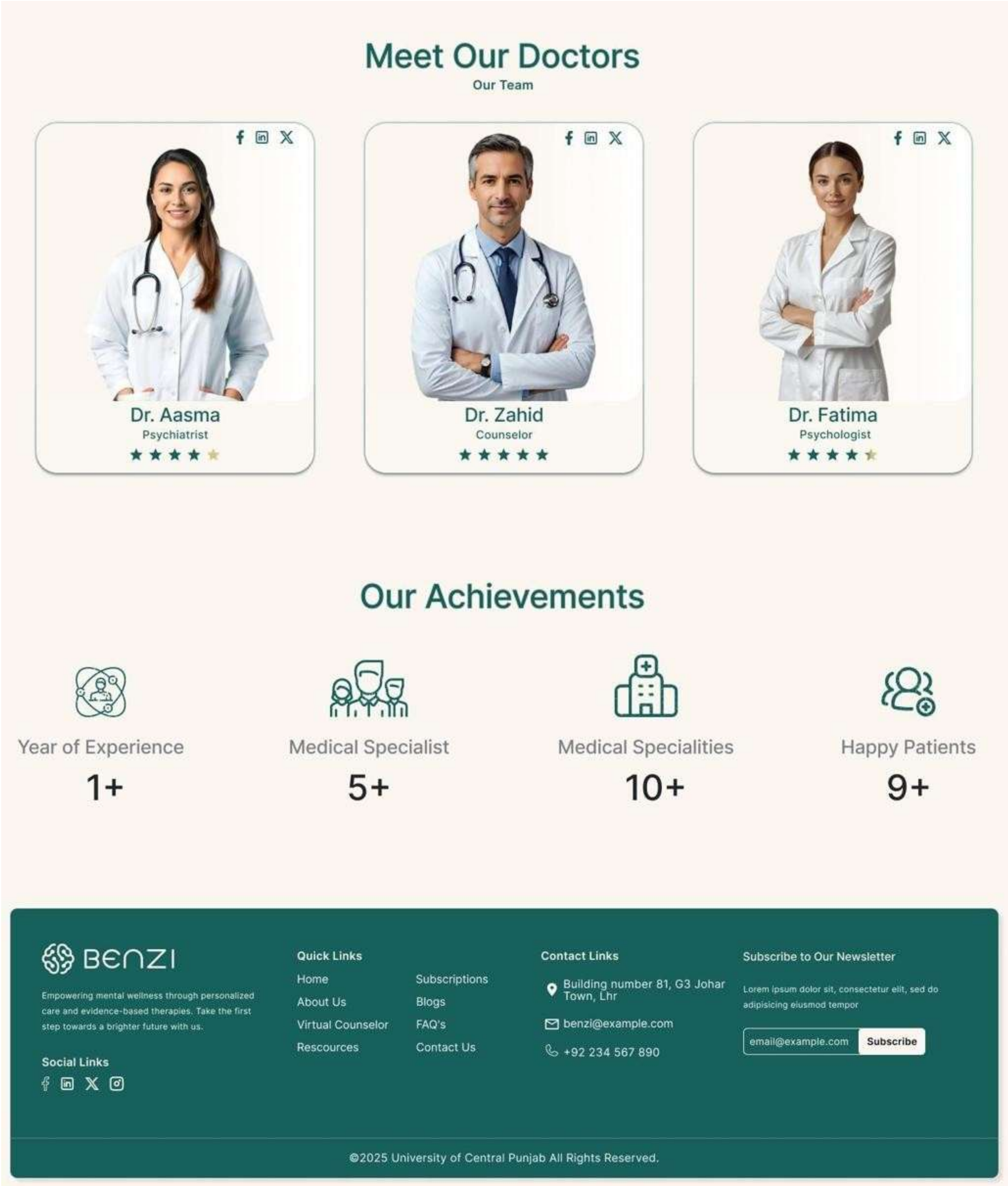

Figure 4.24: About Us   Team and mission information

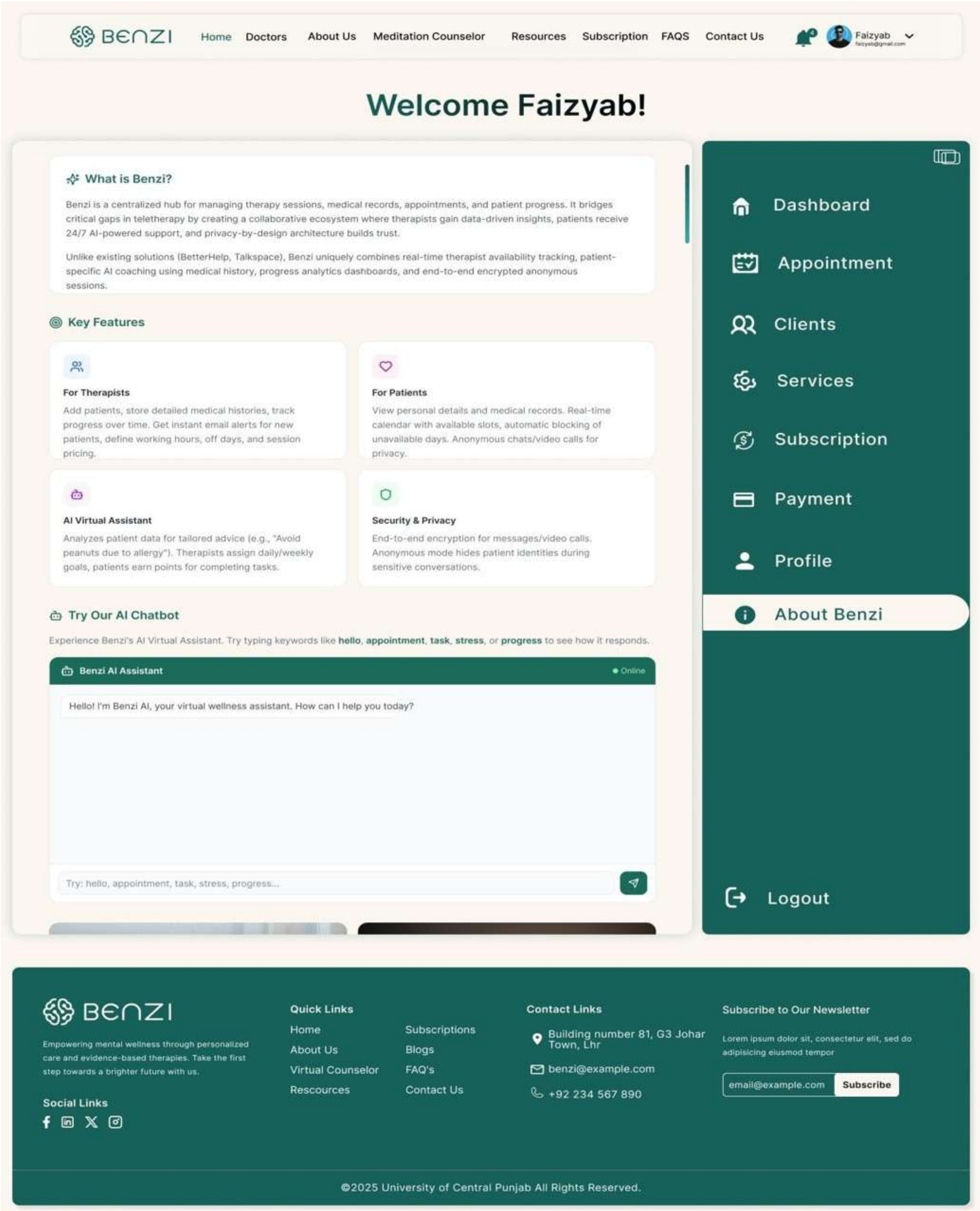

Figure 4.25: About BENZI.AI Product overview page

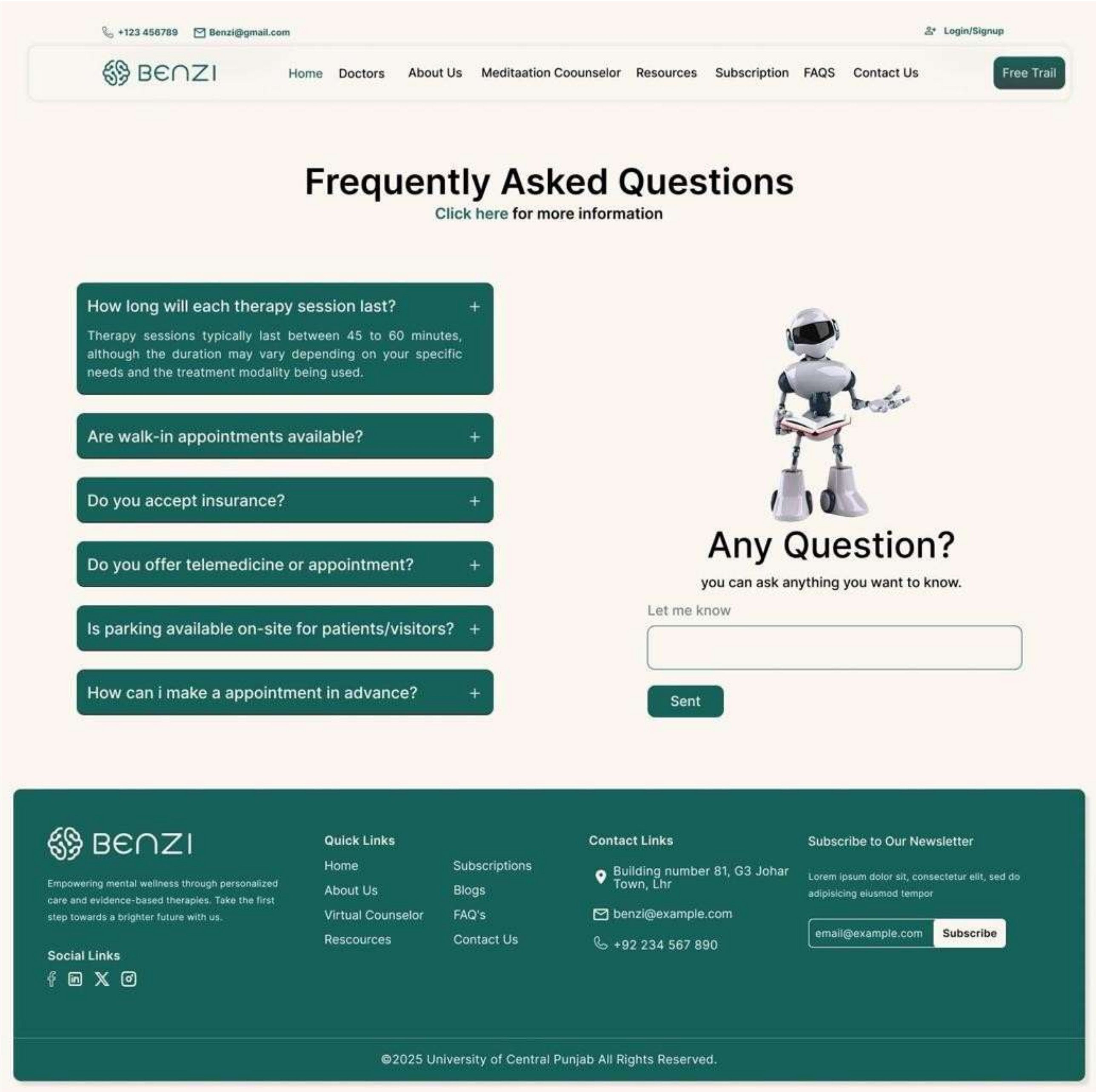

Figure 4.26: FAQs Page   Frequently asked questions

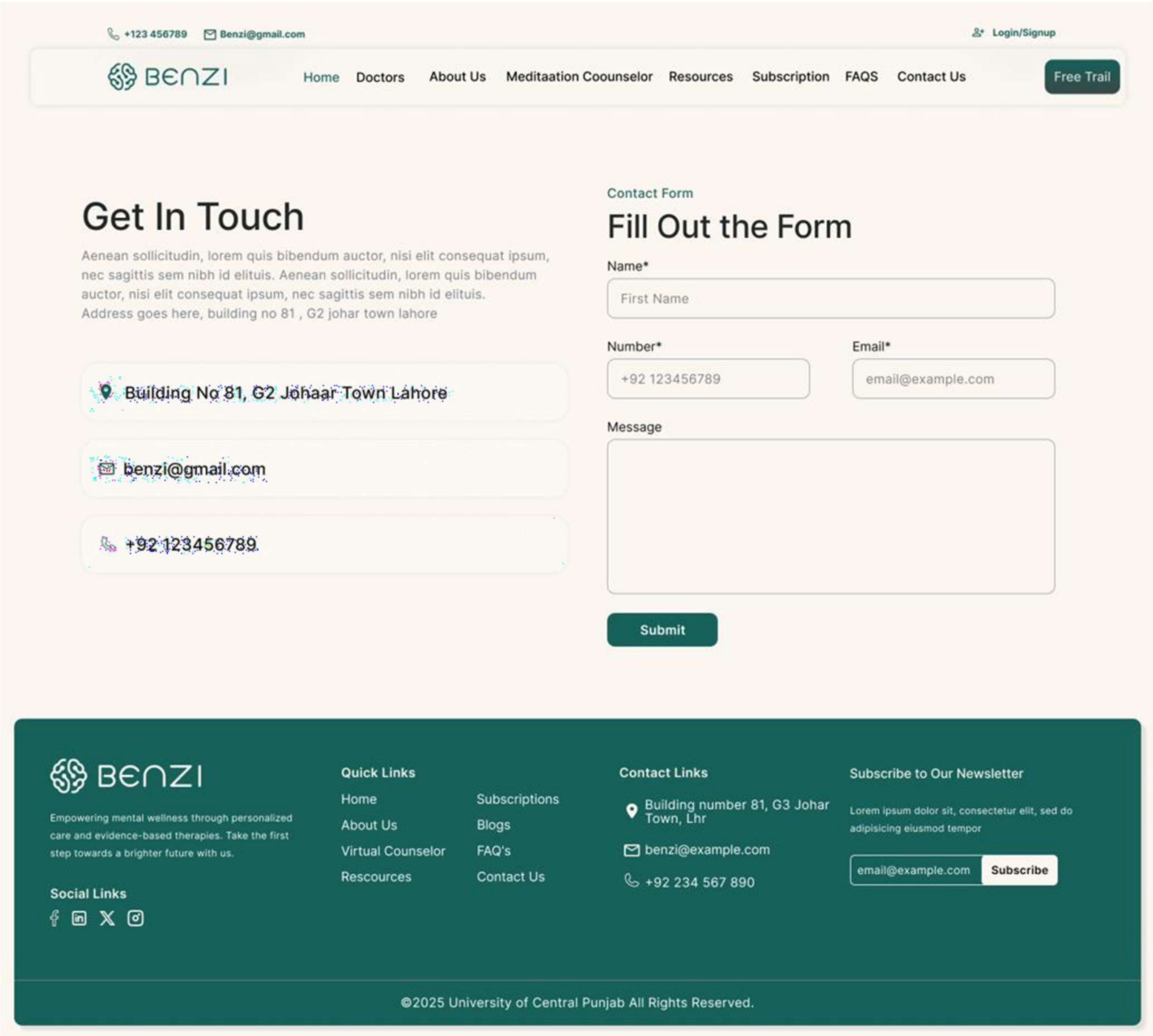

Figure 4.27: Contact Us Support enquiry submission form

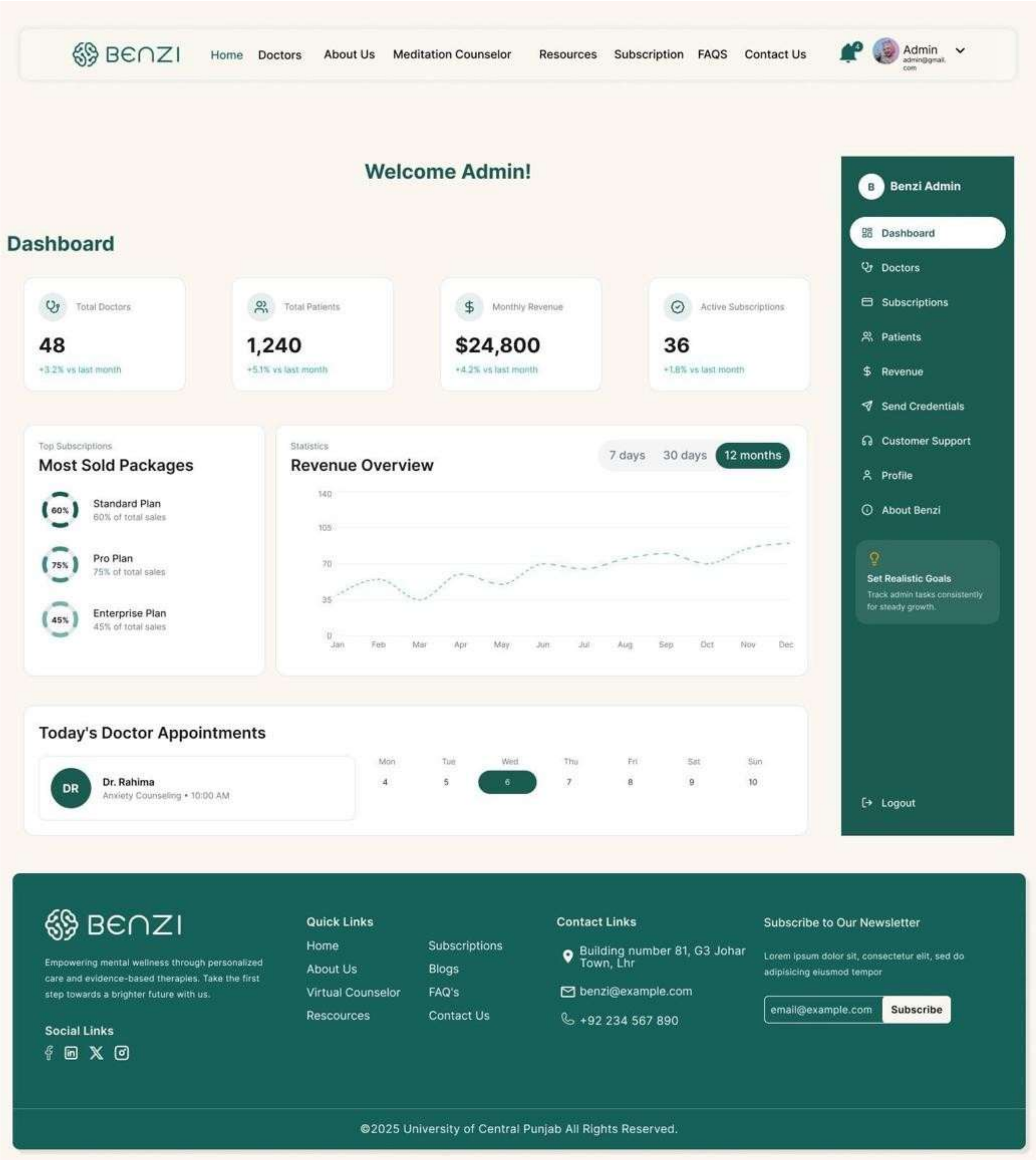

Figure 4.28: Admin Dashboard KPI overview, revenue chart, most sold packages and today's appointments

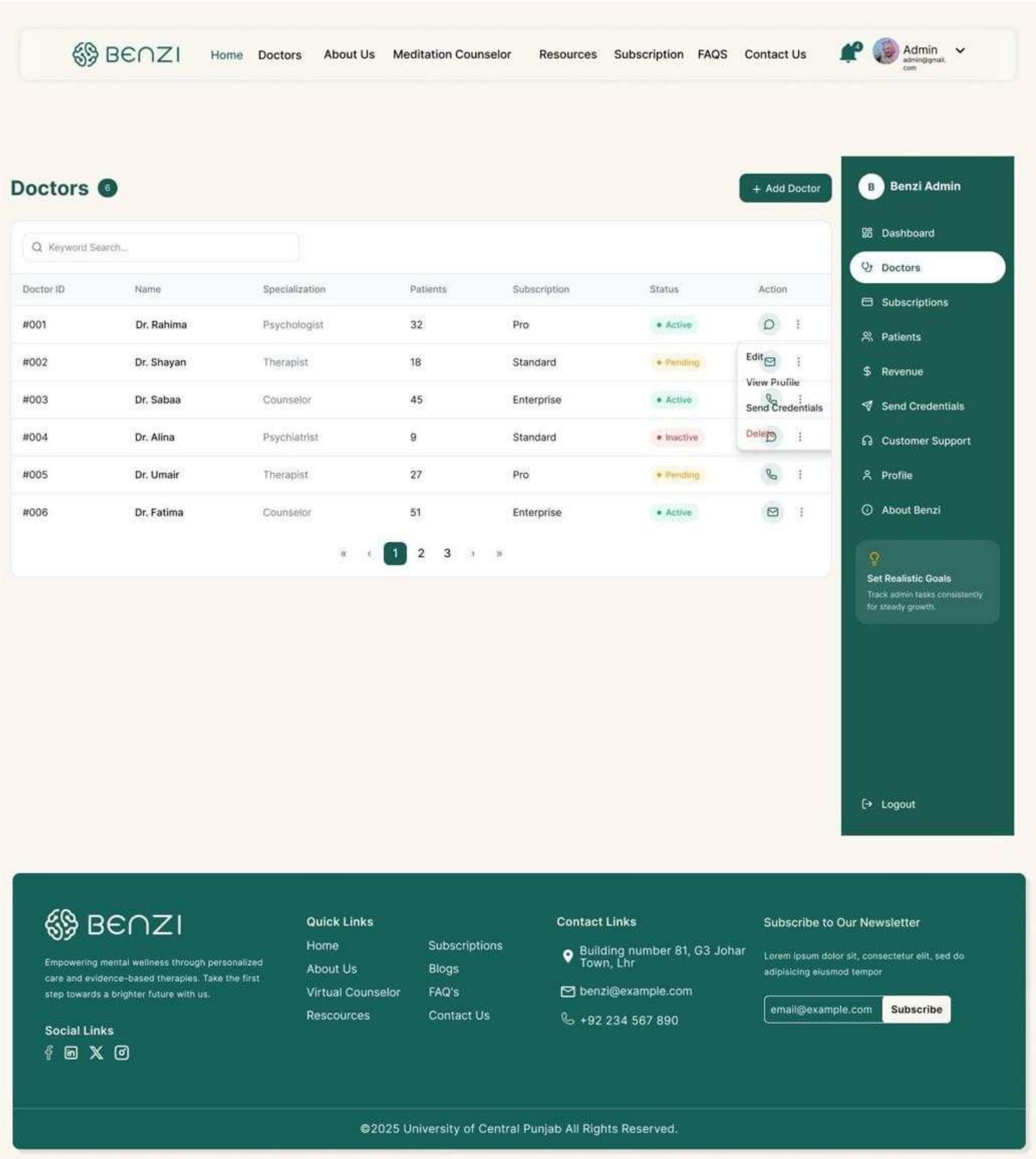

Figure 4.29: Admin Doctors Management — Doctor list with specialization, subscription plan, status and action controls

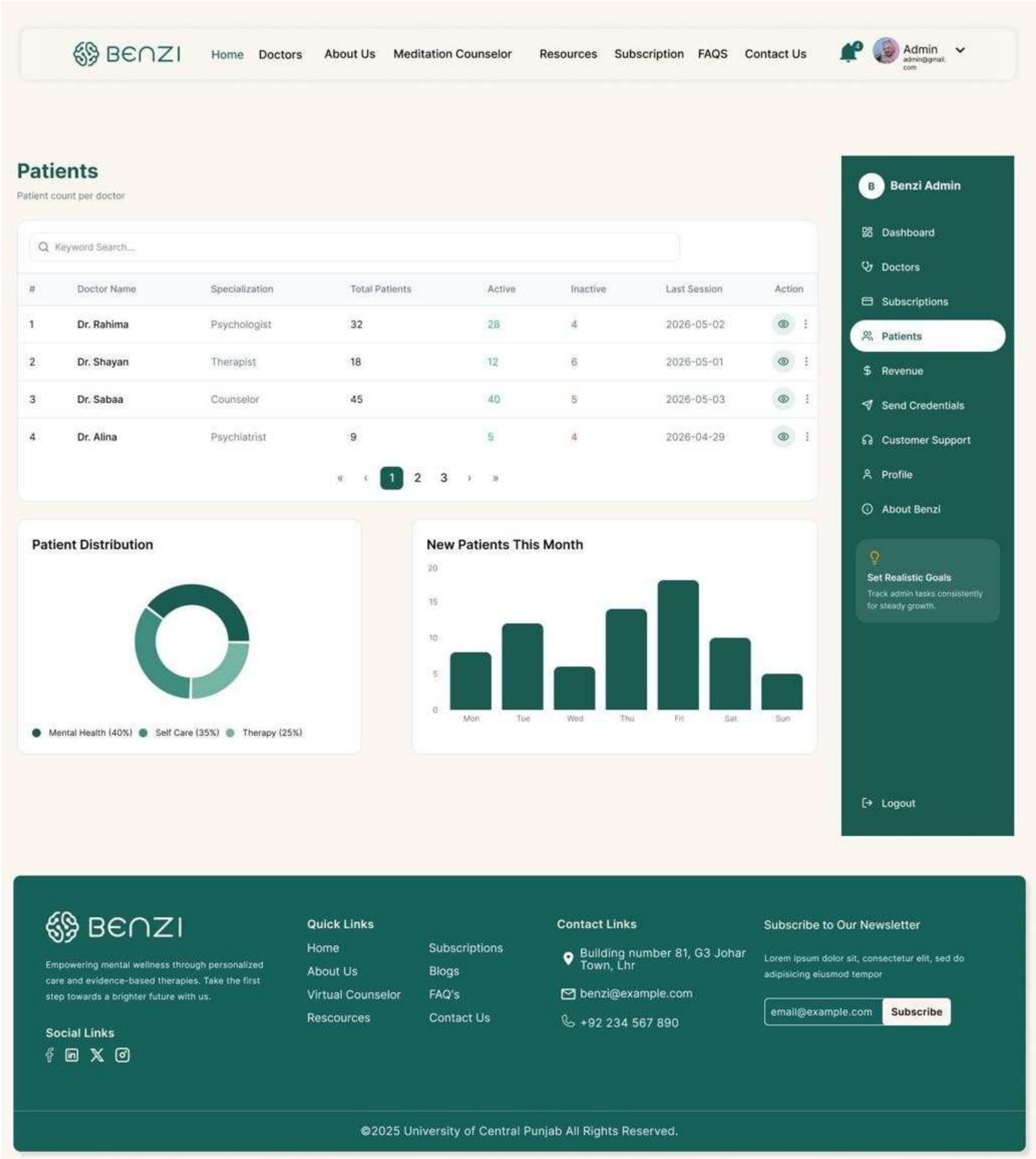

Figure 4.30: Admin Patients Overview Patient counts per doctor, distribution chart and new patient trend

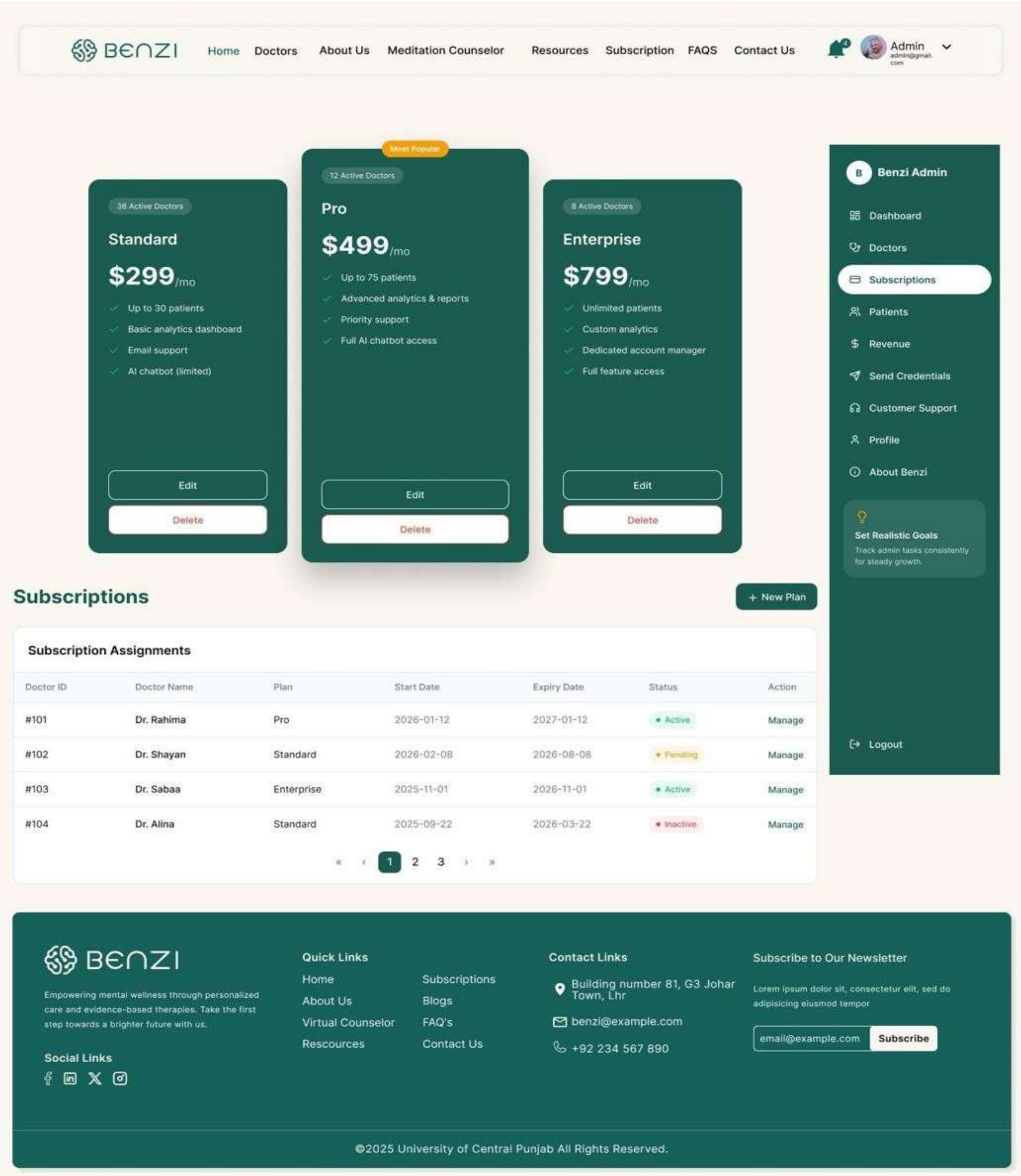

Figure 4.31: Admin Subscriptions Plan management (Standard/Pro/Enterprise) and subscription assignment table

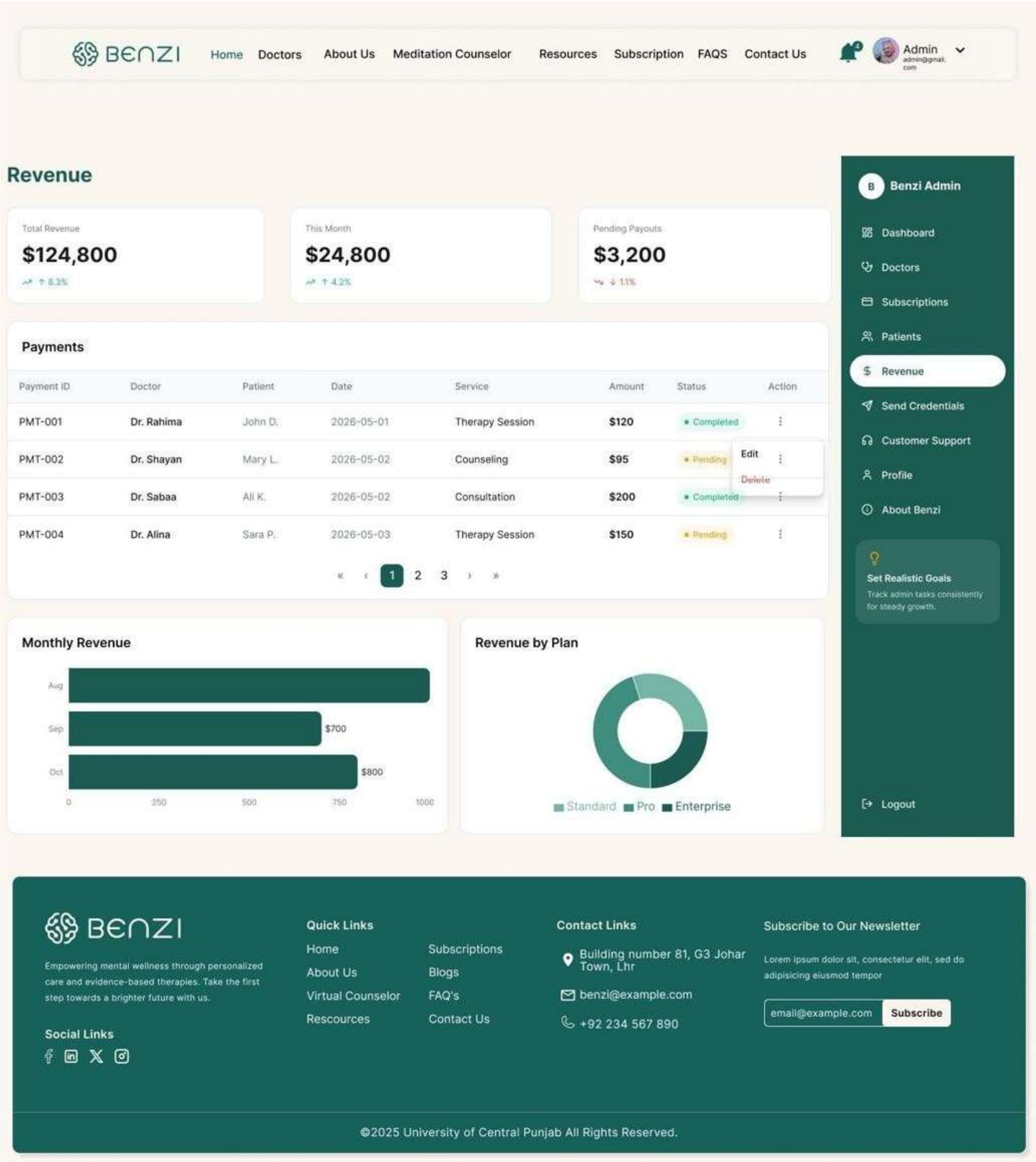

Figure 4.32: Admin Revenue Total revenue KPIs, payments table, monthly bar chart and plan based breakdown

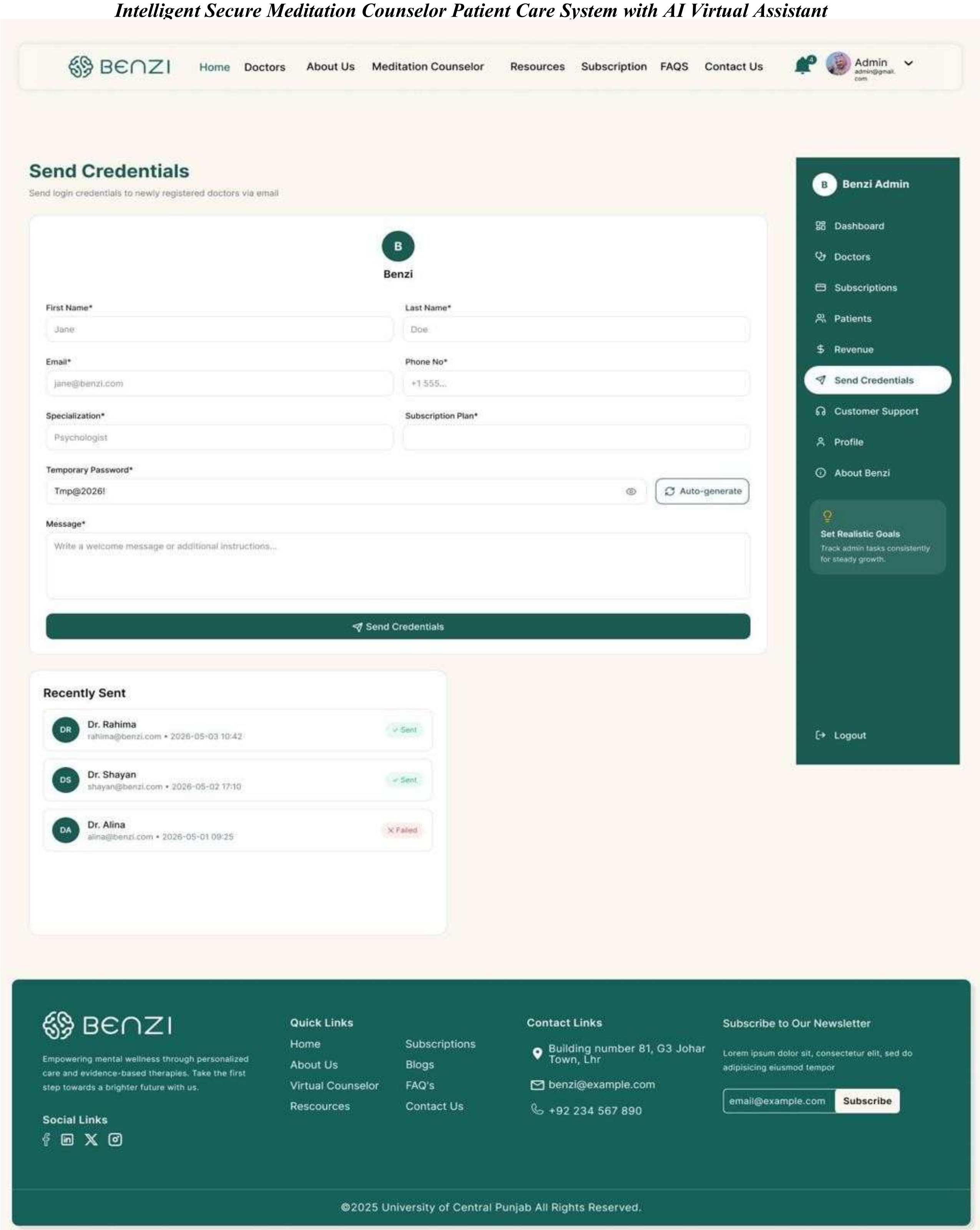

Figure 4.33: Admin Send Credentials (Active State) Send Credentials panel with highlighted navigation state

1.34 Additional Information

The complete interactive prototype of BENZI.AI is accessible via Figma at the following link: https://www.figma.com/design/WUv3qWjLqaIE71mxhA2ReW/Final Year Project
The prototype covers all 27 screens documented in Section 4.2. All Patient Portal screens share a consistent collapsible right side navigation panel containing: Dashboard, Conversations, Goals, Progress, Appointment, Help & Support, Reports, Profile, and Logout. All Therapist Portal screens share a consistent right side panel containing: Dashboard, Appointment, Clients, Services, Subscription, Payment, Profile, About Benzi, and Logout. This persistent navigation pattern eliminates the need for back button navigation and ensures zero depth access to any function from any screen.

The color scheme of deep teal (#0F5C52 approximately) on cream is applied consistently across all authenticated screens. Public screens use the same palette with an off white/cream background to distinguish pre login from post login environments.

1.35 Other Design Details

1.35.1 Security and Privacy Design

BENZI.AI implements a defense in depth security model designed to meet HIPAA and GDPR compliance requirements. All data transmitted between client browsers and the backend is encrypted using TLS 1.2 or higher over HTTPS no plain HTTP communication is permitted in any environment. Session management relies on JWT tokens with a 3600 second expiry, requiring re authentication after timeout. Two Factor Authentication (2FA) is supported for all user roles. All patient healthcare data including session notes, medical history, uploaded reports, and AI interaction logs is encrypted at rest using AES 256, with encryption keys managed through a cloud based Key Management Service (AWS KMS or Azure Key Vault). The Admin role is architecturally blocked from all patient data at the RecordsService middleware level, returning HTTP 403 FORBIDDEN regardless of any token or credential presented.

1.35.2 Security and Privacy Design

The BENZI AI virtual assistant is designed strictly as a supportive guidance tool, not a diagnostic or prescriptive clinical system. The following AI governance rules are enforced by the safetyFilter() module within the AI Engine. First, the AI will never produce a clinical diagnosis, medication prescription, or crisis intervention directive any such query is detected and redirected to professional resources. Second, all AI recommendations are presented transparently with contextual attribution (e.g., "Based on your recent sessions…"). Third, user consent for AI feature activation is explicitly obtained at first use. Fourth, all AI interaction logs are stored in MongoDB and are accessible to the treating Therapist the patient is informed of this at onboarding. Fifth, the AI model is not trained on individually identifiable patient data from the live system; training uses controlled and anonymized datasets only.

1.35.3 Maintainability

The modular service oriented architecture ensures that any individual component the AI Engine, the Notification Service, the Appointment Service can be independently scaled on the cloud platform without redeployment of the full system. MongoDB's horizontal sharding capability supports growth in patient and session data volume. The n8n automation platform supports addition of new workflow types without code changes to the backend. All React.js UI components are independently importable modules, allowing new screens to be added without modifying existing portal layouts.

1.35.4 Accessibility Design

All text elements meet WCAG AA contrast ratio requirements. Font sizes are set at minimum 12pt for body content across all screens, with headings at 16pt or above. All interactive elements buttons, form fields, navigation items are keyboard navigable with visible focus indicators. The mood tracker on the Patient Dashboard uses universally recognizable emoji based icons with text labels to eliminate ambiguity. Color alone is never used as the sole indicator of status all appointment statuses (Confirmed/Pending/Cancelled) additionally use distinct icons alongside color coding.

1.35.5 Real Time Features Design

Live therapy sessions use WebSocket connections maintained between the client browser and the Node.js backend for low latency bidirectional communication. The WebSocket layer implements a three retry auto reconnect protocol with 2 second intervals on connection drop, minimizing session interruption. The Zoom SDK integration creates E2EE encrypted meeting rooms with the session token distributed by the backend the meeting URL is never stored in MongoDB after session completion, only the session metadata (ID, duration, status) is retained.

1.35.6 Gamification Design

The wellness goal system implements a points based engagement mechanism to encourage patient adherence to therapeutic goals. Each goal assigned by a therapist carries a configurable point value. On completion, the addPoints() function credits the patient's profile with the earned points, and a visual counter on the Patient Goals screen animates to reflect the new total. A community sentiment indicator (Negative / Neutral / Positive counters visible on the Goals screen) provides social proof of platform engagement without exposing individual patient data. Goal completion progress is visualized as a pie chart showing Completed (70%), In Progress (20%), and Pending (10%) breakdown across all active goals.

Chapter 6. Test Specification and Results

1.36 Test Case Specification

This section presents test case specifications for the BENZI.AI system covering all major implemented and partially implemented modules. Test cases are designed to validate functional requirements corresponding to the primary use cases defined in the Phase 2 SDS. Modules covered include: User Authentication & Registration, Appointment Booking & Scheduling, Therapist Dashboard & Patient Records, Wellness Goals & Gamification, Notifications, Payment Processing, Admin Panel & Subscription Management, and the AI Virtual Assistant module.

Table 6.1: TC 1

Field	Details
Identifier	TC 1
Related Requirement(s)	UC 6, SRS Section 3.1 User Authentication & RBAC

Short Description	Verify that a registered user can log in with valid credentials and is redirected to the correct role based dashboard
Pre condition(s)	Verified user account exists in MongoDB. System is running. Login page is accessible.
Input Data	Email: testpatient@benzi.ai / Password: Test@1234 / Role: Patient / 2FA Code: Valid OTP
Detailed Steps	1. Open the BENZI.AI login page. 2. Enter registered email and password. Enter valid 2FA code when prompted. 4. Click Login. 5. Observe the redirect destination and dashboard content displayed.
Expected Result(s)	JWT token is issued. User is redirected to the Patient Dashboard. Role specific navigation is shown (appointments, goals, AI chat). No therapist or admin elements are visible.
Post condition(s)	Active JWT session exists. Login event is recorded in system audit log.
Actual Result(s)	Login successful. Patient Dashboard loaded correctly. Role isolation confirmed.
Test Case Result	Pass

Table 6.2: TC 2 New Patient Registration

Field	Details
Identifier	TC 2
Related Requirement(s)	UC 6, SRS Section 3.1 User Authentication & RBAC
Short Description	Verify that a new patient can register successfully and receives an email verification link
Pre condition(s)	No existing account with test email. SendGrid API is operational. Registration page is accessible.
Input Data	Full Name: Sara Khan / Email: sarakhan@test.com / Password:

	Sara@5678 / Role: Patient / Phone: 03001234567
Detailed Steps	1. Navigate to Registration page. 2. Select role as Patient. 3. Fill all required fields. 4. Click Register. 5. Open registered email inbox. 6. Click the verification link received.
Expected Result(s)	Account created in MongoDB with status PENDING_VERIFICATION. Verification email dispatched via SendGrid. After clicking link, status updates to VERIFIED. User redirected to login page with success message.
Post condition(s)	Verified patient account exists. User can log in with registered credentials.
Actual Result(s)	Registration successful. Verification email received and link functional. Account status updated correctly.
Test Case Result	Pass

Table 6.3: TC 3 Book Appointment with Online Payment

Field	Details
Identifier	TC 3
Related Requirement(s)	UC 1, SRS Section 3.1 Appointment Scheduling
Short Description	Verify that a patient can successfully book an available appointment slot using online payment
Pre condition(s)	Patient is logged in. At least one verified therapist with available slots exists. Stripe API operational in test mode.
Input Data	Therapist: Dr. Ahmed (T001) / Date: Next available weekday / Time: 10:00 AM / Payment: Stripe test card 4242 4242 4242 4242
Detailed Steps	1. Navigate to Book Appointment from Patient Dashboard. 2. Select therapist Dr. Ahmed. 3. Pick an available date and time slot. 4. Choose Pay Online. 5. Enter Stripe test card details. 6. Click Confirm Booking. 7. Check confirmation screen and email/SMS inbox.

Expected Result(s)	Stripe payment returns status succeeded. Appointment record inserted in MongoDB with status CONFIRMED. Patient and Therapist receive confirmation email (SendGrid) and SMS (Twilio). Appointment visible on Patient Dashboard. n8n schedules 24 hour reminder.
Post condition(s)	Confirmed appointment in database. Slot marked unavailable for other patients. Reminder workflow active in n8n.
Test Case Result	Pass

Table 6.4: TC 4 Upload and Access Encrypted Patient Record

Field	Details
Identifier	TC 4
Related Requirement(s)	UC 3, SRS Section 3.1 Secure Data Practices
Short Description	Verify that a therapist can upload an encrypted patient record, the patient can view it read only, and the admin is denied access
Pre condition(s)	Therapist is logged in. Linked patient profile exists. Cloud storage (AWS S3 / Azure Blob) is operational.
Input Data	File: session_notes.pdf (valid PDF, under 10MB) / Patient ID: P001 / Therapist ID: T001
Detailed Steps	1. Therapist opens patient P001 profile. 2. Clicks Upload Record and selects session_notes.pdf. 3. Clicks Upload and waits for confirmation. 4. Log in as Patient P001 and navigate to My Records. 5. Attempt to open the uploaded file. 6. Log in as Admin and attempt to access patient records of P001.
Expected Result(s)	File validated, AES 256 encrypted, and stored in cloud storage. File URI persisted in MongoDB. Patient views file in read only mode with no edit or delete options. Admin receives HTTP 403 FORBIDDEN no patient data accessible.
Post condition(s	

)	Encrypted file record exists in MongoDB and cloud storage. RBAC enforcement confirmed for all three roles.
Actual Result(s)	Upload and encryption successful. Patient read only access confirmed. Admin access correctly blocked with 403 response.
Test Case Result	Pass

Table 6.5: TC 5 Assign Wellness Goal and Award Gamification Points

Field	Details
Identifier	TC 5
Related Requirement(s)	UC 4, SRS Section 3.1 Goal Assignment & Progress Tracking
Short Description	Verify that a therapist can assign a wellness goal, a patient can mark it complete, and gamification points are awarded
Pre condition(s)	Therapist and Patient are verified and linked. Both are logged in on separate sessions.
Input Data	Goal: "Practice 10 minutes of mindfulness daily" / Due Date: 7 days from today / Points: 50 / Patient: P001
Detailed Steps	1. Enters goal description, due date, and points value. 2 Clicks Save. 3. Switch to Patient P001 session. 4. Open Wellness Goals and confirm new goal appears with status ACTIVE. 5. Click Mark as Complete. 6. Observe points counter update and notification.
Expected Result(s)	Goal created in MongoDB with status ACTIVE. Goal visible on Patient Dashboard with progress indicator. On completion, status updates to COMPLETED. Points counter increments by 50. Congratulatory notification sent to Patient. Progress update sent to Therapist.
Post condition(s)	Goal status COMPLETED in database. Patient points balance updated. Both dashboards reflect the change.
Actual Result(s)	Goal assigned and visible to patient. Completion and points award functioning correctly. Notifications dispatched successfully.
Test Case Result	Pass

Table 6.6: TC 6 Automated Notification Dispatch

Field	Details
Identifier	TC 6
Related Requirement(s)	UC 1, UC 4, SRS Section 3.1 Notifications & Reminders
Short Description	Verify that automated email and SMS notifications are dispatched correctly for appointment confirmation and goal completion events
Pre condition(s)	Twilio and SendGrid APIs operational. Confirmed appointment (TC 3) and completed goal (TC 5) exist. Notification Service running.
Input Data	Patient Email: sarakhan@test.com / Phone: 03001234567 / Therapist Email: dr.ahmed@benzi.ai / Events: Appointment Confirmation, Goal Completion
Detailed Steps	1. Complete appointment booking (TC 3) and observe email and SMS receipt. 2. Complete goal marking (TC 5) and observe notification. 3. Check SendGrid activity dashboard for delivery status. 4. Check Twilio logs for SMS delivery. 5. Verify n8n workflow shows reminder scheduled 24 hours before appointment.
Expected Result(s)	Confirmation email and SMS received by both Patient and Therapist within 60 seconds of booking. Goal completion notification received by Patient. n8n reminder workflow visible in automation logs with correct trigger time. All delivery statuses show DELIVERED in SendGrid and Twilio dashboards.
Post condition(s	
)	All notification events logged in MongoDB audit trail. No failed delivery entries for the test events.
Actual Result(s)	Email notifications delivered within expected time. SMS delivered with ~45 second delay. n8n reminder job scheduled correctly.

Table 6.7: TC 7 Automated Notification Dispatch

Field	Details
Identifier	TC 7
Related Requirement(s)	UC 1, SRS Section 3.1 Appointment Scheduling, Payment Processing
Short Description	Verify that a failed Stripe payment prevents appointment confirmation and the patient is notified appropriately without creating an orphaned
record	
Pre condition(s)	Patient logged in. Therapist with available slots exists. Stripe test mode active.
Input Data	Therapist: Dr. Ahmed (T001) / Slot: Available / Payment: Stripe test decline card 4000 0000 0000 0002
Detailed Steps	1. Patient selects available slot for Dr. Ahmed. 2. Chooses Pay Online and enters Stripe decline test card. 3. Clicks Confirm Booking. 4. Observe on screen response. 5. Check database for any new appointment record.
6. Check patient email for failure notification.	
Expected Result(s)	Stripe returns payment failure event. No appointment record inserted in MongoDB. Selected slot remains available for other patients. On screen error message displayed: "Payment unsuccessful. Please try a different payment method." Failure notification email sent via SendGrid. No SMS dispatched.
Post condition(s	Database state unchanged. Slot still available. No orphaned appointment record exists.
Actual Result(s)	Payment failure handled correctly. No record created. Slot remained available. Error message displayed as expected. Failure
Test Case Result	Pass

Table 6.8: TC 8 Automated Notification Dispatch

Field	Details
Identifier	TC 8
Related Requirement(s)	UC 1, UC 4, SRS Section 3.1 Notifications & Reminders
Related Requirement(s)	UC 6, SRS Section 3.1 Admin Panel & Subscription Management
Short Description	Verify that admin can verify a pending therapist account and assign a subscription plan via Stripe
Pre condition(s)	Admin account active. Therapist registration with status PENDING exists. Stripe test mode active.
Input Data	Admin: admin@benzi.ai / Therapist ID: T002 (PENDING) / Plan: Pro Monthly
Detailed Steps	1. Admin logs in and opens Admin Panel. 2. Views pending therapist list and selects T002. 3. Reviews credentials and clicks Approve. 4. Confirms action in dialog. 5. Assigns Pro Monthly plan to T002 via Subscription Management. 6. Verify T002 receives welcome email. 7. Log in as T002 and verify dashboard access.
Expected Result(s)	T002 status updated to VERIFIED in MongoDB with timestamp. Stripe subscription created with status active. Welcome email dispatched to T002. Therapist T002 can log in and access full Therapist Dashboard. Admin blocked from accessing any patient record throughout (HTTP 403 confirmed).
Post condition(s)	T002 fully onboarded. Subscription record in Stripe referenced in MongoDB. Admin verification action recorded in system audit log.
Test Case Result	Pass

Table 6.9: TC 9 Automated Notification Dispatch

Field	Details
Identifier	TC 9
Related Requirement(s)	UC 5, SRS Section 3.1 AI Powered Virtual Assistant
Short Description	Verify that the AI Virtual Assistant returns a contextually relevant, guardrail compliant response based on the patient's uploaded report and message, and correctly rejects out of scope clinical queries
Pre condition(s)	Patient logged in. Therapist has uploaded a patient report (PDF) which has been processed through the RAG pipeline. AI module (Python/FastAPI) and LLM (Llama 3 / Mistral / Phi 3 via Ollama) are running. Guardrails configured with topic limits and crisis detection rules.
Input Data	Patient Message 1 (valid): "What relaxation techniques have been recommended for me?" / Patient Message 2 (out of scope): "Do I have anxiety disorder?" / Patient Report: Uploaded PDF containing therapist notes and treatment plan
Detailed Steps	1. Patient opens the AI Virtual Assistant chat. 2. Submits Message 1 asking about relaxation techniques. 3. Observe AI response and verify it references content from the uploaded patient report. 4. Submit Message 2 asking for a clinical diagnosis. 5. Observe AI response and verify guardrail behaviour. 6. Optionally save AI response to personal notes. 7. Check MongoDB for logged interaction entries.
Expected Result(s)	For Message 1: AI retrieves relevant chunks from the RAG processed patient report, generates a contextually accurate supportive response, and returns it within the topic boundaries set by the clinician. Response does not include any clinical diagnosis. For Message 2: Guardrail layer detects out of scope clinical query and returns a polite redirect response (e.g., "I'm not able to provide a diagnosis please speak with your therapist directly.").

	No LLM generated diagnosis is returned. Interaction log for both messages stored in MongoDB with timestamp, patientId, message, and response fields.
Post condition(s)	Both interaction records exist in MongoDB AI logs. No clinically diagnostic content present in any stored response. Patient notes updated if save action was triggered.
Actual Result(s)	Context aware response for Message 1 correctly referenced report content. Guardrail for Message 2 triggered successfully redirect response returned. Interaction logs persisted in database.
Test Case Results	Pass

1.37 Summary of Test Results

Table 6.2: Summary of All Test Results

Module Name	Test cases run	Number of defects found	Number of defects corrected so far	Number of defects still need to be corrected
User Authentication & Registration	TC 1, TC 2	1	1	0
Appointment Booking & Scheduling	TC 3	1	1	0
Therapist Dashboard & Patient Records	TC 4	1	0	1
Wellness Goals & Gamification	TC 5	0	0	0
Notifications (Email / SMS)	TC 6	1	1	1
Payment Processing (Stripe)	TC 7	0	0	0
Admin Panel & Subscription Management	TC 8	1	1	0
AI Virtual Assistant	TC 9	1	0	1
Complete System	TC 1 to TC 9	6	4	2

Chapter 7. Project Completion Status/Conclusion

The development of BENZI.AI has been successfully completed. All planned modules, features, and system components identified during the requirements and design phases have been implemented, integrated, and tested. The system includes Patient, Therapist, and Administrator portals with role based access control and secure authentication mechanisms.
Major completed features include patient and therapist registration, therapist verification, appointment scheduling, encrypted communication channels, AI powered virtual assistant, secure medical record management, goal assignment and tracking, subscription management, payment processing through Stripe, analytics dashboards, and notification services. The AI virtual assistant has been implemented using a fine tuned Ollama Llama 3.5:3B model integrated with a context aware Retrieval Augmented Generation (RAG) pipeline to provide personalized assistance and recommendations.
Comprehensive testing was performed on all modules, including authentication, appointment management, therapist verification workflows, AI interactions, payment processing, and security controls. Test results confirmed that all core functionalities operate according to the specified requirements. The project has achieved its intended objectives and is ready for deployment and future scalability enhancements.

Table 7.1: Project Completion Status

Module Name	**Status** (Complete, Partially Implemented, Not Implemented)
User Authentication & Authorization Module (JWT, RBAC, 2FA)	Complete
Patient Management Module	Complete
Therapist Verification Module	Complete
Appointment Scheduling Module	Complete
Secure Communication & Chat Module	Complete
AI Virtual Assistant Module (Fine Tuned Ollama + RAG)	Complete
Medical Records Management Module	Complete
Goal Assignment & Progress Tracking Module	Complete
Payment Processing Module (Stripe)	Complete
Notification & Reminder Module	Complete
Analytics & Reporting Module	Complete

Admin Management Module	Complete
Video Consultation Module (Google Meet / Jitsi Meet)	Complete
Complete System	Complete
Overall Completion Percentage	100%

Table 7.2: Objective(s)/Target(s) Status

Target/Objective	**Status** (Completed, Partially Completed, Not Completed)	**Reason(s)**
Develop a Secure Therapist Patient Management Platform	Completed	Successfully implemented with role based portals and secure data management.
Implement Automated Appointment Scheduling	Completed	Real time appointment booking and scheduling functionality implemented.
Integrate Secure Communication and Consultation Channels	Completed	Secure chat and encrypted online consultation features implemented.
Incorporate an AI Powered Virtual Assistant	Completed	Fine tuned Ollama Llama model integrated with context aware RAG pipeline.
Enable Goal Assignment and Progress Tracking	Completed	Therapists can assign goals and patients can track progress through dashboards.
Establish Secure Medical Record Storage	Completed	Medical records stored securely using MongoDB Atlas and encryption mechanisms.
Implement Robust Authentication and Access Control	Completed	JWT authentication and role based access control successfully implemented.
Ensure Comprehensive System Security	Completed	AES 256 encryption, secure APIs, protected authentication, and access control implemented.
Automate Notifications and	Completed	Automated notifications and appointment reminders implemented.

Reminders		
Provide Data Driven Progress Analytics	Completed	Analytics dashboards and progress reports available for therapists and patients.
Enhance Accessibility and User Experience	Completed	Responsive web interface developed for multiple devices and user roles.

References

- React.js Documentation https://react.dev
- Node.js Documentation https://nodejs.org
- Express.js Documentation https://expressjs.com
- MongoDB Atlas Documentation https://www.mongodb.com
- Ollama Documentation https://ollama.com
- JSON Web Token (JWT) Documentation https://jwt.io
- Stripe API Documentation https://stripe.com/docs
- Google Meet Developer Documentation https://developers.google.com
- Jitsi Meet Documentation https://jitsi.org
- AES 256 Encryption Standard Documentation
- HIPAA Security Guidelines
- GDPR Data Protection Regulation Documentation.

Appendix A Glossary

Term	**Definition**
AI	Artificial Intelligence
RAG	Retrieval Augmented Generation
JWT	JSON Web Token

Term	**Definition**
RBAC	Role Based Access Control
AES 256	Advanced Encryption Standard (256 bit Encryption)
API	Application Programming Interface
MERN Stack	MongoDB, Express.js, React.js, and Node.js
MongoDB Atlas	Cloud based MongoDB Database Service
Therapist	Healthcare professional providing counselling services
Patient	Individual receiving therapy or counselling services
Administrator	User responsible for managing the platform
AI Virtual Assistant	Intelligent chatbot providing personalized support
Appointment	Scheduled consultation between therapist and patient
Medical Record	Securely stored patient health information
Goal Tracking	Monitoring patient progress through assigned goals
Stripe	Online payment processing platform
Google Meet	Video conferencing platform

Term	**Definition**
Jitsi Meet	Secure video communication platform
HTTPS	HyperText Transfer Protocol Secure
Dashboard	Interface displaying system information and analytics
Encryption	Process of securing data from unauthorized access
Authentication	Process of verifying user identity
Authorization	Process of granting access permissions
Telemedicine	Delivery of healthcare services through digital technologies
Analytics	Data driven insights and progress reporting

Appendix B IV & V Report

(Independent verification & validation)

IV & V Resource

Name Signature

S#	Defect Description	Origin Stage	Status	Fix Time	
				Hours	Minutes
1					
2					
3					
…					

Table B.2: List of non trivial defects

Appendix C Deployment/Installation Guide

Deployment/Installation Guide

BENZI.AI is a web based MERN stack application consisting of a React.js frontend, Node.js and Express.js backend, MongoDB Atlas cloud database, and an AI engine powered by a fine tuned Ollama Llama model integrated with a context aware Retrieval Augmented Generation (RAG) pipeline. The system is currently deployed in a local development environment, while the frontend is prepared for deployment on Vercel.

1.38 System Requirements

- Operating System: Windows 10/11, Linux, or macOS
- Node.js Version 18 or above
- MongoDB Atlas Account
- Ollama Installed and Configured
- Modern Web Browser (Chrome, Edge, Firefox, Safari)
- Internet Connection

1.39 Installation Steps

1.39.1 Frontend Setup

1. Clone the frontend repository.
2. Open terminal in the frontend directory.
3. Install dependencies:

npm install

4. Start the frontend application:

npm run dev

1.39.2 Backend Setup

1. Clone the backend repository.
2. Open terminal in the backend directory.
3. Install dependencies:

npm install

4. Configure environment variables including MongoDB Atlas URI, JWT Secret Key, Stripe API Keys, and AI service settings.
5. Start the backend server:

npm start

1.39.3 Database Configuration

1. Create a MongoDB Atlas cluster.
2. Obtain the database connection string.
3. Add the connection string to the backend environment configuration.
4. Run the application to automatically establish database connectivity.

1.39.4 AI Engine Configuration

1. Install Ollama on the host machine.
2. Load the fine tuned Llama model.
3. Configure the RAG pipeline and knowledge base.
4. Connect the AI service with the backend API endpoints.

1.40 10.3 Deployment

The frontend application can be deployed on Vercel, while the backend services can be hosted on cloud platforms such as AWS, Azure, or Google Cloud Platform. MongoDB Atlas serves as the cloud database. After deployment, users can access the system through a web browser and utilize all platform features including appointment management, AI assistance, secure communication, and therapist patient collaboration.

Appendix D User Manual

Patients can create an account through the registration page and are automatically assigned the Patient role. After logging in, patients can browse available therapists, book appointments, communicate with therapists through dedicated chat modules, join online therapy sessions, interact with the AI virtual assistant, view progress analytics, and track assigned therapeutic goals.Therapists create accounts by submitting professional information, medical certifications, academic degrees, and supporting documents. After verification and approval by an administrator, therapists gain access to the Therapist Dashboard where they can manage appointments, maintain patient records, assign goals, review analytics, communicate with patients, and monitor treatment progress.Administrators access the Admin Panel to verify therapist applications, approve or reject registration requests, manage platform subscriptions, monitor revenue and payments, supervise system activity, and maintain platform operations. The role based architecture ensures that each user can access only the features and information relevant to their responsibilities.